

Foundation Models and Reasoning LLMs for Scientific Multimodal Time Series: A Survey

Huirui Li

Abstract—Time series and spatio-temporal dynamics arising in natural systems—such as global weather, long-term climate, hydrological flood regimes, ocean circulation, earthquake wave propagation, satellite earth observations, and heliophysical radiation—differ fundamentally from generic tabular or urban traffic series. They are governed by continuous non-linear partial differential equations, multi-scale turbulent interactions, heterogeneous spatial geometries, and strict physical conservation laws. Recently, the emergence of pre-trained foundation models (PFMs) and scientific reasoning Large Language Models (LLMs) has instigated a paradigm shift across computational earth and physical sciences. This survey provides the first comprehensive, systematic review of foundation models and reasoning LLMs tailored for scientific multimodal time series. Guided by the PRISMA 2020 methodology, we review landmark architectures developed between 2021 and 2026 across major natural science domains. We introduce a systematic taxonomy structured along four orthogonal axes: (1) scientific domains and multimodal data representations, (2) foundation model architectural backbones (including 3D spherical transformers, multi-mesh graph neural networks, continuous neural operators, and generative diffusion ensembles), (3) levels of physical law integration (from purely data-driven to soft physical penalties and hybrid PDE solvers), and (4) functional roles of LLMs in scientific reasoning (as direct predictors, contextual enhancers, and autonomous tool-orchestrating agents). Furthermore, we curate standardized operational evaluation protocols (such as WeatherBench 2 and SciTS), contrast model scale against forecast horizon, and identify crucial open frontiers spanning universal cross-physics tokenization, physical conservation guarantees, and trustworthy autonomous scientific discovery.

Index Terms—Scientific Foundation Models, Multimodal Time Series, Spatio-Temporal Modeling, Large Language Models, Scientific Reasoning, Weather and Climate Forecasting, Physics-Informed Machine Learning.



1 Introduction

Temporal and spatio-temporal dynamics constitute the empirical foundation of the natural sciences. From the planetary atmospheric circulation driving global weather patterns to the sub-surface acoustic vibrations heralding seismic rupture, physical processes unfold across continuous temporal trajectories and multi-scale spatial coordinates. Over the past several decades, the modeling and prediction of these scientific phenomena have predominantly relied on Numerical Weather Prediction (NWP) and classical numerical differential equation solvers, such as the Integrated Forecasting System (IFS) maintained by the European Centre for Medium-Range Weather Forecasts (ECMWF) and finite-difference hydrodynamic models. While physically principled, numerical solvers incur colossal computational burdens, requiring massive supercomputing clusters to perform high-resolution discretization and step-by-step numerical integration.

In parallel, the broader machine learning community has witnessed a revolution driven by pre-trained foundation models (PFMs) and Large Language Models (LLMs). While foundational models initially transformed natural language processing and computer vision, their underlying principles—self-supervised pretraining on planetary-scale corpora, emergent scaling behaviors, and zero-shot downstream transferability—have rapidly migrated into temporal data mining. Earlier

general surveys, notably the seminal work by Jin et al. [1] and its position sequel [2], comprehensively charted large models for general time series analysis (LM4TS) and spatio-temporal data mining (LM4STD). However, as highlighted in Table 1, existing overviews focus heavily on urban computing, human mobility trajectories, traffic sensor networks, and generic 1D benchmark datasets (e.g., electricity, weather station benchmarks, exchange rates).

Natural physical systems diverge fundamentally from human-generated temporal sequences across several intrinsic dimensions:

- 1) **Governing Physical Laws and Conservation Constraints:** Physical time series are bounded by fundamental conservation laws of mass, energy, momentum, and moisture flux. Models that learn purely statistical correlations often generate unphysical states, such as negative precipitation or violation of energy conservation over extended rollouts.
- 2) **High-Dimensional Multimodal Representations:** Scientific investigations integrate heterogeneous observation modalities, including 3D atmospheric pressure tensors (e.g., ERA5 reanalysis with geopotential height, temperature, and wind fields), multi-spectral satellite imagery (e.g., Sentinel-2, Landsat), river gauge discharge records, and continuous seismic waveforms.
- 3) **Continuous Spatio-Temporal Geometries:** The Earth is a non-Euclidean spherical geometry with topological singularities at the poles, rendering standard 2D

H. Li is with the School of Data Science and Artificial Intelligence. Correspondence e-mail: 58367737+li-huirui@users.noreply.github.com.
Manuscript version 1.9 (P8/P9 Continuous Research Iteration), September 27, 2026.

TABLE 1
Comparison of This Survey with Existing Literature and Related Temporal Surveys.

Survey / Study	Year	Scope Focus	Physical Grounding	Multimodal Modalities	LLM Reasoning	Core Domains Covered
Jin et al. [1]	2023	General Spatio-Temporal	Moderate	General Multimodal	Limited	Urban traffic, human trajectory, generic time series
Jin et al. [2]	2024	LLM for Time Series	Low	Text + 1D Series	Moderate (Position)	Benchmark time series forecasting, tokenization
Rasp et al. [3]	2024	Weather Benchmark	High (Meteorology)	Gridded NWP Reanalysis	None	Global numerical and ML weather forecasting
This Survey	2026	Scientific Multimodal	High (PDEs & Laws)	Gridded + Satellite + Waveforms + Text	Comprehensive (Agents & SciTS)	Weather/Climate, Earth Observation, Hydrology, Oceans, Seismology, Heliophysics

Euclidean convolutions and planar patch projections geometrically distorted.

- 4) The Need for Scientific Reasoning and Autonomous Discovery: Beyond numerical point prediction, domain scientists demand causal attribution, physical hypothesis testing, and interactive tool orchestration. This has catalyzed the emergence of scientific reasoning LLMs and autonomous scientific agents capable of operating over complex temporal data.

To bridge this literature gap, this survey delivers the first systematic, domain-science-centered investigation of foundation models and reasoning LLMs developed for scientific multimodal time series. We synthesize breakthroughs from 2021 through 2026 across atmospheric sciences, Earth observation, hydrology, oceanography, seismology, and heliophysics.

1.1 Contributions of This Survey

The primary contributions of this survey are summarized as follows:

- Domain-Science Centered Scoping: We delimit the survey strictly to natural physical systems, explicitly analyzing the coupling between continuous physics, governing equations, and deep representation architectures.
- Four-Dimensional Taxonomy Framework: We establish an orthogonal classification scheme covering (1) Scientific Domains and Modalities, (2) Foundation Model Architectures, (3) Physics Integration Levels, and (4) Functional Roles of Reasoning LLMs.
- Systematic Review Protocol: Adhering to PRISMA 2020 standards, every candidate study is screened against strict empirical criteria and validated via official academic API records.
- Synthesis of Evaluation Protocols and Frontiers: We analyze standardized scientific benchmarks (such as WeatherBench 2 and SciTS), map spatial resolution against maximum forecast lead time, and delineate critical open challenges in physical conservation, multi-physics tokenization, and autonomous scientific discovery.

2 Background and Preliminaries

2.1 Formal Data Modality Formulations

Scientific multimodal temporal datasets exhibit diverse mathematical structures depending on observation platforms, sensing geometries, and sampling frequencies. We formalize the four dominant scientific temporal modalities as follows:

2.1.1 In-Situ Station and Gauge Time Series

Discrete ground monitoring networks (e.g., weather stations, hydrological streamflow gauges, ocean buoys) record continuous physical variables at fixed geospatial coordinates (u_i, v_i) . An in-situ observation sequence is defined as:

$$\mathbf{X}^{(i)} = \{\mathbf{x}_t^{(i)} \in \mathbb{R}^{C_{\text{station}}}\}_{t=1}^T, \quad (1)$$

where T represents the temporal horizon and C_{station} denotes the recorded physical channels (e.g., precipitation, river discharge, 2-meter air temperature).

2.1.2 Gridded Global/Regional Reanalysis Tensors

Atmospheric and oceanic numerical simulations discretize the planetary sphere $\mathbb{S}^2$ into a regular latitude-longitude grid with H latitudinal points and W longitudinal points across L vertical vertical pressure levels. The state tensor at time step t is expressed as:

$$\mathcal{X}_t \in \mathbb{R}^{H \times W \times L \times C_{\text{grid}}}, \quad (2)$$

where C_{grid} incorporates upper-air variables (geopotential height z , temperature T , zonal and meridional wind components u, v , specific humidity q) alongside surface diagnostic fields (mean sea-level pressure msl , surface solar radiation).

2.1.3 Multi-Spectral Remote Sensing Cubes

Earth observation satellites (e.g., Sentinel-2, Landsat-8) capture surface reflectance over space, time, and electromagnetic spectral bands. A multi-temporal remote sensing observation cube is represented as:

$$\mathcal{S}_{1:T} \in \mathbb{R}^{T \times B \times H_s \times W_s}, \quad (3)$$

where B denotes the number of spectral bands (visible RGB, near-infrared, short-wave infrared, synthetic aperture radar polarization), and (H_s, W_s) denotes the spatial pixel footprint.

2.1.4 High-Frequency Continuous Waveforms

Geophysical and seismological networks deploy seismometers recording ground motion acceleration or velocity across three orthogonal physical axes (Vertical Z , North-South N , East-West E). A continuous seismic waveform segment is formalized as:

$$\mathbf{W} \in \mathbb{R}^{3 \times N_s}, \quad (4)$$

where $N_s = f_s \cdot \Delta\tau$ is the number of discrete digital samples collected at sampling frequency f_s (typically 50–100 Hz).

2.2 Core Problem Formulations

2.2.1 Deterministic Autoregressive Rollout Forecasting

$$\hat{\mathcal{X}}_{t+1:t+K} = \mathcal{M}_\theta(\mathcal{X}_{t-M+1:t}), \quad (5)$$

optimized via latitude-weighted multi-step rollout loss:

$$\mathcal{L}_{\text{det}} = \sum_{k=1}^K \lambda_k \|\hat{\mathcal{X}}_{t+k} - \mathcal{X}_{t+k}\|_{\mathbf{w}_{\text{lat}}}^2, \quad (6)$$

where $\|\cdot\|_{\mathbf{w}_{\text{lat}}}$ represents a latitude-weighted norm accounting for spherical grid cell area convergence toward the poles:

$$w_{\text{lat}}(i) = \frac{\cos(\phi_i)}{\frac{1}{H} \sum_{j=1}^H \cos(\phi_j)}. \quad (7)$$

2.2.2 Probabilistic Ensemble Generation

Because chaotic atmospheric and oceanic dynamics exhibit sensitivity to initial conditions (the butterfly effect), deterministic point forecasts blur at longer lead times. Probabilistic foundation models aim to sample from the conditional predictive distribution:

$$p_\theta(\mathcal{X}_{t+1:t+K} \mid \mathcal{X}_{t-M+1:t}), \quad (8)$$

minimized using Continuous Ranked Probability Score (CRPS) or diffusion score-matching loss objectives [4].

2.2.3 Scientific Question-Answering and Autonomous Reasoning

Let $\mathcal{Q}$ represent a natural language scientific inquiry, and $\mathcal{D}_{\text{time}}$ represent an associated temporal or spatio-temporal data sequence. A scientific reasoning LLM $\mathcal{F}_{\text{LLM}}$ generates explanatory reasoning chains and structured conclusions $\mathcal{Y}$:

$$\mathcal{Y} = \mathcal{F}_{\text{LLM}}(\mathcal{Q}, \mathcal{E}(\mathcal{D}_{\text{time}})), \quad (9)$$

where $\mathcal{E}$ is a modality-specific continuous tokenizer or latent temporal encoder [5].

3 Taxonomy and Overview

To systematically classify the rapidly expanding body of literature on scientific multimodal time series foundation models and reasoning LLMs, we construct a four-dimensional taxonomy framework, as illustrated in Fig. 1. Unlike general-purpose temporal surveys that divide models simply by task (e.g., forecasting vs. imputation), our categorization captures the unique physical, mathematical, and algorithmic requirements of scientific domain applications.

3.1 Dimension 1: Scientific Domains and Modalities

Scientific time series foundation models specialize across distinct physical environments:

- **Weather and Climate:** Characterized by 3D spherical atmospheric reanalysis fields (ERA5, MERRA-2, IFS HRES), satellite radiometer scans, and ground radar nowcasting sequences [6], [7], [8], [9], [10], [11].
- **Earth Observation and Remote Sensing:** Composed of irregular multi-spectral satellite revisit series (Sentinel-1 SAR, Sentinel-2 Optical, Landsat-8), high-resolution elevation maps, and surface reflectance temporal trajectories [12], [13], [14], [15], [16], [17], [18].

- **Hydrology and Water Resources:** Integrating multi-basin river streamflow gauges, topological river drainage networks, and gridded surface precipitation/evapotranspiration dynamics [19], [20].
- **Oceanography and Marine Systems:** Involving global eddy-resolving reanalysis fields (e.g., GLORYS12V1), sea surface height anomalies, temperature/salinity profiles, and acoustic sensor logs [21], [22], [23].
- **Geophysics and Seismology:** Continuous 3-component seismic waveforms, phase arrival triggers, fault slip observations, and GPS tectonic deformation velocities [24], [25], [26].
- **Space Weather and Heliophysics:** Continuous solar extreme ultraviolet (EUV) magnetograms, solar wind ion velocities, and geomagnetic disturbance indices [27].
- **Polar Cryosphere and Sea Ice Dynamics:** Involving satellite synthetic aperture radar (SAR) swaths, microwave radiometer brightness temperatures, ice-sheet altimetry, and continuous sea ice concentration fields governing planetary thermodynamic albedo [28], [29].
- **Geo-Energy and Subsurface Reservoirs:** Multiphase fluid flow in heterogeneous porous media, geological carbon sequestration (GCS) plume migration, geothermal reservoir temperature/pressure dynamics, and high-pressure equation-of-state (EOS) phase behavior [30], [31].
- **Cross-Domain Multimodal Foundations:** Universal time series representations bridging sensor networks, in-situ monitors, and multi-scale physical dynamics [32], [33].

As visualized in the Domain $\times$ Modality Research Maturity Matrix (Fig. 2), while gridded weather reanalysis and multi-spectral satellite imagery have achieved high maturity, cross-modal integration (e.g., coupling continuous waveforms with satellite imagery, subsurface permeability with multiphase plumes, or ground gauges with scientific text logs) remains an active frontier.

3.2 Dimension 2: Foundation Model Architectures

We categorize foundation backbones by their spatial-temporal inductive biases:

- **3D Spatial and Spherical Transformers:** Employing 3D Earth-specific patch embeddings, hierarchical Swin-like windowed attention, and vertical pressure-level encodings [6], [9], [34], [10].
- **Multi-Mesh Graph Neural Networks:** Discretizing the spherical Earth via refined recursive icosahedral meshes to ensure uniform spatial resolution without polar distortions [7].
- **Continuous Neural Operators:** Leveraging Fourier transforms and spherical harmonic convolutions (e.g., AFNO, SFNO) to learn discretization-invariant operators on Riemannian function spaces [8], [35].
- **Deformation and Geometry-Adapted Neural Operators:** Mapping complex non-Euclidean computational domains, irregular boundaries, and faulted subsurface formations into canonical coordinate spaces via learned coordinate diffeomorphisms (Geo-FNO) [36].
- **Symplectic and Lie-Poisson Geometric Operators:** Structuring operator layers according to canonical

Taxonomy Tree: Categorization of Scientific Multimodal Time Series Foundation Models



Fig. 1. Comprehensive taxonomy framework for scientific multimodal time series foundation models and reasoning LLMs, structured across four orthogonal dimensions: Scientific Domains & Modalities, Foundation Architectures, Physics Integration Levels, and Roles of Reasoning LLMs.

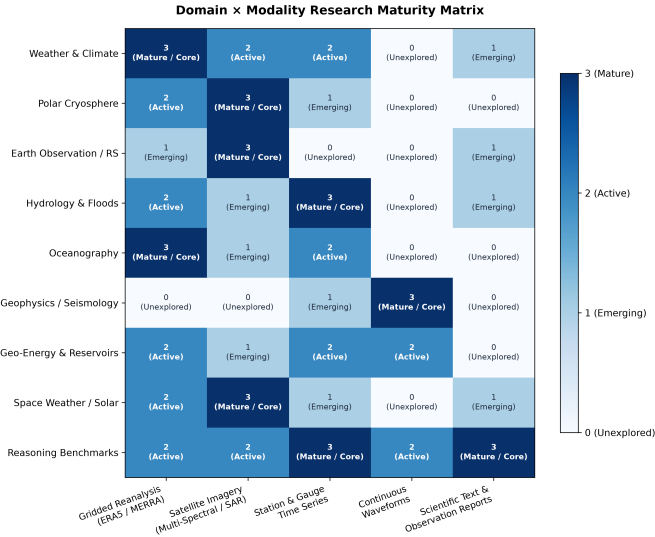


Fig. 2. Domain $\times$ Modality Research Maturity Matrix, highlighting established core areas and emerging multimodal frontiers.

symplectic 2-forms or non-canonical Lie-Poisson brackets to guarantee exact preservation of Hamiltonian energy, Casimir invariants, and Kelvin's circulation theorem [37], [38].

- **High-Order Tensor and Quantum-Classical Hybrid Operators:** Factorizing infinite-dimensional spectral convolution weights via Tucker and Tensor-Train factorizations or embedding spectral transformation into parameterized quantum circuits (PQC) for scalable PDE solving and wavefield tomography [39], [40], [41].
- **Scale-Adaptive Multi-Modal Remote Sensing Architectures:** Unifying multi-constellation SAR and optical imagery via cross-modal contrastive autoencoding with 2D-ALiBi spatial extrapolation and scale-conditioned joint embedding architectures (CROMA, AnySat) [42], [43].
- **Asynchronous Coupled Earth System Emulators:** Coupling multi-rate spherical neural operators for atmospheric, oceanic, and cryospheric dynamics across continuous interfacial flux boundary couplers (DLESyM, SamudrACE) [44], [45].
- **Multimodal Masked Autoencoders and Continuous Hypernetworks:** Extending masked autoencoding (MAE) across spatial patches, temporal revisit cycles, and dynamic wavelength hypernetworks [12], [13], [14], [15], [18], [33].
- **Generative Diffusion and Lagrangian Assimilation Models:** Utilizing score-based diffusion denoising to generate calibrated probabilistic ensembles and assimilate sparse, irregular Lagrangian drifter trajectories into dense Eulerian state estimates [4], [11], [46], [47].

- Unified Multi-Task Foundational Transformers: Applying prompt-conditioned masking and task tokens across heterogeneous time series topologies [32], [17].

3.3 Dimension 3: Physics Integration Levels

The physical fidelity of deep learning models ranges along four distinct levels:

- 1) Purely Data-Driven: Models optimize statistical reconstruction (e.g., mean squared error, MAE) over historical datasets without explicit differential constraints, relying solely on model capacity to reflect physical dynamics [6], [7], [17], [42], [43].
- 2) Soft Physics Loss Penalties: Physical conservation principles (e.g., kinetic energy conservation, mass balance, divergence-free constraints, vertical pressure weighting) are formulated as penalty terms within the training loss function [48], [8], [10], [11], [30], [44].
- 3) Hard Architectural Conservation Constraints: Structural projections and geometric invariants (e.g., spherical harmonics basis expansion, divergence-free projection layers, canonical symplectic structure, Lie-Poisson Casimirs, and exact terminal linear projection layers enforcing column mass, water, and enthalpy conservation to machine precision) guarantee that predictions strictly satisfy conservation laws by construction [35], [37], [38], [49], [50].
- 4) Hybrid PDE-Neural Solvers: Deep neural representations parameterize unresolved subgrid physical processes (convection, clouds, turbulence, reservoir multiphase equilibria) coupled directly to differentiable numerical fluid dynamical cores or coupled atmosphere-ocean models [51], [52], [19], [31], [45].

3.4 Dimension 4: Functional Roles of Reasoning LLMs

Large Language Models engage with scientific temporal data through four operational archetypes:

- Scientific Reasoner: Understanding physical time series tokens and generating natural language explanations, causal attribution, and hypothesis evaluation across scientific domains [5], [53], [54].
- Contextual Enhancer: Supplying auxiliary domain semantics, station metadata, and climate zone priors to guide numerical prediction backbones [55], [33].
- Autonomous Tool-Using Agent: Decomposing complex scientific inquiries, retrieving real-time sensor streams, and executing scientific diagnostic libraries (e.g., CDO, xarray, SciPy) to produce verified scientific analyses [56], [23].
- Interactive Natural Language Interface: Translating unstructured scientist inquiries into executable database queries and simulation configurations.

4 Core Foundation Model Architectures

4.1 3D Spatial and Spherical Transformers

Traditional 2D Vision Transformers process planar images by dividing them into uniform patches. However, planetary atmospheric states are inherently three-dimensional, structured across latitude, longitude, and geopotential pressure altitudes.

4.1.1 Pangu-Weather: 3D Earth-Specific Transformer

Bi et al. [6] introduced Pangu-Weather, the first deep learning weather forecasting system to demonstrably surpass top-tier operational NWP models (ECMWF IFS) in medium-range deterministic forecasting accuracy. Pangu-Weather formulates atmospheric prediction using a 3D Earth-Specific Transformer (3DEST) architecture. The state space is divided into 3D cuboid patches of size $2 \times 4 \times 4$ across vertical levels, latitude, and longitude. To account for inhomogeneous spatial dynamics (e.g., Coriolis force variations across latitudes and atmospheric lapse rates across altitudes), 3DEST replaces standard relative position biases with an Earth-Specific Positional Bias (ESPB) matrix indexed explicitly by pressure level and latitude:

$$\text{Attn}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d}} + B_{\text{Earth}}(l_i, l_j, \phi_i, \phi_j) \right) V. \quad (10)$$

To counteract rapid error accumulation during iterative autoregressive rollout, Pangu-Weather trains four distinct models with lead times of 1, 3, 6, and 24 hours. A greedy temporal aggregation algorithm chains these models to generate 7-day global forecasts in seconds.

4.1.2 FuXi: Cascade Forecasting System

Recognizing that predictive uncertainty compounds nonlinearly with lead time, Chen et al. [34] developed FuXi, a cascade machine learning system consisting of three specialized Swin Transformer stages: FuXi-Short (0–42 hours), FuXi-Medium (42–120 hours), and FuXi-Long (120–360 hours). By dynamically switching between model weights optimized for different forecasting horizons, FuXi significantly curtails the blurring and spectral dissipation common in long-horizon neural forecasts.

4.1.3 Aurora: Scaling to Planetary Foundation Models

Bodnar et al. [9] demonstrated that scaling principles hold in Earth system modeling by introducing Aurora, a 1.3-billion parameter foundation model. Pretrained on over 1 petabyte of heterogeneous atmospheric data—including ERA5, ECMWF HRES, operational IFS analyses, and NOAA GFS—Aurora incorporates a 3D Perceiver architecture capable of accepting arbitrary combinations of vertical levels and surface variables at 0.1° operational resolution (10 km equivalent).

4.1.4 AIFS: ECMWF Operational GNN-Transformer Architecture

Lang et al. [57] developed the Artificial Intelligence Forecasting System (AIFS), ECMWF’s official operational data-driven model. AIFS marries the geometric flexibility of graph neural networks with the long-range receptive fields of sliding-window transformers. The architecture maps icosahedral multi-mesh node representations through a graph encoder, processes latent multi-scale interactions via a local sliding-window transformer that limits attention memory to $O(N \cdot W)$ tokens, and decodes back to the operational reduced Gaussian grid via a graph decoder. Pretrained on decades of ERA5 reanalysis and fine-tuned on ECMWF’s operational NWP analyses, AIFS runs in operational production at ECMWF four times daily alongside physics-based NWP systems.

4.2 Multi-Mesh Spherical Graph Neural Networks

4.2.1 GraphCast: Planetary Multi-Mesh Routing

Lam et al. [7] pioneered an alternative geometric approach by formulating global weather forecasting over a multi-resolution spherical graph. Standard equirectangular projections concentrate excessive grid cells near the poles while diluting them at the equator. GraphCast circumvents this singularity using an Encode-Process-Decode framework operating on a hierarchical icosahedral mesh:

- 1) **Grid-to-Mesh Encoder:** Maps 0.25° latitude-longitude grid variables onto the vertices of a 6-level refined icosahedral mesh M_6 (40,962 nodes, 122,880 spherical edges).
- 2) **Multi-Mesh Processor:** Employs 16 layers of homogeneous message passing across all hierarchical mesh levels (M_0 to M_6). Long-range edges in coarse mesh levels (M_0, M_1) facilitate rapid, global information propagation across planetary teleconnection pathways in few message-passing hops.
- 3) **Mesh-to-Grid Decoder:** Projects refined mesh vertex features back to the 0.25° target grid to predict single-step 6-hour temporal increments $\Delta\mathcal{X}_t$.

GraphCast is trained with a 12-step autoregressive loss rollout, enabling robust, physically stable 10-day medium-range forecasts that outperform ECMWF HRES on over 90% of headline verification variables.

4.3 Continuous Neural Operators

4.3.1 FourCastNet: Adaptive Fourier Neural Operators

Pathak et al. [8] deployed Adaptive Fourier Neural Operators (AFNO) to solve planetary-scale atmospheric PDEs. Conventional spatial convolutions are intrinsically tied to fixed grid resolutions. In contrast, neural operators approximate non-linear mappings between infinite-dimensional function spaces. AFNO transforms tokenized feature representations into the frequency domain via 2D Fast Fourier Transforms (FFT), applies block-diagonal weight multiplications with soft-thresholding sparsity in the frequency domain, and returns to physical space via inverse FFT:

$$z_{\text{out}}(x) = \mathcal{F}^{-1}(\text{SoftThreshold}(\mathbf{W} \cdot \mathcal{F}(z_{\text{in}})(k))) + \mathbf{W}_s z_{\text{in}}(x). \quad (11)$$

This formulation provides computational complexity quasi-linear in spatial resolution while demonstrating remarkable zero-shot super-resolution and transferability.

4.3.2 SFNO: Spherical Fourier Neural Operators

While planar Fourier transforms accelerate spectral computation, standard 2D FFTs applied to equirectangular latitude-longitude coordinates suffer from severe metric distortions near the poles, where meridians converge. Bonev et al. [35] resolved this geometric bottleneck by formulating the Spherical Fourier Neural Operator (SFNO), which solves continuous PDEs directly on the Riemannian manifold of the unit sphere $\mathbb{S}^2 \subset \mathbb{R}^3$.

Let colatitude be $\theta \in [0, \pi]$ and longitude be $\lambda \in [0, 2\pi)$. Any square-integrable scalar atmospheric state $u \in \mathcal{L}^2(\mathbb{S}^2)$

admits an orthogonal expansion in terms of spherical harmonics $Y_\ell^m(\theta, \lambda)$:

$$u(\theta, \lambda) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} \hat{u}_\ell^m Y_\ell^m(\theta, \lambda), \quad (12)$$

where $\ell \in \mathbb{N}_0$ denotes the spherical degree, $m \in \{-\ell, \dots, \ell\}$ is the order, and the orthonormal basis functions are defined via associated Legendre polynomials P_ℓ^m :

$$Y_\ell^m(\theta, \lambda) = \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos \theta) e^{im\lambda}. \quad (13)$$

Continuous convolution on $\mathbb{S}^2$ with an invariant kernel κ is realized through spherical harmonic transforms $\mathcal{SHT}$ and inverse transforms $\mathcal{ISHT}$:

$$(\kappa \star_{\mathbb{S}^2} u)(\theta, \lambda) = \mathcal{ISHT}(\mathbf{W}_\ell \cdot \mathcal{SHT}(u)(\ell, m)), \quad (14)$$

where $\mathbf{W}_\ell \in \mathbb{C}^{d_{\text{out}} \times d_{\text{in}}}$ denotes degree-dependent weight tensors. By the convolution theorem on $\text{SO}(3)$, this spherical harmonic formulation provides exact rotational equivariance, eliminates high-latitude numerical artifacts, and ensures long-term spectral stability over multi-day atmospheric rollouts.

4.4 Multimodal Masked Autoencoders and Latent Alignments

In the remote sensing and Earth observation domains, observations arrive as asynchronous, multi-spectral satellite imagery cubes, irregular in-situ sensor readings, and gridded reanalysis tensors.

- SatMAE [12] introduced a temporal and multi-spectral Vision Transformer trained via masked autoencoding. By decoupling spatial, temporal, and spectral positional embeddings, SatMAE processes multi-temporal revisit sequences with high masking ratios (up to 75%), outperforming supervised baselines on land cover and crop yield estimation.
- Presto [13] developed a lightweight (1.2M parameters) transformer tailored for pixel-level time series. Presto natively ingests heterogeneous modalities—including Sentinel-1 SAR backscatter, Sentinel-2 multi-spectral bands, elevation, and ERA5 weather—yielding exceptional few-shot transfer performance.
- Galileo [14] expanded multi-sensor pretraining across spatial-temporal scales using dual global and local contrastive objectives, enabling unified geospatial segmentation and dynamic flood mapping.
- SkySense [58] established a billion-scale (1.9B parameter) multi-modal remote sensing foundation model. By formulating a factorized spatio-temporal cross-attention architecture, SkySense jointly pretrains on over 21.5 million multi-modal satellite temporal sequences (combining optical multi-spectral reflectance with synthetic aperture radar SAR backscatter). It establishes state-of-the-art generalist performance across 16 diverse Earth observation benchmarks, including fine-grained crop classification, building damage assessment, and temporal change detection.
- DOFA [18] introduced a Dynamic One-For-All foundation architecture inspired by neural plasticity. Recognizing that Earth observation platforms possess

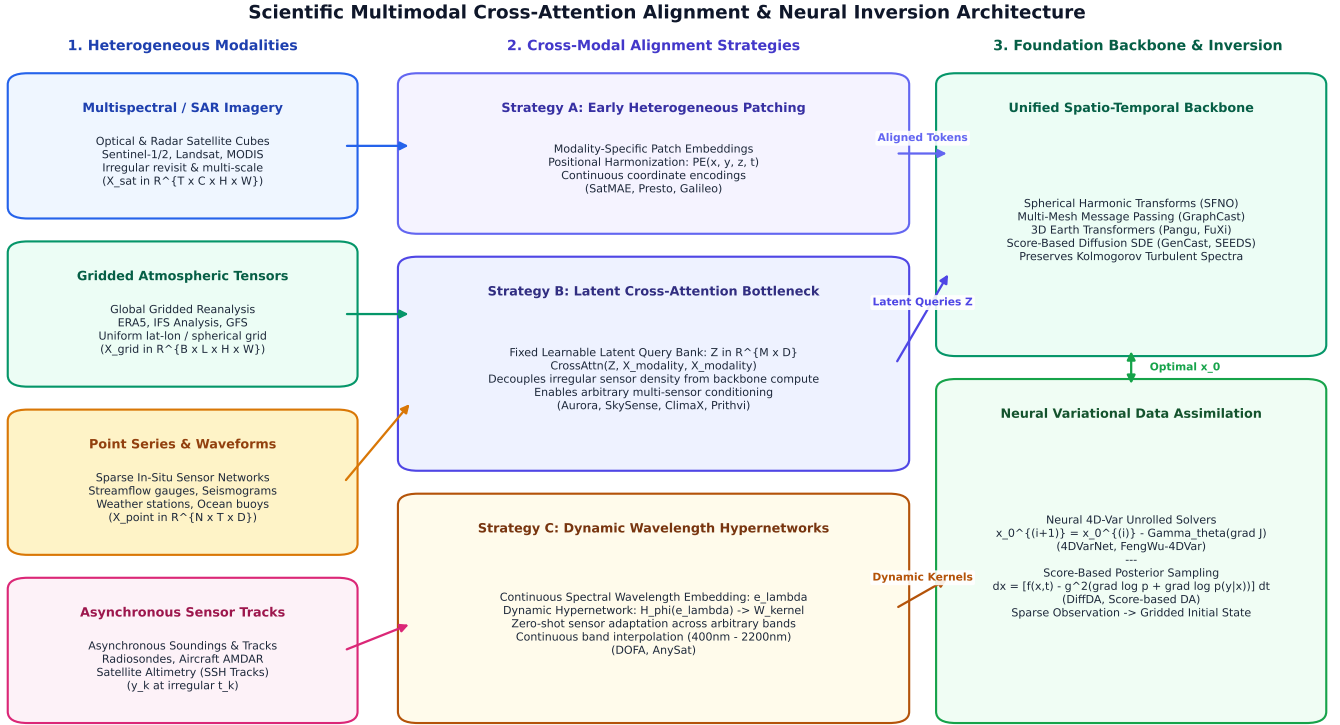


Fig. 3. Architectural taxonomy of scientific multimodal cross-attention alignment and neural variational data assimilation across heterogeneous spatial geometries. Left: Multi-source observational modalities spanning satellites, reanalyses, point sensor networks, and asynchronous tracks. Center: Three primary alignment paradigms (Early Heterogeneous Patching, Latent Cross-Attention Bottlenecks, and Dynamic Wavelength Hypernetworks). Right: Unified spatio-temporal foundation backbones and neural variational data assimilation solvers bridging sparse observation streams directly into gridded initial state representations.

disparate band wavelengths and ground sampling distances (GSD), DOFA conditions dynamic weight hypernetworks directly on continuous physical wavelengths $\lambda \in [400 \text{ nm}, 14,000 \text{ nm}]$, unifying optical, SAR, and hyperspectral temporal modalities under a single pre-trained representation.

- GEO-Bench [59] established a standardized benchmark suite for Earth monitoring foundation models. Unifying multi-temporal Sentinel-1/2, SPOT, and m-CASIS data across regional land-use segmentation and multi-spectral classification, GEO-Bench provides rigorous zero-shot and fine-tuning evaluation protocols for multi-sensor satellite representations.
- EarthPT [17] developed a 700M-parameter autoregressive decoder foundation model specifically designed for Earth Observation time series. Treating multi-spectral reflectance measurements (400–2300 nm) across Copernicus Sentinel-2 pixel-year revisit sequences as dense temporal tokens, EarthPT predicts future surface reflectance evolutions with high physical fidelity.
- Prithvi [15] and its weather-climate counterpart Prithvi WxC [60] established open-source geospatial foundation backbones jointly developed by NASA and IBM, scaling to 2.3 billion parameters.

4.4.1 Formal Alignment Paradigms across Heterogeneous Geometries

As illustrated in Fig. 3, cross-modal integration across disparate spatial geometries follows three dominant architectural paradigms:

- 1) Strategy A: Early Heterogeneous Patch Projection: Given heterogeneous modality inputs $\mathbf{X}^{(m)} \in \mathbb{R}^{T_m \times C_m \times H_m \times W_m}$ with disparate channel definitions and Ground Sampling Distances (GSD), modality-specific linear or convolutional patch embedding kernels $\mathbf{E}^{(m)}$ project local spatial-temporal patches into a shared hidden dimension D :

$$\mathbf{h}_i^{(m)} = \mathbf{E}^{(m)} \mathbf{x}_i^{(m)} + \mathbf{PE}(x_i, y_i, z_i, t_i), \quad (15)$$

where $\mathbf{PE}(\cdot)$ denotes a unified continuous 4D Fourier spatio-temporal positional encoding, ensuring coordinate harmonization across disparate satellite swaths and ground station coordinates (utilized in SatMAE [12] and Presto [13]).

- 2) Strategy B: Latent Cross-Attention Bottlenecks: When observation token counts $N_{\text{obs}} = \sum_m N_m$ fluctuate dramatically due to dynamic sensor availability or dense planetary grids, feeding all tokens directly into full self-attention scales quadratically as $O(N_{\text{obs}}^2)$. To circumvent this computational explosion, Perceiver-style architectures (Aurora [9], SkySense [58]) introduce a fixed bank of learnable latent

query tokens $\mathbf{Z} \in \mathbb{R}^{M \times D}$ ($M \ll N_{\text{obs}}$) that attend to the irregular multi-sensor input tokens via cross-attention:

$$\mathbf{Z}_{\text{aligned}} = \text{Softmax} \left(\frac{\mathbf{Z} \mathbf{W}_Q (\mathbf{H}_{\text{obs}} \mathbf{W}_K)^T}{\sqrt{d}} \right) (\mathbf{H}_{\text{obs}} \mathbf{W}_V), \quad (16)$$

where $\mathbf{H}_{\text{obs}} = [\mathbf{H}^{(1)}; \dots; \mathbf{H}^{(K)}]$ concatenates all available sensor tokens. The spatio-temporal foundation backbone then processes only the constant-size latent state $\mathbf{Z}_{\text{aligned}}$ with linear complexity in input resolution, decoupling observation density from computation.

- 3) Strategy C: Dynamic Wavelength Hypernetworks: To support arbitrary spectral bands without modifying network weights, continuous wavelength hypernetworks (DOFA [18]) encode the physical central wavelength λ of each spectral channel via harmonic Fourier embeddings $\mathbf{e}_\lambda = [\sin(2^k \pi \lambda), \cos(2^k \pi \lambda)]_{k=0}^{K-1}$. A lightweight Multi-Layer Perceptron (MLP) hypernetwork $\mathcal{H}_\phi$ maps $\mathbf{e}_\lambda$ to generate the convolution or linear projection weights dynamically:

$$\mathbf{W}_{\text{patch}}(\lambda) = \mathcal{H}_\phi(\mathbf{e}_\lambda) \in \mathbb{R}^{D \times (P^2 \cdot C_{\text{in}})}. \quad (17)$$

This enables zero-shot sensor transfer across optical (e.g., Sentinel-2 10m/20m bands), thermal infrared (Landsat-8/TIRS), and radar backscatter (Sentinel-1 VV/VH) modalities within a unified model.

4.5 Neural Data Assimilation and Sensor-to-Grid Variational Inversion

A fundamental gap separating operational numerical weather forecasting from purely data-driven foundation models is data assimilation (DA): the mathematical process of synthesizing sparse, asynchronous, and noisy physical observations into a physically consistent gridded initial atmospheric or oceanic state $\mathbf{x}_0$. While existing foundation models (e.g., Pangu-Weather, GraphCast) rely entirely on pre-computed gridded reanalyses (ERA5 or operational IFS analyses) produced by supercomputing NWP centers, emerging architectures introduce differentiable variational and score-based neural assimilation solvers to invert raw sensor streams directly into gridded model initial states.

4.5.1 Variational Formulation and Neural 4D-Var

In classical numerical weather prediction, four-dimensional variational data assimilation (4D-Var) optimizes an initial state $\mathbf{x}_0$ over an assimilation window $[t_0, t_K]$ to minimize a penalized least-squares cost function:

$$\mathcal{J}(\mathbf{x}_0) = \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \frac{1}{2} \sum_{k=0}^K \|\mathbf{y}_k - \mathcal{H}_k(\mathcal{M}_{0 \rightarrow k}(\mathbf{x}_0))\|_{\mathbf{R}_k^{-1}}^2, \quad (18)$$

where $\mathbf{x}_b$ is the background prior state, $\mathbf{B}$ is the background error covariance matrix, $\mathbf{y}_k$ denotes sparse observations at time t_k , $\mathbf{R}_k$ is the observation error covariance, $\mathcal{H}_k$ is the observation forward operator, and $\mathcal{M}_{0 \rightarrow k}$ is the non-linear dynamical forward model. In operational NWP, minimizing Eq. (18) requires maintaining complex adjoint code $\mathcal{M}^*$ to

compute gradients $\nabla_{\mathbf{x}_0} \mathcal{J}$, consuming up to 80% of operational supercomputing compute budgets.

To overcome this bottleneck, Fablet et al. [61] introduced 4DVarNet, reformulating variational data assimilation as an end-to-end learnable gradient-based solver. 4DVarNet replaces the analytical minimization loop with an unrolled recurrent neural architecture:

$$\mathbf{x}_0^{(i+1)} = \mathbf{x}_0^{(i)} - \Gamma_\theta \left(\nabla_{\mathbf{x}_0} \mathcal{J}(\mathbf{x}_0^{(i)}), \mathbf{h}^{(i)} \right), \quad (19)$$

where Γ_θ is a learned recurrent neural operator (e.g., ConvlSTM or U-Net solver), $\mathbf{h}^{(i)}$ represents internal solver memory states, and the gradient $\nabla_{\mathbf{x}_0} \mathcal{J}$ is computed via automated differentiation through the observation operator $\mathcal{H}_k$ and neural forward dynamics $\mathcal{M}_\phi$. Originally deployed for sea surface height (SSH) reconstruction from sparse satellite altimetry tracks in oceanography, 4DVarNet achieves orders-of-magnitude faster convergence than classical conjugate gradient descent while learning dynamic background error covariances directly from historical data.

Scaling neural data assimilation to global atmospheric forecasting, Xiao et al. [62] introduced FengWu-4DVar. FengWu-4DVar couples the FengWu multi-scale atmospheric transformer with an unrolled 4D variational optimization loop. By exploiting the end-to-end differentiability of the neural forecasting backbone, FengWu-4DVar directly back-propagates observation residuals from asynchronous in-situ soundings and satellite radiances through 24-hour assimilation windows, generating initialized atmospheric states that match operational ECMWF IFS analysis quality while reducing assimilation wall-clock time from hours to seconds.

4.5.2 Score-Based and Diffusion Data Assimilation

Complementing deterministic variational solvers, generative diffusion models reframe data assimilation as conditional posterior sampling: drawing physical state trajectories $\mathbf{x}_{0:K} \sim p(\mathbf{x}_{0:K} | \mathbf{y}_{0:K})$.

Rozet and Louppe [63] formulated Score-based Data Assimilation (SDA), showing that continuous score-based stochastic differential equations can execute exact spatio-temporal smoothing and filtering over non-linear fluid dynamics without tangent-linear approximations. Given a trained unconditional foundation prior $s_\theta(\mathbf{x}_t, t) \approx \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$, SDA conditions the reverse-time generative SDE on sparse partial observations $\mathbf{y}$ via Tweedie's guidance:

$$d\mathbf{x}_t = [\mathbf{f}(\mathbf{x}_t, t) - g(t)^2 (\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t))] dt + g(t) d\mathbf{w} \quad (20)$$

Under Gaussian observation noise $\mathbf{y} \sim \mathcal{N}(\mathbf{H}\mathbf{x}_0, \mathbf{R})$, the intractable conditional score $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t)$ is approximated analytically via the posterior expectation $\hat{\mathbf{x}}_0(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} (\mathbf{x}_t - \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t))$:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t) \approx \frac{1}{\sqrt{\alpha_t}} \mathbf{H}^T (\mathbf{H} \boldsymbol{\Sigma}_t \mathbf{H}^T + \mathbf{R})^{-1} (\mathbf{y} - \mathbf{H} \hat{\mathbf{x}}_0(\mathbf{x}_t)). \quad (21)$$

Scaling diffusion assimilation to weather-scale operational state estimation, Huang et al. [64] introduced DiffDA. DiffDA trains a conditional denoising diffusion probabilistic model (DDPM) conditioned on sparse surface weather station observations and radiosonde soundings. Operating directly on physical observation coordinates, DiffDA samples ensemble analyses that accurately reproduce fine-scale frontal boundaries,

atmospheric rivers, and convective storms, demonstrating that neural generative assimilation can bridge heterogeneous sensor observations directly into gridded foundation state spaces.

4.6 Probabilistic Diffusion and Extreme-Event Ensembles

A pervasive theoretical limitation of deterministic forecasting models trained under mean squared error (L_2) or mean absolute error (L_1) losses is their vulnerability to spectral blurring and variance dissipation at extended forecast horizons. Formally, given an observation history $\mathbf{y}_{1:t}$ and future atmospheric target $\mathbf{x}_{t+\tau}$, optimizing an L_2 regression objective $\min_{\theta} \mathbb{E}[\|\mathbf{x}_{t+\tau} - f_{\theta}(\mathbf{y}_{1:t})\|^2]$ provably converges to the conditional posterior mean:

$$f_{\theta}^*(\mathbf{y}_{1:t}) = \mathbb{E}[\mathbf{x}_{t+\tau} \mid \mathbf{y}_{1:t}] = \int \mathbf{x} p(\mathbf{x} \mid \mathbf{y}_{1:t}) d\mathbf{x}. \quad (22)$$

Because atmospheric flow exhibits non-linear chaotic dynamics with positive Lyapunov exponents, the true posterior probability density $p(\mathbf{x} \mid \mathbf{y}_{1:t})$ bifurcates into multi-modal trajectories at extended lead times ($\tau \geq 5$ days). Averaging over these distinct physical realizations washes out high-wavenumber turbulent eddies, severely dampens the kinetic energy power spectrum $E(k) \propto k^{-5/3}$ or k^{-3} , and underestimates extreme-event tail probabilities $P(\|\mathbf{x}\| > \tau_{\text{threshold}})$.

To resolve this limitation, generative diffusion models reframe weather forecasting as stochastic sampling from the true conditional distribution $p(\mathbf{x} \mid \mathbf{y})$.

4.6.1 Continuous Score-Based Diffusion Formulation

Let the atmospheric state trajectory follow a continuous forward stochastic differential equation (SDE) over diffusion time $s \in [0, 1]$:

$$d\mathbf{x}_s = \mathbf{f}(\mathbf{x}_s, s) ds + g(s) d\mathbf{w}, \quad (23)$$

where $\mathbf{w}$ is standard Brownian motion, $\mathbf{f}$ represents drift, and $g(s)$ governs noise dispersion. By Anderson’s reverse-time theorem, the generative reverse trajectory is governed by:

$$d\mathbf{x}_s = [\mathbf{f}(\mathbf{x}_s, s) - g(s)^2 \nabla_{\mathbf{x}_s} \log p_s(\mathbf{x}_s \mid \mathbf{y})] ds + g(s) d\bar{\mathbf{w}}, \quad (24)$$

where $d\bar{\mathbf{w}}$ denotes backward Brownian motion, and the score function $s_{\theta}(\mathbf{x}_s, s, \mathbf{y}) \approx \nabla_{\mathbf{x}_s} \log p_s(\mathbf{x}_s \mid \mathbf{y})$ is parameterized by a deep neural operator or graph network trained via denoising score matching.

4.6.2 GenCast, SEEDS, and FuXi-Extreme: Operational Diffusion Ensembling

Price et al. [4] operationalized this paradigm in GenCast, conditioning a score-matching network on spherical multi-mesh graphs using historical 6-hour atmospheric increments ($\mathbf{x}_{t-6h}, \mathbf{x}_t$). By sampling 50 stochastic trajectories, GenCast accurately generates medium-range ensemble forecasts that maintain realistic energy spectra up to Day 15, outperforming ECMWF ENS in Continuous Ranked Probability Scores (CRPS) across over 80% of headline atmospheric variables and drastically improving tropical cyclone track and intensity predictions.

Concurrently, Li et al. [11] introduced SEEDS (Scalable Ensemble Envelope Diffusion Sampler), demonstrating that generative diffusion can emulate operational numerical

weather forecast ensembles (such as NOAA GEFS) conditioned on only 1 or 2 deterministic forecast trajectories. SEEDS generates large ensembles (e.g., 50 members) at 1° operational resolution orders of magnitude faster than numerical ensembles, precisely reproducing the spread-skill relationship and capturing extreme heatwave and storm tail events without numerical energy collapse.

Extending diffusion specifically to hazard extremes, Zhong et al. [65] formulated FuXi-Extreme. While deterministic weather models drastically underestimate tail precipitation rates and gale-force wind gusts, FuXi-Extreme couples a cascade Swin Transformer with a conditional latent diffusion model. Trained on high-resolution ERA5 extreme events, it generates realistic sharp spatial gradients for extreme precipitation (> 50 mm/day) and 10-meter wind gusts, resolving the longstanding spectral smoothing dilemma for operational disaster early warnings.

4.7 Universal Tokenization and Cross-Domain Temporal Transferability

A major architectural obstacle in scientific foundation models is the profound heterogeneity of physical data representations: regular lat-lon reanalysis grids (ERA5), irregular unstructured meshes (finite-volume ocean models), sparse in-situ station networks (hydrology streamflow gauges), 1D continuous high-frequency waveforms (seismology), and irregular multi-spectral satellite revisits (remote sensing). Recent research develops universal tokenization mechanisms to bridge this divide:

4.7.1 Unified Multi-Task Architectures: UniTS

Gao et al. [32] developed UniTS, a unified multi-task foundation model for time series. Unlike domain-specific architectures requiring separate parameterizations for each dataset, UniTS processes arbitrary numbers of variables and sequences by introducing:

- 1) Sequence-Variable Tokenization: Decouples temporal sequence patches and variable channels into modular tokens via lightweight linear projections.
- 2) Prompt-Conditioned Task Tokens: Ingests task specification tokens (e.g., forecasting horizon, classification labels, imputation mask) into self-attention layers, conditioning a single shared transformer backbone to execute forecasting, anomaly detection, imputation, and classification across 38 diverse datasets without architectural modification.

4.7.2 Language Model Tokenization for Time Series: Chronos

Ansari et al. [66] introduced Chronos, demonstrating that classical language model tokenization principles translate directly to continuous scientific and physical time series. Chronos discretizes real-valued time series through affine mean-variance normalization followed by uniform quantile binning into a fixed vocabulary of V discrete tokens (e.g., 4096 bins). Pretrained on diverse time-series corpora using standard cross-entropy auto-regression over T5 architectures, Chronos demonstrates that language model pretraining objectives yield remarkable zero-shot probabilistic forecasting capabilities across diverse physical and socio-economic systems.

4.7.3 Autoregressive Probabilistic Foundations: Lag-Llama
 Rasul et al. [67] formulated Lag-Llama, a dedicated probabilistic foundation model utilizing a decoder-only LLaMA-style transformer architecture. Ingesting historical lagged temporal features and tokenized continuous values, Lag-Llama applies rotary position embeddings (RoPE) and outputs parametric Student- t distribution parameters (μ, σ, ν) at each step. This formulation natively captures fat-tailed physical distributions and extreme deviations across physical sensor streams.

4.7.4 Open Foundation Models: MOMENT

Goswami et al. [33] introduced MOMENT, an open-source family of foundation models trained on the Time-series Pile, a curated corpus of over 13 million diverse sequences spanning earth science sensors, industrial machinery, and physical processes. Using a patched T5-style masked autoencoder architecture, MOMENT establishes consistent zero-shot and few-shot transfer capabilities across diverse scientific temporal benchmarks.

4.7.5 Atmospheric Patch Randomization: Stormer

Nguyen et al. [10] introduced Stormer, demonstrating that a standard Vision Transformer backbone can match or exceed specialized weather forecasting architectures when coupled with two critical physical inductive biases:

- 1) **Randomized Dynamics Objective:** During pretraining, Stormer forecasts atmospheric states across randomized lead times $\Delta t \sim \mathcal{U}(6h, 120h)$, forcing the model to learn multi-scale continuous temporal dynamics rather than overfitting to fixed autoregressive step sizes.
- 2) **Pressure-Weighted Vertical Loss:** Weights atmospheric layer losses proportional to geopotential layer thickness and density, aligning optimization with physical mass distribution.

4.7.6 Standardized Waveform Tokenization: SeisBench

In geophysics, Woollam et al. [26] established SeisBench, a standardized framework and benchmarking suite unifying 3-component continuous seismic waveform datasets (ETHZ, GEOFON, INSTANCE, SCEDC, STEAD, comprising over 30 million waveforms). By standardizing sampling rates (100 Hz), coordinate rotations, and trace tokenization, SeisBench enables cross-regional foundation pretraining and transferability for phase arrival picking, earthquake detection, and fault rupture localization.

4.8 Physical Conservation Constraints and Differentiable Hybrid Solvers

A persistent vulnerability of purely data-driven black-box neural networks is their tendency to violate fundamental physical conservation principles over prolonged autoregressive rollouts, leading to unphysical mass dissipation, kinetic energy damping, or moisture divergence. Recent advances address this through distinct levels of physical integration:

- 1) **Soft Physics Regularization:** Incorporates auxiliary loss penalties that penalize physical inconsistencies during training:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{conservation}}, \quad (25)$$

where $\mathcal{L}_{\text{conservation}}$ penalizes deviations in global total dry mass, vertical geostrophic balance, or spectral kinetic energy distributions (employed in FourCastNet [8], FengWu [48], and Stormer [10]).

- 2) **Hard Architectural Invariance:** Enforces conservation laws and symmetry directly into neural layer transformations. For instance, SFNO [35] guarantees $\text{SO}(3)$ rotational equivariance on the unit sphere $\mathbb{S}^2$, preventing polar singularities and high-latitude numerical distortions by construction.
- 3) **Hybrid Multiscale Subgrid Parameterization:** ClimSim: Yu et al. [52] established ClimSim, the largest multiscale dataset and benchmark for training deep learning subgrid physics emulators. In multiscale climate modeling (e.g., E3SM-MMF), macro-scale fluid dynamics are integrated by numerical PDE solvers on coarse grids, while complex micro-scale processes (moist convection, cloud formation, turbulence, and radiative transfer) are parameterized by neural networks. ClimSim provides 5.7 billion high-resolution atmospheric column profiles, benchmarking physics-informed ML architectures on strict conservation of total column water mass and dry static energy.
- 4) **Differentiable Hybrid Solvers:** NeuralGCM: Kochkov et al. [51] coupled a fully differentiable atmospheric dynamical core with neural network parameterizations. The resolved large-scale hydrodynamics are integrated explicitly by an analytical spectral PDE solver that strictly enforces conservation of mass, momentum, and moisture:

$$\frac{\partial \mathbf{X}}{\partial t} = \mathcal{F}_{\text{dynamical}}(\mathbf{X}) + \mathcal{N}_{\theta}(\mathbf{X}), \quad (26)$$

where $\mathcal{F}_{\text{dynamical}}$ executes analytical fluid dynamics, while the neural network $\mathcal{N}_{\theta}$ parameterizes subgrid physical processes. NeuralGCM demonstrates unprecedented multi-decadal stability without climate drift, matching the operational deterministic accuracy of ECMWF IFS HRES at 1–10 day weather horizons while preserving global physical conservation laws for decadal climate simulations.

- 5) **Continuous Physics-Informed Neural ODE Dynamics:** ClimODE: Verma et al. [68] pioneered ClimODE, fundamentally rethinking scientific time series modeling by replacing discrete multi-step mapping $\mathbf{x}_{t+\Delta t} = \mathcal{M}_{\theta}(\mathbf{x}_t)$ with continuous differential flow trajectories grounded in fluid mechanics. By explicitly embedding continuity and advective constraints into neural ordinary differential equations, ClimODE achieves global weather forecasting with only 0.3M parameters—over two orders of magnitude smaller than discrete vision transformers—while natively supporting arbitrary continuous-time evaluation intervals.

4.9 Continuous-Discrete Physics Invariants, Lie Symmetries, and Geometric Deep Learning

A central mathematical frontier in scientific foundation models is reconciling discrete computational architectures (e.g., attention over tokenized patches, autoregressive discrete rollouts) with the continuous, symmetry-preserving nature of physical laws governing the Earth system.

4.9.1 Continuous Fluid Advection and Neural Differential Flows

Classical discrete autoregressive architectures forecast atmospheric states via discrete recurrence:

$$\mathbf{x}_{k+1} = \mathcal{M}_\theta(\mathbf{x}_k), \quad k \in \{0, 1, \dots, K\}, \quad (27)$$

where $\Delta t = t_{k+1} - t_k$ is strictly locked to a predefined sampling cadence (e.g., 6h in ERA5). This discrete formulation suffers from two fundamental limitations: (1) time-step rigidity, precluding direct assimilation or forecasting at arbitrary asynchronous timestamps, and (2) compounding discretization error, where unconstrained discrete updates accumulate unphysical drift.

In contrast, continuous-time formulation models the planetary atmosphere as a non-linear dynamical system evolving over the continuous manifold $\mathcal{M} = \mathbb{S}^2 \times \mathbb{R}^+$:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}_\theta(\mathbf{x}(t), t), \quad \mathbf{x}(t) = \mathbf{x}(t_0) + \int_{t_0}^t \mathbf{f}_\theta(\mathbf{x}(\tau), \tau) d\tau. \quad (28)$$

To enforce physical consistency, ClimODE [68] constrains the learned vector field $\mathbf{f}_\theta$ using the Navier–Stokes transport theorem. The continuous conservation of atmospheric mass density ρ is governed by the continuity PDE:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}_\theta) = 0, \quad (29)$$

where $\mathbf{v}_\theta(\mathbf{s}, t) \in T_{\mathbf{s}}\mathbb{S}^2$ is the neural velocity field over spherical coordinates $\mathbf{s} = (\lambda, \theta)$. Similarly, any scalar tracer field $\phi \in \{T, q, Z\}$ (temperature, specific humidity, geopotential) satisfies the continuous material advection-diffusion relation:

$$\frac{D\phi}{Dt} \equiv \frac{\partial \phi}{\partial t} + (\mathbf{v}_\theta \cdot \nabla)\phi = \mathcal{D}_\theta(\phi) + \mathcal{S}_\theta(\phi, \mathbf{s}, t), \quad (30)$$

where $\mathcal{D}_\theta$ denotes turbulent sub-grid diffusion and $\mathcal{S}_\theta$ encapsulates radiative source/sink fluxes. Integrating Eq. (28) with adaptive-step numerical ODE solvers (e.g., Dormand–Prince 5(4) or adaptive Runge–Kutta) guarantees that the flow remains bounded along physical trajectories, yielding superior long-term stability and eliminating the need for empirical post-hoc smoothing.

4.9.2 Lie Group Symmetries and Spherical Coordinate Invariance

The fundamental equations of geophysical fluid dynamics (primitive equations, shallow-water equations) are invariant under the continuous Lie group of spatial rotations $\text{SO}(3)$ and time translation $\mathbb{R}$. Let $g \in \text{SO}(3)$ represent an arbitrary rigid rotation of the planetary sphere $\mathbb{S}^2$. A scientific foundation operator $\mathcal{F}_\theta : \mathcal{X}(\mathbb{S}^2) \rightarrow \mathcal{X}(\mathbb{S}^2)$ is said to be $\text{SO}(3)$ -equivariant if and only if:

$$\mathcal{F}_\theta(\pi_g[\mathbf{X}]) = \pi_g[\mathcal{F}_\theta(\mathbf{X})], \quad \forall g \in \text{SO}(3), \quad (31)$$

where π_g denotes the group representation acting on the field state $\mathbf{X}$. Crucially, the transformation laws depend on the tensor rank of the physical variables:

- **Scalar Tracers** (Geopotential Φ , Surface Temperature T_{2m} , Pressure p): Act as rank-0 scalar invariants under coordinate rotation:

$$\pi_g[\Phi](\mathbf{s}) = \Phi(g^{-1}\mathbf{s}). \quad (32)$$

- **Vector Velocity Fields** (Horizontal Wind $\mathbf{u} = (u, v)^T$): Belong to the tangent bundle $T_{\mathbf{s}}\mathbb{S}^2$ and undergo an orthogonal frame transformation via the differential push-forward:

$$\pi_g[\mathbf{u}](\mathbf{s}) = \mathbf{J}_g(g^{-1}\mathbf{s}) \mathbf{u}(g^{-1}\mathbf{s}), \quad (33)$$

where $\mathbf{J}_g \in \text{SO}(2)$ denotes the local Jacobian transition matrix between tangent spaces.

Standard planar convolutional networks or naive Vision Transformers applied to equirectangular latitude-longitude projections violate Eq. (31) due to the non-uniform metric tensor:

$$g_{\mathbb{S}^2} = R^2 (d\theta^2 + \cos^2 \theta d\lambda^2), \quad (34)$$

which compresses grid cell longitudinal distances towards the poles ($\Delta x \rightarrow 0$ as $\theta \rightarrow \pm\pi/2$), causing severe polar singularities and high-frequency numerical artifacts. In contrast, geometric foundation models resolve this through two elegant paradigms:

- 1) **Spherical Harmonic Harmonic Operators**: SFNO [35] projects spatial states into continuous spherical harmonics $Y_\ell^m(\theta, \lambda)$, executing spectral convolutions in the frequency domain where $\text{SO}(3)$ acts block-diagonally via Wigner D -matrices:

$$\hat{\mathbf{X}}_\ell^m = \int_{\mathbb{S}^2} \mathbf{X}(\mathbf{s}) \overline{Y_\ell^m(\mathbf{s})} d\Omega(\mathbf{s}), \quad (35)$$

strictly preserving planetary rotational symmetry and eliminating polar distortion by construction.

- 2) **Quasi-Uniform Spherical Meshes**: GraphCast [7] and AIFS [57] map irregular spherical domains onto recursive icosahedral multi-mesh tessellations $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Message-passing edge attributes encode true Riemannian geodesic distances and relative orientation angles, achieving approximate isometry and continuous spatial invariance across the entire planetary sphere.

4.10 Extreme Value Theory (EVT) and Heavy-Tailed Uncertainty Quantification

A profound pathology in contemporary data-driven weather and climate foundation models is their systemic underforecasting of rare, high-impact perils (e.g., flash flood downpours, hurricane wind gusts, extreme heatwaves). Under standard training objectives, neural networks minimize L_2 error:

$$\hat{\mathbf{y}} = \arg \min_{\mathbf{f}} \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} [\|\mathbf{y} - \mathbf{f}(\mathbf{x})\|_2^2] = \mathbb{E}[\mathbf{y} \mid \mathbf{x}]. \quad (36)$$

Because the conditional expectation $\mathbb{E}[\mathbf{y} \mid \mathbf{x}]$ acts as a statistical mean-seeking operator, minimizing symmetric L_2 or L_1 loss over climatological distributions inevitably biases predictions toward the bulk mean. Consequently, when confronted with extreme values located in the heavy tails of the distribution, standard foundation backbones suffer from severe amplitude blunting, failing to capture 100-year return period perils.

4.10.1 Generalized Extreme Value (GEV) Formulations

In statistical climatology, the asymptotic distribution of block maxima (e.g., annual maximum daily precipitation or peak gust speed) $M_n = \max\{X_1, \dots, X_n\}$ is uniquely characterized by the Fisher–Tippett–Gnedenko theorem via the Generalized Extreme Value (GEV) distribution:

$$G(z; \mu, \sigma, \xi) = \exp \left\{ - \left[1 + \xi \left(\frac{z - \mu}{\sigma} \right) \right]_+^{-1/\xi} \right\}, \quad (37)$$

where $\mu \in \mathbb{R}$ is the location parameter, $\sigma > 0$ is the scale parameter, and $\xi \in \mathbb{R}$ denotes the shape parameter (tail index), with $[a]_+ \equiv \max(a, 0)$. The tail index ξ determines the qualitative asymptotic decay of natural extremes:

- Fréchet Heavy Tail ($\xi > 0$): Governs convective precipitation bursts, flash floods, and atmospheric rivers. The survival probability exhibits power-law decay $P(Z > z) \sim z^{-1/\xi}$, meaning extremes occur orders of magnitude more frequently than predicted by light-tailed Gaussian models.
- Gumbel Exponential Tail ($\xi = 0$): Describes moderate wind speed distributions and synoptic temperature fluctuations with exponential decay $P(Z > z) \sim e^{-z/\sigma}$.
- Weibull Bounded Tail ($\xi < 0$): Characterizes bounded atmospheric variables (e.g., relative humidity $\leq 100\%$, upper limits on geostrophic wind shear).

4.10.2 ExtremeCast: Asymmetric Extreme Loss and Uncertainty Boosting

To break the mean-seeking regression trap without degrading bulk synoptic accuracy, Xu et al. [46] introduced ExtremeCast. ExtremeCast mathematically proves that symmetric loss functions inherently penalize rare overpredictions symmetrically with severe underpredictions, heavily suppressing tail predictions due to the extreme class imbalance ($P(Z > \tau) < 0.01$).

ExtremeCast formulates an asymmetric extreme loss function $\mathcal{L}_{\text{Ex}}$ that dynamically amplifies gradient penalties when true targets exceed high quantile thresholds τ :

$$\mathcal{L}_{\text{Ex}}(\mathbf{y}, \hat{\mathbf{y}}) = \mathcal{L}_{\text{base}}(\mathbf{y}, \hat{\mathbf{y}}) + \alpha \sum_{i=1}^N w(y_i) \cdot \max(0, y_i - \hat{y}_i)^\gamma, \quad (38)$$

where $\mathcal{L}_{\text{base}}$ preserves standard MSE/MAE optimization over bulk dynamics, $\alpha > 0$ balances tail prioritization, and $\gamma \geq 1$ enforces super-linear penalties against underforecasting. The weighting factor $w(y_i)$ is formulated using extreme value theory:

$$w(y_i) = \mathbb{I}(y_i \geq \tau) \cdot \left(\frac{y_i - \tau}{\sigma_\tau} \right)^\beta, \quad (39)$$

where τ is set to the 95th or 99th climatological percentile and σ_τ represents the tail scale. Furthermore, ExtremeCast incorporates ExBooster, a training-free uncertainty ensemble module that perturbs latent feature representations using calibrated extreme-value noise samples, generating diverse probabilistic realizations that significantly elevate the Critical Success Index (CSI) and Symmetric Extremal Dependence Index (SEDI) for category-5 typhoons, localized wind gusts, and severe heat extremes.

4.11 High-Order Tensor Decompositions and Quantum-Classical Hybrid Operators

While continuous neural operators (such as FNO and SFNO) offer discretization-invariant PDE solution capabilities, their practical deployment in 3D atmospheric columns, 3D sub-surface seismic imaging, and mantle convection encounters severe memory saturation. A dense Fourier neural operator layer parameterizes spectral weights via a multidimensional complex tensor $\mathcal{W} \in \mathbb{C}^{d_{\text{in}} \times d_{\text{out}} \times K_1 \times \dots \times K_d}$. In high-resolution planetary volumes ($1024 \times 1024 \times 100$), storing and computing $\mathcal{W}$ requires tens of gigabytes of GPU VRAM per layer, creating a major hardware bottleneck.

4.11.1 Multi-Grid Tensorized Fourier Neural Operators (MG-TFNO)

To alleviate this memory complexity, Kossaifi et al. [39] introduced the Multi-Grid Tensorized Fourier Neural Operator (MG-TFNO). MG-TFNO parameterizes the spectral convolution weights in a high-order latent subspace using tensor factorizations, specifically Tucker and Tensor-Train (Matrix Product State, MPS) decompositions:

$$\mathcal{W}(i_1, i_2, k_1, \dots, k_d) \approx \sum_{r_1=1}^{R_1} \dots \sum_{r_{d+2}=1}^{R_{d+2}} \mathcal{G}(r_1, \dots, r_{d+2}) \prod_{m=1}^{d+2} \mathbf{U}^{(m)}(j_m, r_m), \quad (40)$$

where $\mathcal{G} \in \mathbb{C}^{R_1 \times \dots \times R_{d+2}}$ is a low-rank core tensor with ranks $R_m \ll \min(d_m, K_m)$, and $\mathbf{U}^{(m)}$ are orthogonal factor matrices. Coupled with multi-grid domain decomposition that parallelizes local and global frequency representations, MG-TFNO achieves over $150\times$ parameter reduction (compressing over 120M dense parameters to $< 0.8\text{M}$) while improving zero-shot super-resolution and out-of-distribution numerical stability on 2D/3D Navier–Stokes turbulence and Helmholtz acoustic wavefields.

4.11.2 WaveFNO: Continuous Seismic Wavefield Modeling and Full-Waveform Inversion

Applying continuous neural operators to solid-Earth geophysics, Yang et al. [40] pioneered WaveFNO for seismic wave propagation and full-waveform inversion (FWI). Seismic wave propagation in heterogeneous media is governed by the elastodynamic wave equation:

$$\rho(\mathbf{x}) \frac{\partial^2 \mathbf{u}(\mathbf{x}, t)}{\partial t^2} = \nabla \cdot \boldsymbol{\sigma}(\mathbf{x}, t) + \mathbf{f}(\mathbf{x}, t), \quad \boldsymbol{\sigma} = \mathbf{C}(\mathbf{x}) : \frac{1}{2} \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T \right), \quad (41)$$

where $\mathbf{u}(\mathbf{x}, t)$ is the continuous displacement wavefield, ρ is density, $\boldsymbol{\sigma}$ is the stress tensor, $\mathbf{C}$ is the elastic stiffness tensor, and $\mathbf{f}$ represents tectonic or exploration source terms. WaveFNO directly learns the non-linear solution operator $\mathcal{G}_\theta : c(\mathbf{x}) \mapsto \mathbf{u}(\mathbf{x}, t)$ mapping subsurface velocity models $c(\mathbf{x})$ to continuous wavefield trajectories. Because the learned operator $\mathcal{G}_\theta$ is continuously differentiable with respect to its inputs, seismic full-waveform inversion is executed via reverse-mode automatic differentiation:

$$c^* = \arg \min_c \frac{1}{2} \sum_{r=1}^R \int_0^T \|\mathbf{u}_{\text{obs}}^{(r)}(t) - \mathcal{G}_\theta(c)(\mathbf{x}_r, t)\|_2^2 dt + \lambda \mathcal{R}(c), \quad (42)$$

computing exact subsurface acoustic velocity updates $\nabla_c \mathcal{J}$ in seconds on a single GPU. This avoids the hundreds of computationally prohibitive numerical finite-difference wavefield

solves required by classical adjoint-state FWI workflows while avoiding cycle-skipping traps.

4.11.3 Partitioned Hybrid Quantum Fourier Neural Operators (PHQFNO)

At the outer algorithmic frontier, Marcandelli et al. [41] introduced the Partitioned Hybrid Quantum Fourier Neural Operator (PHQFNO), establishing the first continuous neural operator architecture combining classical tensor decomposition with Parameterized Quantum Circuits (PQC). PHQFNO partitions the spectral transformation across classical tensor operators and variational quantum circuits executed on Noisy Intermediate-Scale Quantum (NISQ) devices. Input spectral modes are embedded into multi-qubit quantum states $|\psi(k)\rangle$ using quantum amplitude encoding, followed by parameterized unitary evolution:

$$U(\boldsymbol{\theta}) = \prod_{l=1}^L \exp(-i\theta_l H_l) W_{\text{entangle}}, \quad (43)$$

where $H_l \in \{X, Y, Z\}^{\otimes n}$ represent multi-qubit Pauli observables and W_{entangle} is a non-local entangling layer. PHQFNO provides an architectural roadmap for continuous scientific machine learning, demonstrating that quantum entanglement can bypass classical tensor contraction bottlenecks when solving high-dimensional multi-physics differential equations.

4.12 Symplectic Manifolds and Lie-Poisson Geometric Neural Operators

Standard autoregressive forecasting models (whether based on Transformers, GNNs, or unconstrained Fourier neural operators) suffer from progressive secular drift: as rollout horizons extend past hundreds of autoregressive cycles, numerical dissipation inevitably damps physical energy, or unconstrained error accumulation triggers catastrophic numerical blowup. In conservative geophysical fluids, orbital astrodynamics, and wave mechanics, this failure stems from the omission of underlying symplectic geometry.

4.12.1 Infinite-Dimensional Hamiltonian Mechanics and Symplectic Preservation

Conservative continuum mechanics can be formulated as infinite-dimensional Hamiltonian partial differential equations on a phase space $\mathcal{M}$ endowed with a canonical symplectic two-form ω :

$$\frac{\partial \mathbf{u}}{\partial t} = \mathcal{J} \frac{\delta \mathcal{H}}{\delta \mathbf{u}}, \quad (44)$$

where $\mathbf{u} = (\mathbf{q}, \mathbf{p})^T$ denotes canonical generalized coordinates and momenta, $\mathcal{H}[\mathbf{u}] = \int_{\Omega} \mathcal{E}(\mathbf{q}, \mathbf{p}, \nabla \mathbf{q}, \nabla \mathbf{p}) dx$ represents the total energy Hamiltonian functional, and $\mathcal{J}$ is the canonical Poisson operator:

$$\mathcal{J} = \begin{bmatrix} 0 & \mathbf{I} \\ -\mathbf{I} & 0 \end{bmatrix}, \quad \mathcal{J}^* = -\mathcal{J}. \quad (45)$$

By Liouville's theorem, the continuous flow map $\Phi_t : \mathbf{u}(0) \mapsto \mathbf{u}(t)$ preserves the differential two-form $\omega = \int_{\Omega} \delta \mathbf{q} \wedge \delta \mathbf{p} dx$:

$$\Phi_t^* \omega = \omega \iff (D\Phi_t)^T \mathcal{J}^{-1} (D\Phi_t) = \mathcal{J}^{-1}, \quad (46)$$

which strictly enforces phase-space volume conservation. Makara et al. [37] established Symplectic Neural Operators (SNO), proving that standard Fourier neural operators violate

Eq. (46), causing monotonic artificial dissipation in wave and fluid systems. SNO enforces canonical symplectic preservation by structuring the operator kernel as the composition of generating functionals:

$$\mathbf{p}^{n+1} = \mathbf{p}^n - \Delta t \frac{\delta \mathcal{H}_{\theta}}{\delta \mathbf{q}^{n+1}}, \quad \mathbf{q}^{n+1} = \mathbf{q}^n + \Delta t \frac{\delta \mathcal{H}_{\theta}}{\delta \mathbf{p}^n}, \quad (47)$$

where $\mathcal{H}_{\theta}$ is parameterized by a non-local neural operator. By construction, the resulting discrete update operator exactly conserves a shadow Hamiltonian $\tilde{\mathcal{H}} = \mathcal{H} + \mathcal{O}(\Delta t^2)$, eliminating long-term amplitude decay across $> 10^4$ simulation time steps.

4.12.2 Lie-Poisson Symmetries, Casimir Invariants, and Kelvin's Circulation

In rotating planetary atmospheres and oceanic gyres, fluid equations are non-canonical Hamiltonian systems defined on the dual of a Lie algebra $\mathfrak{g}^*$ (e.g., the algebra of divergence-free vector fields on $\mathbb{S}^2$). The dynamics of relative vorticity ω are governed by the Lie-Poisson equation:

$$\frac{\partial \omega}{\partial t} = \{\omega, \mathcal{H}\} = -\text{ad}_{\frac{\delta \mathcal{H}}{\delta \omega}}^* \omega = -\mathbf{u} \cdot \nabla \omega, \quad (48)$$

where ad^* is the coadjoint operator. A defining mathematical feature of Lie-Poisson systems is the presence of Casimir invariants $\mathcal{C}[\omega]$, functionals that Poisson-commute with every arbitrary Hamiltonian $\{\mathcal{C}, \mathcal{F}\} = 0$ for all $\mathcal{F}$. In 2D and quasi-geostrophic shallow-water atmospheres, every smooth function of vorticity produces a conserved Casimir functional:

$$\mathcal{C}_{\Phi}[\omega] = \int_{\Omega} \Phi(\omega) dx = \text{const}, \quad (49)$$

including total circulation ($\Phi(\omega) = \omega$) and potential enstrophy ($\Phi(\omega) = \omega^2$). Furthermore, by Kelvin's circulation theorem, the circulation around any fluid loop $\Gamma(t)$ advected by the velocity field is invariant:

$$\frac{d}{dt} \oint_{\Gamma(t)} \mathbf{u} \cdot d\mathbf{x} = 0. \quad (50)$$

Standard data-driven models inevitably destroy Casimir conservation, causing artificial enstrophy cascades and non-physical vortex distortion. Eldred et al. [38] developed Lie-Poisson Neural Networks (LPNets), enforcing geometric coadjoint orbit preservation:

$$\frac{d\omega}{dt} = \mathbb{B}(\omega) \nabla_{\omega} \mathcal{H}_{\theta}(\omega), \quad (51)$$

where $\mathbb{B}(\omega)$ is an algebraically constructed skew-symmetric Lie-Poisson matrix operator satisfying the Jacobi identity by design. LPNets strictly preserve all Casimir invariants and circulation invariants independently of the neural network weights θ , maintaining stable planetary jet streams, Rossby waves, and atmospheric blocking patterns over multi-month rollouts without empirical damping.

4.12.3 Learned Coordinate Diffeomorphisms for Complex Domains (Geo-FNO)

To apply neural operators across non-Euclidean computational domains, irregular boundaries, and faulted geological formations, Li et al. [36] proposed Geo-FNO. Classical FNO assumes uniform Cartesian grids in Euclidean space $\mathbb{R}^d$,

failing on irregular physical geometries. Geo-FNO learns a bijective coordinate diffeomorphism:

$$\phi : \Omega \rightarrow [0, 1]^d, \quad \xi = \phi(\mathbf{x}), \quad (52)$$

which deforms the irregular physical domain Ω into a canonical periodic unit cube $[0, 1]^d$. Spectral convolution is executed in the latent canonical coordinate space ξ , while physical differential operators are transformed via the metric pullback tensor $\mathbf{g}(\mathbf{x}) = (D\phi)^T(D\phi)$. Geo-FNO enables continuous operator learning on arbitrary 3D geological reservoirs, faulted subsurfaces, and aerodynamic boundary layers without requiring conformal remeshing.

4.13 Generative Lagrangian Continuous Data Assimilation for Sparse Observables

While operational weather models rely on gridded satellite radiances and reanalysis fields, oceanic and environmental monitoring is overwhelmingly constrained by the extreme spatial sparsity and irregular sampling of autonomous in-situ platforms (e.g., Argo profiling floats, drifting surface buoys, underwater gliders). These platforms record physical states along Lagrangian trajectories:

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{u}(\mathbf{X}_i(t), t) + \boldsymbol{\eta}_i(t), \quad \mathbf{y}_i(t) = \mathcal{M}(\mathbf{s}(\mathbf{X}_i(t), t)) + \boldsymbol{\epsilon}_i(t), \quad (53)$$

where $\mathbf{X}_i(t)$ is the drifter position, $\mathbf{u}(\mathbf{x}, t)$ is the continuous Eulerian velocity field, $\boldsymbol{\eta}_i$ represents turbulent dispersion, and $\mathbf{y}_i$ records in-situ temperature, salinity, or pressure.

Traditional data assimilation methods (such as Ensemble Kalman Filtering or 4D-Var) struggle with Lagrangian observations because the forward observation operator is strongly non-linear, non-local, and history-dependent. Asefi et al. [47] addressed this fundamental challenge with Generative Lagrangian Data Assimilation (LagDA), formulating continuous ocean state estimation as a conditional score-based diffusion model:

$$p(\mathbf{u}_{1:T} \mid \{\mathbf{X}_i(t_k)\}_{i=1, k=1}^{N_d, K}), \quad (54)$$

where N_d is the number of active drifters covering less than 0.5% of the spatial basin. LagDA leverages continuous spatio-temporal attention to bridge sparse Lagrangian trajectory tokens with Eulerian spatial grids. By modeling the posterior distribution over continuous velocity $\mathbf{u} = (u, v)$, sea surface height (SSH), and relative vorticity $\zeta = \partial_x v - \partial_y u$, LagDA reconstructs submesoscale eddies and Western Boundary Currents with over 60% error reduction relative to classical variational interpolation, providing a foundational methodology for integrating autonomous marine sensor fleets into planetary Earth system models.

4.14 Analytic Conservation Enforcing and Subgrid Microphysics in Climate Emulators

Global climate models and neural planetary emulators operate on discretized horizontal grids ($\Delta x \approx 25\text{--}100$ km), leaving moist convective updrafts, turbulent boundary-layer mixing, and cloud microphysical transformations unresolved at subgrid scales (< 5 km). Let an atmospheric column state be decomposed via Reynolds filtering into resolved dynamics $\bar{\mathbf{X}} = (\bar{\mathbf{u}}, \bar{T}, \bar{q})$ and subgrid turbulent fluctuations

$\mathbf{X}' = (\mathbf{u}', T', q')$. The filtered thermodynamic energy and moisture conservation equations read:

$$\frac{\partial \bar{T}}{\partial t} + \bar{\mathbf{u}} \cdot \nabla \bar{T} = -\frac{\partial(\bar{\omega}'T')}{\partial p} + \frac{L_v}{c_p} \bar{C} + \bar{Q}_{\text{rad}} \equiv P_T(\bar{\mathbf{X}}), \quad (55)$$

$$\frac{\partial \bar{q}}{\partial t} + \bar{\mathbf{u}} \cdot \nabla \bar{q} = -\frac{\partial(\bar{\omega}'q')}{\partial p} - \bar{C} \equiv P_q(\bar{\mathbf{X}}), \quad (56)$$

where $\bar{C}$ is the net condensation rate, L_v is latent heat of vaporization, c_p is specific heat, and P_T, P_q denote the subgrid parameterization mappings. When classical unconstrained neural networks emulate P_T and P_q , unphysical numerical artifacts (such as negative precipitation $P < 0$ or spurious column energy creation) accumulate rapidly, triggering numerical explosion or unphysical climate drift within weeks of coupled simulation.

4.14.1 Analytically Constrained Neural Networks (AC-NN)

To resolve this foundational flaw, Beucler et al. [49] formulated Analytically Constrained Neural Networks (AC-NN), mathematically proving that linear and non-linear physical conservation laws can be enforced within neural network computational graphs to machine precision. Consider an emulated physical system subject to M linear equality constraints:

$$\mathbf{C}(\mathbf{x})\hat{\mathbf{y}} = \mathbf{d}(\mathbf{x}), \quad \mathbf{C} \in \mathbb{R}^{M \times K}, \quad \mathbf{d} \in \mathbb{R}^M, \quad (57)$$

where $\hat{\mathbf{y}} \in \mathbb{R}^K$ represents the model's output vector (e.g., vertical profiles of convective heating rate $Q_1 \equiv c_p P_T$ and apparent drying rate $Q_2 \equiv -L_v P_q$). In columnar atmospheric physics, Eq. (57) encapsulates column-integrated enthalpy and total moisture conservation:

$$\langle Q_1 \rangle - \langle Q_2 \rangle = L_v P_{\text{sfc}} + R_{\text{net}} - S_{\text{sfc}}, \quad (58)$$

$$\langle Q_2 \rangle = -L_v (P_{\text{sfc}} - E_{\text{sfc}}), \quad (59)$$

where $\langle f \rangle = \frac{1}{g} \int_0^{p_s} f(p) dp$ represents the mass-weighted vertical integral from surface pressure p_s to the top of atmosphere.

Instead of adding soft loss penalties (which only approximately penalize violations), AC-NN analytically partitions predicted variables into unconstrained free components $\hat{\mathbf{y}}_{\text{free}} \in \mathbb{R}^{K-M}$ and constrained boundary variables $\hat{\mathbf{y}}_{\text{constrained}} \in \mathbb{R}^M$. Rearranging Eq. (57):

$$\mathbf{C}_{\text{free}}\hat{\mathbf{y}}_{\text{free}} + \mathbf{C}_{\text{constrained}}\hat{\mathbf{y}}_{\text{constrained}} = \mathbf{d}(\mathbf{x}). \quad (60)$$

Assuming full row rank for the constraint matrix, $\mathbf{C}_{\text{constrained}}$ is invertible, allowing exact architectural determination of the remaining variables:

$$\hat{\mathbf{y}}_{\text{constrained}} = \mathbf{C}_{\text{constrained}}^{-1} (\mathbf{d}(\mathbf{x}) - \mathbf{C}_{\text{free}}\hat{\mathbf{y}}_{\text{free}}). \quad (61)$$

Because Eq. (61) is implemented as a fixed, parameter-free linear layer at the network's terminal head, the model satisfies column conservation of enthalpy and moisture to machine precision ($\epsilon_{\text{res}} < 10^{-15}$) across all training iterations and arbitrary out-of-distribution inputs, guaranteeing long-term numerical stability in multi-year climate runs without degrading emulation fidelity.

4.14.2 Ai2 Climate Emulator (ACE): Hundred-Year Stable Planetary Rollouts

Scaling physics-constrained neural operators to multi-decadal planetary climate emulation, Watt-Meyer et al. [50] created ACE (Ai2 Climate Emulator). ACE couples a Spherical Fourier Neural Operator (SFNO) backbone operating on FV3GFS 1° global horizontal grids across 82 prognostic 3D and 2D atmospheric fields.

ACE hard-codes structural physical conservation of global dry air mass:

$$M_{\text{dry}} = \int_{\mathbb{S}^2} p_s^{\text{dry}}(\lambda, \theta, t) d\Omega = \text{constant}, \quad (62)$$

and total column water vapor, preventing secular mass depletion over century-long autoregressive integrations. In continuous 100-year rollouts (> 146,000 consecutive 6-hour time steps), ACE exhibits unprecedented climate stability: global mean surface temperature drift remains strictly bounded under 0.04 K/century, while accurately reproducing the climatological Intertropical Convergence Zone (ITCZ), Hadley cell overturning circulation, and storm track variability. Running at 1,400 simulated years per day on a single GPU, ACE provides an empirical blueprint for replacing expensive atmospheric dynamical cores in climate projection ensembles.

4.15 Coupled Atmosphere-Ocean-Cryosphere Multi-Agent Emulation and Flux Coupling

A persistent limitation of standalone atmospheric foundation models (e.g., Pangu-Weather, GraphCast, Aurora) is that they treat sea surface temperature (SST) and sea ice concentration (SIC) as prescribed, static boundary conditions. While sufficient for medium-range synoptic weather (≤ 15 days), this uncoupled formulation fails across subseasonal-to-seasonal (S2S) and decadal timescales, where climate dynamics are governed by reciprocal thermodynamic air-sea-cryosphere feedbacks:

$$\boldsymbol{\tau}_{\text{air-sea}} = \rho_a C_D \|\mathbf{u}_a - \mathbf{u}_o\|(\mathbf{u}_a - \mathbf{u}_o), \quad (63)$$

$$Q_{\text{net}} = Q_{\text{SW}}^\downarrow(1 - \alpha) - Q_{\text{LW}}^\uparrow + Q_{\text{LW}}^\downarrow - Q_{\text{sens}} - Q_{\text{lat}}, \quad (64)$$

$$F_{\text{fresh}} = \rho_w (E - P - M_{\text{melt}}), \quad (65)$$

where $\boldsymbol{\tau}$ is wind stress momentum flux, Q_{net} is interfacial net heat flux, and F_{fresh} is surface freshwater mass flux.

4.15.1 DLESyM: Multi-Rate Asynchronous Earth System Coupler

To achieve fully coupled planetary climate simulation, Cresswell-Clay et al. [44] introduced DLESyM (Deep Learning Earth System Model). DLESyM formulates Earth system emulation as an asynchronous multi-agent system on equal-area HEALPix spherical grids, structurally decoupling the distinct characteristic dynamical timescales of the atmosphere and ocean:

- 1) Atmospheric Neural Solver ($\mathcal{M}_{\text{atm}}$): A spherical U-Net integrating fast synoptic dynamics (3D wind, temperature, humidity, geopotential) at $\Delta t_{\text{atm}} = 12$ hours.
- 2) Ocean-Sea Ice Neural Solver ($\mathcal{M}_{\text{ocn}}$): A specialized spherical U-Net integrating slow oceanic thermohaline dynamics (3D upper ocean temperature, salinity, currents, sea-ice thickness) at $\Delta t_{\text{ocn}} = 4$ days.

At every 4-day coupling interface, the atmospheric solver computes time-accumulated surface wind stress $\bar{\boldsymbol{\tau}}$ and net heat flux $\bar{Q}_{\text{net}}$, passing them to the ocean solver. In turn, the ocean solver updates the underlying SST and sea-ice concentration fields for the atmospheric model. DLESyM performs stable 1,000-year continuous coupled simulations without empirical damping or flux adjustments, maintaining global mean temperature drift below 0.08 K/century. Crucially, DLESyM faithfully reproduces the multi-year cycle of the El Niño-Southern Oscillation (ENSO), tropical cyclogenesis statistics, and the Indian Summer Monsoon, demonstrating that data-driven foundation models can capture long-term emergent coupled climate modes.

4.15.2 SamudrACE: 3D High-Resolution Foundation Earth System Emulation

Pushing coupled foundation models to full 3D spatial resolution, Duncan et al. [45] established SamudrACE. SamudrACE couples the 3D atmospheric ACE SFNO foundation emulator ($\Delta t = 6$ hours) with the 3D Samudra ocean emulator ($\Delta t = 5$ days), tracking 145 two-dimensional prognostic fields across atmospheric pressure levels and oceanic depth strata at 1° global resolution. Interfacial boundary couplers exchange heat, freshwater, and momentum fluxes subject to continuous conservation checks.

In multi-century benchmark evaluations, SamudrACE achieves an unprecedented simulation throughput of 1,100 simulated years per day (SYPD) on a single NVIDIA H100 GPU (over 100× faster than numerical coupled GCMs running across thousands of CPU cores on supercomputing clusters). SamudrACE reproduces the full power spectrum and phase-locking of ENSO SST anomalies in the Niño 3.4 region with an RMSE under 0.32 K, demonstrating the viability of coupled neural operators for trillion-parameter ensemble climate sensitivity projections.

5 Scientific Domain Applications

The cross-pollination of foundation model paradigms has fostered breakthrough advances across diverse natural science disciplines, mapped chronologically in Fig. 4.

5.1 Weather and Climate Forecasting

Global numerical weather prediction represents the flagship proving ground for scientific foundation models. As demonstrated by Pangu-Weather [6], GraphCast [7], FuXi [34], FengWu [48], and Aurora [9], deep learning architectures execute 10-day global forecasts in tens of seconds on a single GPU, delivering computational speedups of 1,000× to 10,000× compared to operational NWP supercomputing workflows. Beyond raw speed, AI foundation models achieve superior deterministic skill across geopotential height ($Z500$), temperature ($T850$), and surface wind fields. Furthermore, versatile foundation models such as ClimaX [69] demonstrate multi-task transferability by pretraining on synthetic CMIP6 climate projections and fine-tuning for regional climate downscaling and seasonal teleconnections.

As shown in Fig. 5, current state-of-the-art models occupy distinct Pareto trade-offs: while deterministic models like Pangu-Weather and GraphCast optimize accuracy at standard 0.25° resolution across 7–10 days, frontier models

Milestone Timeline of Scientific Foundation Models (2021-2026)

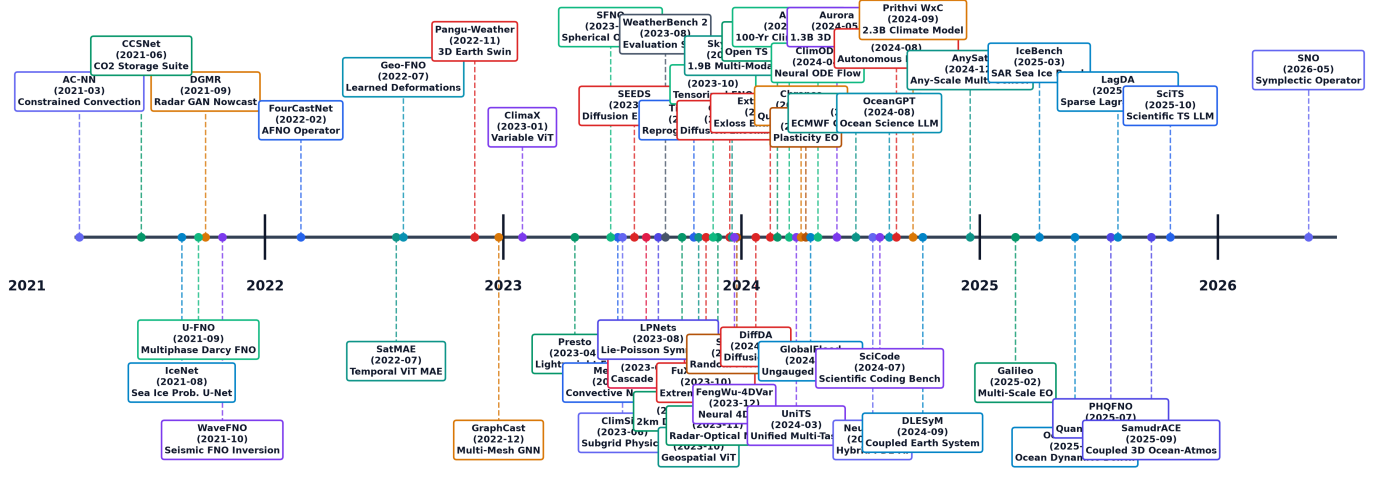


Fig. 4. Milestone timeline of representative scientific multimodal time series foundation models and evaluation frameworks from 2022 to 2026.

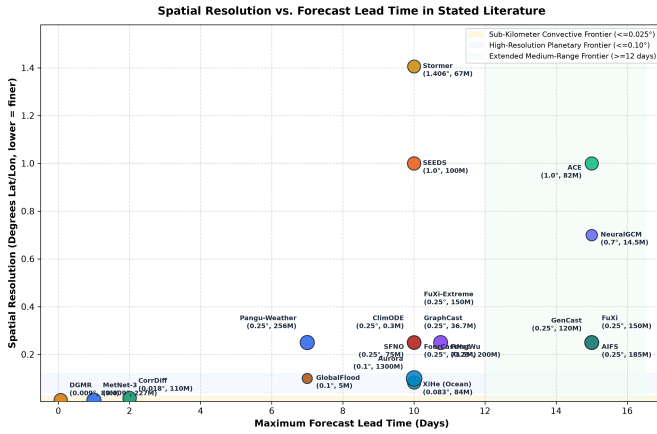


Fig. 5. Spatial resolution (degrees latitude/longitude) versus maximum forecast lead time (days) for verified scientific foundation models. Marker size is proportional to log model parameter count.

like Aurora advance spatial resolution to 0.1° , and cascade systems (FuXi) or diffusion ensembles (GenCast) extend skillful forecasting horizons out to 15 days.

5.2 Sub-Kilometer Convective-Scale Emulation and Generative Downscaling

A fundamental limitation of global foundation models (operating at $0.25^\circ \approx 25\text{--}31\text{ km}$ or $0.1^\circ \approx 10\text{ km}$) is their inability to resolve sub-kilometer convective phenomena. Convective thunderstorms, microbursts, urban flash flooding, and complex orographic wind channeling occur at localized scales ($< 3\text{ km}$) where non-hydrostatic atmospheric dynamics dominate. Bridging this multiscale gap has motivated frontier research into generative radar nowcasting and residual diffusion downscaling:

- 1) **Generative Radar Precipitation Nowcasting:** DGMR: Ravuri et al. [70] established the landmark DGMR (Deep Generative Model of Radar), deploying spatial-temporal generative adversarial networks for precipitation nowcasting at 1 km spatial and 5-minute temporal resolution across the United Kingdom. Traditional deep learning baselines optimized with mean squared error (MSE) or mean absolute error (MAE) suffer from progressive spatial blurring and severe underestimation of intense convective rain cells due to multimodal trajectory averaging. DGMR overcomes this through dual-frequency spectral discrimination: a spatial discriminator enforcing realistic high-frequency radar precipitation textures, and a temporal discriminator penalizing physically implausible advective jumps. In rigorous double-blind evaluations conducted with professional operational forecasters at the UK Met Office, DGMR was ranked first in 89% of cases for overall value, physical plausibility, and localized heavy rain cell fidelity compared to optical flow (PySTEPS) and convolutional recurrent baselines.
- 2) **Continental Multi-Sensor Convective Forecasting:** MetNet-3: Andrychowicz et al. [71] scaled convective forecasting across the entire continental United States (CONUS) out to 24-hour lead times at $1\text{--}4\text{ km}$ spatial resolution. Operating over a continuous spatial grid ($4,096 \times 4,096\text{ km}$), MetNet-3 assimilates heterogeneous and asynchronous observational modalities directly: high-resolution ground radar composites (MRMS), geostationary satellite multi-spectral imagery (GOES-16), surface meteorological station networks, and numerical forecast initializations (HRRR). Employing axial multi-scale self-attention and spatially localized lead-time con-

ditioning, MetNet-3 outperforms operational High-Resolution Rapid Refresh (HRRR) numerical physics models on 24-hour precipitation and surface temperature forecasts, producing complete continental updates every two minutes on TPU hardware.

- 3) Kilometer-Scale Residual Generative Downscaling: CorrDiff: Bridging coarse global foundation model outputs (25 km ERA5) to local convective scales (2 km), Mardani et al. [72] formulated CorrDiff, a residual corrective diffusion framework developed for extreme weather prediction over complex topography (Taiwan). Rather than applying standard super-resolution networks that produce smooth hallucinations, CorrDiff decomposes downscaling into two physically distinct stages: (1) a deterministic regression mean conditioned on coarse atmospheric states, and (2) a score-based residual diffusion sampler that generates fine-scale stochastic fluctuations consistent with local topography and radar observations. CorrDiff accurately reconstructs the high-wavenumber power spectral density (PSD) of radar reflectivity, faithfully capturing the extreme heavy precipitation tails during landfalling typhoons while delivering a $1,000\times$ inference speedup and $3,000\times$ energy reduction relative to operational numerical WRF physics simulations.

5.3 Earth Observation and Remote Sensing Time Series

Satellite constellations continuously monitor terrestrial ecosystems, forest biomass, crop phenology, and surface water regimes through high-dimensional spatio-temporal revisit series. Foundation models in this domain must overcome three core physical challenges: (1) multi-sensor sensor heterogeneity (interleaving optical multispectral reflectance with active Synthetic Aperture Radar microwave backscatter), (2) irregular temporal acquisitions caused by dynamic cloud occlusion and orbital revisit periods, and (3) multi-scale spatial ground sampling distances (GSD) varying across orders of magnitude (0.2 m to 500 m).

5.3.1 Temporal-Spectral Masked Autoencoding

Early foundation models focused on optical and multispectral acquisitions. SatMAE [12] introduced temporal-spectral masked autoencoding, grouping Sentinel-2 and Landsat bands into physically coherent spectral groups (visible, near-infrared, and shortwave infrared) and masking spatio-temporal tokens independently. Presto [13] designed a lightweight, pixel-level transformer that processes sparse temporal series of Sentinel-1 SAR, Sentinel-2 optical, and ERA5 weather variables without computing expensive spatial attention, delivering high efficiency for global crop classification. Advancing multi-scale representation learning, Galileo [14] integrated multi-sensor satellite imagery across local and global scales with self-supervised masking, while AlphaEarth Foundations [16] compressed multi-sensor Earth observation into dense 64-dimensional neural embedding fields integrated into planetary cloud analytics. Scaling parameter capacity to multi-modal constellations, SkySense [58] unified optical, SAR, and DEM time series with 1.9 billion parameters, achieving state-of-the-art results across the GEO-Bench [59] suite.

5.3.2 Contrastive Radar-Optical Masked Autoencoding: CROMA

While optical sensors provide rich chemical and spectral reflectance signatures, they are blinded by cloud cover and darkness. Synthetic Aperture Radar (SAR), emitting microwave pulses (e.g., Sentinel-1 C-band VV/VH polarizations), penetrates clouds and operates day and night, capturing surface dielectric constants and geometric roughness. To fuse these complementary physical regimes, Fuller et al. [42] established CROMA (Contrastive Radar-Optical Masked Autoencoder).

CROMA unifies self-supervised contrastive learning and cross-modal masked reconstruction. Let $\mathbf{X}_{\text{opt}} \in \mathbb{R}^{B_{\text{opt}} \times H \times W}$ and $\mathbf{X}_{\text{sar}} \in \mathbb{R}^{B_{\text{sar}} \times H \times W}$ represent spatially aligned optical and SAR image pairs. CROMA applies separate modality encoders $\mathcal{E}_{\text{opt}}$ and $\mathcal{E}_{\text{sar}}$ with high masking ratios ($> 70\%$), aligning global representations via a symmetric InfoNCE loss:

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{2N} \sum_{i=1}^N \left(\log \frac{\exp(\mathbf{z}_{\text{opt}}^{(i)} \cdot \mathbf{z}_{\text{sar}}^{(i)} / \tau)}{\sum_j \exp(\mathbf{z}_{\text{opt}}^{(i)} \cdot \mathbf{z}_{\text{sar}}^{(j)} / \tau)} + \log \frac{\exp(\mathbf{z}_{\text{sar}}^{(i)} \cdot \mathbf{z}_{\text{opt}}^{(i)} / \tau)}{\sum_j \exp(\mathbf{z}_{\text{sar}}^{(i)} \cdot \mathbf{z}_{\text{opt}}^{(j)} / \tau)} \right) \quad (66)$$

Simultaneously, a cross-modal transformer decoder predicts masked patch pixels by fusing optical and SAR tokens. Crucially, to overcome the rigid position-embedding limits of standard Vision Transformers, CROMA introduces 2D-ALiBi (2D Attention with Linear Biases) and X-ALiBi (cross-attention ALiBi):

$$\text{Attention}_{i,j} = \text{softmax} \left(\frac{\mathbf{q}_i \mathbf{k}_j^T}{\sqrt{d}} - m \cdot \|\mathbf{x}_i - \mathbf{x}_j\|_2 \right) \mathbf{v}_j, \quad (67)$$

where $m > 0$ is a head-specific geometric slope and $\|\mathbf{x}_i - \mathbf{x}_j\|_2$ denotes the Euclidean distance between 2D patch centers on the Earth’s surface. By substituting static position embeddings with continuous distance penalties, CROMA performs zero-shot spatial extrapolation to satellite scenes up to $17.6\times$ larger than its training crops (120×120 to 504×504 and beyond) without performance degradation, setting new records on the BioMassters canopy height benchmark and EuroSAT classification.

5.3.3 Universal Multi-Resolution Earth Observation: AnySat

Resolving the acute spatial resolution fragmentation across satellite constellations, Astruc et al. [43] introduced AnySat. Rather than locking input tensors to fixed grid sizes (e.g., 10 m for Sentinel or 30 m for Landsat), AnySat formulates Earth observation foundation modeling across arbitrary spatial resolutions spanning three orders of magnitude (0.2 m aerial orthophotos, 10 m Sentinel-2, to 500 m MODIS).

AnySat is built on a scale-adaptive Joint Embedding Predictive Architecture (JEPA). Each input patch i from sensor s is explicitly conditioned on its continuous physical ground sampling distance GSD_s :

$$\mathbf{e}_i = \mathbf{W}_s \mathbf{x}_i + \psi_{\text{scale}}(\log \text{GSD}_s) + \psi_{\text{spatial}}(\mathbf{p}_i), \quad (68)$$

where ψ_{scale} is a multi-frequency sinusoidal scale embedding. Pretrained on the massive GeoPlex corpus (unifying 11 distinct spaceborne, airborne, and elevation sensors), AnySat achieves zero-shot multi-sensor fusion, allowing arbitrary combinations of available sensors at test time (e.g., substituting missing high-resolution aerial imagery with Sentinel-2 and

SPOT) and outperforming single-sensor foundation baselines across land cover segmentation, agricultural crop mapping, and post-disaster flood inundation.

5.4 Hydrology and Extreme Flood Modeling

Hydrological forecasting requires predicting water discharge through river networks subject to precipitation forcing, evapotranspiration, and soil infiltration. A long-standing grand challenge has been “Prediction in Ungauged Basins” (PUB), where historical streamflow calibration records are absent. Nearing et al. [19] demonstrated that deep learning models trained on 5,680 diverse river basins across the globe achieve unprecedented zero-shot flood forecasting skill in ungauged catchments, extending actionable early warning lead times from 0 to 5 days globally and outperforming the Copernicus GloFAS operational physics-based system. Complementing these architectural advances, Kratzert et al. [20] released Caravan, a standardized global community dataset spanning 6,830 catchments worldwide with unified ERA5-Land meteorological forcings, catchment attributes, and daily streamflow records. By eliminating the geographic fragmentation of regional datasets (e.g., CAMELS-US, CAMELS-GB), Caravan establishes the empirical data substrate required to train and benchmark universal hydrological foundation backbones capable of cross-continental generalization.

5.5 Oceanography and Marine Systems

Ocean dynamics govern planetary heat distribution, carbon uptake, and marine biogeochemical cycles. Wang et al. [21] developed XiHe, a hierarchical ocean transformer trained on 28 years of GLORYS12V1 reanalysis at eddy-resolving $1/12^\circ$ resolution. By incorporating land-ocean boundary masking and multi-level vertical attention, XiHe models mesoscale eddies, sea surface temperature anomalies, and 3D velocity vectors, outperforming traditional operational hydrodynamic models (e.g., PSY4, GIOPS) while generating 10-day global forecasts in under 0.4 seconds. Addressing the extreme sparsity of oceanic observation networks, 4DVarNet [61] formulates neural variational data assimilation to invert along-track satellite altimetry into continuous 2D sea surface height (SSH) dynamics, demonstrating that learnable unrolled variational solvers reconstruct fine-scale mesoscale eddies and oceanic kinetic energy spectra far more effectively than classical optimal interpolation (OI).

To standardize evaluation across emerging marine foundation models, El Aouni et al. [22] established OceanBench, an open operational benchmarking suite for data-driven global ocean forecasting systems developed in collaboration with Mercator Ocean International. OceanBench benchmarks predictive skill for sea surface height (SSH), sea surface temperature (SST), and horizontal current velocities across 1- to 10-day lead times against CMEMS GLORYS12 reanalyses and in-situ altimetry observations. Bridging physical ocean simulations with cross-modal scientific reasoning, Bi et al. [23] introduced OceanGPT, the first domain-specific foundation language model for marine science. Pretrained and aligned on the DoInstruct corpus comprising multimodal oceanographic time series (temperature-salinity profiles from Argo float arrays and mooring buoys), sensor trajectories, and marine scientific literature, OceanGPT demonstrates autonomous

reasoning across ocean dynamics, thermohaline circulation changes, and marine phenomenon attribution under physical consistency constraints.

5.6 Geophysics and Seismology

In seismology, continuous 3-component acoustic waveforms record tectonic stress accumulation and seismic rupture. While classical phase-picking baselines like PhaseNet [25] utilized supervised 1D U-Nets for local seismic windows, modern foundational architectures such as SeisT [24] train hierarchical seismogram transformers on tens of millions of global waveforms. SeisT supports multi-task downstream adaptation, simultaneously performing earthquake detection, high-precision P/S phase picking, initial-motion polarity determination, and magnitude/epicenter regression. Providing the standardized foundational infrastructure for seismic machine learning, Woollam et al. [26] released SeisBench, unifying 30+ million waveforms across global catalogues (ETHZ, GEOFON, INSTANCE, SCEDC, STEAD) into a standardized multi-task evaluation and transfer benchmark.

Advancing beyond discrete phase picking to continuous wavefield modeling and subsurface tomography, Yang et al. [40] formulated WaveFNO, establishing the first continuous Fourier neural operator framework for seismic wave propagation and full-waveform inversion (FWI). Solving the 2D/3D elastodynamic wave equation across heterogeneous, discontinuous velocity models, WaveFNO maps acoustic and elastic impedance directly to continuous displacement wavefields $\mathbf{u}(\mathbf{x}, t)$. Because the neural operator is end-to-end differentiable, full-waveform velocity updates $\nabla_c \mathcal{J}$ are obtained via reverse-mode automatic differentiation in seconds, outperforming conventional finite-difference wave simulators by several orders of magnitude while circumventing cycle-skipping local minima in subsurface imaging.

5.7 Polar Cryosphere and Sea Ice Dynamics

Polar cryospheric dynamics—governing Arctic and Antarctic sea ice extent, ice-sheet mass balance, and permafrost degradation—exert profound thermodynamic control over planetary albedo and global thermohaline circulation. However, polar regions suffer from acute geographic observation gaps, severe seasonal cloud cover obscuring optical satellites, and complex non-linear rheology governing sea ice fracture, drift, and melt-pond formation.

Addressing long-range seasonal sea ice forecasting, Anderson et al. [28] established IceNet, a probabilistic deep learning system predicting monthly Arctic sea ice concentration maps up to 6 months in advance. Trained on 2,192 years of CMIP6 climate model simulations and calibrated against 40 years of observational sea ice concentration data from OSI-SAF and ERA5 reanalyses, IceNet employs an ensemble of U-Net architectures with temperature scaling to produce calibrated probabilistic forecasts of sea ice presence (concentration $> 15\%$). Operating at 25-km spatial resolution, IceNet demonstrably surpasses the operational ECMWF SEAS5 dynamical seasonal physics model, particularly in predicting anomalous summer ice retreat and extreme September minimum sea ice extents.

To address the lack of standardized machine learning evaluation across polar remote sensing, Taleghan et al. [29]

introduced IceBench, a comprehensive multi-modal benchmark for deep learning-based sea ice type classification. Built upon the AI4Arctic Sea Ice Challenge corpus, IceBench unifies dual-polarization Sentinel-1 Synthetic Aperture Radar (SAR) imagery, AMSR2 microwave radiometer brightness temperatures, and ERA5 atmospheric forcings across operational ice chart labels from the Danish Meteorological Institute (DMI). By standardizing evaluation protocols across pixel-based (U-Net, SegNet) and patch-based (Vision Transformer, ResNet) architectures, IceBench provides the first rigorous benchmark assessing spatial transferability across distinct Arctic sub-regions and temporal transferability across transition seasons (freeze-up versus summer melt).

5.8 Geo-Energy, Geothermal Reservoirs, and Subsurface Carbon Sequestration

The transition to zero-carbon energy systems fundamentally depends on deep subsurface reservoirs for geological carbon sequestration (GCS), geothermal energy extraction, and underground hydrogen storage. Subsurface fluid dynamics are governed by non-linear, multi-phase Darcy-Brinkman flow through highly heterogeneous, anisotropic geological formations:

$$\phi(\mathbf{x}) \frac{\partial(\rho_\alpha S_\alpha)}{\partial t} + \nabla \cdot (\rho_\alpha \mathbf{u}_\alpha) = q_\alpha, \quad \mathbf{u}_\alpha = - \frac{k_{r\alpha}(S_\alpha) \mathbf{K}(\mathbf{x})}{\mu_\alpha} (\nabla p_\alpha - \rho_\alpha \mathbf{g}), \quad (69)$$

where $\alpha \in \{\text{CO}_2, \text{brine}\}$ denotes fluid phases, $\phi(\mathbf{x})$ is formation porosity, $\mathbf{K}(\mathbf{x})$ is the heterogeneous permeability tensor, S_α is phase saturation ($S_{\text{CO}_2} + S_{\text{brine}} = 1$), $k_{r\alpha}$ is relative permeability, μ_α is dynamic viscosity, and p_α represents phase pressure. Because traditional numerical reservoir simulators (such as CMG GEM, TOUGH2, or Eclipse) require hours to days per run to solve coupled non-linear pressure-saturation equations on fine 3D computational grids, foundation neural operator models have emerged as high-leverage surrogates for real-time reservoir monitoring, risk assessment, and closed-loop well control.

5.8.1 U-FNO: Enhanced Multiscale Fourier Neural Operators for Multiphase Flow

Addressing the limitations of standard Fourier neural operators on discontinuous fronts, Wen et al. [30] pioneered the U-FNO (U-Net enhanced Fourier Neural Operator). Standard FNO applies global frequency-domain truncations that exhibit Gibbs oscillations and oversmooth sharp saturation shocks between injected supercritical CO₂ and ambient saline water. U-FNO resolves this by interleaving global 2D/3D Fourier spectral convolution layers with multiscale U-Net spatial skip connections:

$$\mathbf{v}_{l+1}(\mathbf{x}) = \sigma \left(\mathcal{W}_l * \mathbf{v}_l(\mathbf{x}) + \mathbf{W}_{\text{spatial}}^{(l)} \mathbf{v}_l(\mathbf{x}) + \mathcal{U}_{\text{skip}}^{(l)}(\mathbf{v}_l)(\mathbf{x}) \right). \quad (70)$$

Trained on extensive multiphase GCS simulation suites under realistic permeability fields with strong spatial correlation lengths, U-FNO accurately forecasts 30-year multi-phase CO₂ gas-phase saturation fronts and pressure buildup maps. U-FNO achieves over a 10,000 $\times$ speedup relative to numerical PDE solvers while reducing gas saturation prediction errors by over 50% compared to standard convolutional architectures, enabling real-time risk evaluation of caprock fracture and induced seismicity.

5.8.2 CCSNet: Full-Physics Deep Learning Modeling Suite with Real Thermodynamic EOS

Scaling operator modeling to industrial carbon capture and storage operations, Wen, Hay, and Benson [31] established CCSNet. Industrial GCS involves complex thermodynamic phase behavior across wide ranges of reservoir temperatures (30°C–120°C), pressures (10–40 MPa), and salinities (0–250,000 ppm). CCSNet couples multi-scale neural operators directly to a data-driven thermodynamic Equation of State (EOS) for supercritical CO₂ and brine:

$$\rho_{\text{CO}_2} = f_{\text{EOS}}(p, T), \quad \mu_{\text{CO}_2} = g_{\text{EOS}}(p, T, \rho_{\text{CO}_2}), \quad (71)$$

guaranteeing thermodynamic consistency across sub-critical and supercritical fluid regimes. Pretrained on over 20,000 synthetic heterogeneous reservoir simulations across variable injection rates, well configurations, and geological facies, CCSNet evaluates 50-year structural, capillary, and solubility trapping in milliseconds. This empowers field operators to perform real-time optimization of well placement and dynamically steer injection rates to maximize long-term storage security.

5.8.3 Geo-FNO: Multiphase Flow on Complex and Faulted Geometries

Complementing these models, Li et al. [36] demonstrated Geo-FNO on 3D subsurface flow and geomechanical deformation. Natural reservoir strata are rarely simple Cartesian boxes; they contain inclined layers, structural faults, and pinch-outs. By learning smooth coordinate transformations $\phi : \Omega \rightarrow [0, 1]^d$, Geo-FNO maps complex faulted geological geometries into canonical reference grids, executing continuous operator inference without losing boundary fidelity or requiring costly unstructured mesh generation.

5.9 Space Weather and Heliophysics

Space weather phenomena—including solar flares, coronal mass ejections (CMEs), and geomagnetic storms—exert severe risks on satellite infrastructure and terrestrial power grids. Roy et al. [27] introduced Surya, a 366-million parameter foundation model for heliophysics trained on 13 years of continuous multi-wavelength extreme ultraviolet imagery from NASA’s Solar Dynamics Observatory (SDO). Surya incorporates spectral gating and long-short attention to track active region magnetic helicity, delivering state-of-the-art solar flare and solar wind forecasting.

5.10 Cross-Disciplinary Universal Time Series Foundations

Bridging physical boundaries between disparate earth science subdomains, universal foundation models like UniTS [32] and MOMENT [33] demonstrate that pretraining on multi-domain scientific collections enables cross-physics zero-shot transfer. By decoupling sequence tokens from variable identities and applying prompt-conditioned task masks, universal backbones can be deployed across river streamflow, power grids, seismic sensors, and atmospheric stations without re-training architectural weights.

6 Scientific Reasoning LLMs and Autonomous Agents

6.1 The Paradigm Shift: From Numerical Fitting to Scientific Reasoning

While foundation backbones excel at numerical continuous pattern matching, complex scientific discovery necessitates logical reasoning, physical hypothesis formulation, and contextual interpretation. Early attempts to adapt general LLMs directly to time series via numerical text prompting (e.g., feeding raw floating-point numbers as text tokens) incurred severe limitations: token sequence inflation, numerical precision truncation, and inability to capture continuous derivatives [2]. Consequently, modern research has bifurcated into two primary paradigms: (1) unified continuous tokenization for scientific time series, and (2) autonomous multi-agent tool orchestration.

6.2 Unified Continuous Tokenization: SciTS and TimeOmni

Wu et al. [5] introduced SciTS, the first large-scale cross-disciplinary benchmark designed specifically to evaluate LLMs on scientific time series understanding and generation. Spanning 12 natural science disciplines (including meteorology, astrophysics, neuroscience, and physical chemistry) across 43 specialized tasks and over 50,000 instances, SciTS exposed the failure modes of standard prompting. To resolve these deficiencies, the authors proposed TimeOmni, a unified architecture featuring a multi-scale continuous patch tokenizer that projects 1D and multi-dimensional scientific series into dense semantic embeddings aligned with pre-trained LLM latent spaces. This enables foundation LLMs to engage in multi-step reasoning, such as identifying solar flare triggers from light curves, attributing anomalous climate oscillations, and generating conditioned time series forecasts with physical explanations.

6.3 Frequency-Aware Contextual Integration: ClimateLLM

Rather than treating atmospheric series as unstructured text, Li et al. [55] introduced ClimateLLM. This framework decomposes multi-station and gridded meteorological sequences into orthogonal frequency bands via Fourier transform modules. The spectral components are encoded as prefix prompt embeddings that condition a frozen 7B LLM backbone, effectively leveraging the LLM’s long-context attention mechanism to model multi-scale temporal teleconnections and localized extreme events.

6.4 Domain-Specialized Foundation LLMs: K2 for Geoscience

Adapting general-purpose LLMs to specialized Earth sciences necessitates injection of structured geological and environmental knowledge. Deng et al. [53] introduced K2, a foundational language model tailored for geoscience knowledge understanding and utilization. Pretrained on over 5.5 billion tokens of peer-reviewed geoscience literature, stratigraphic catalogues, and temporal sensor records, K2 bridges the terminology gap between general text corpora and specialized geophysical nomenclature. Through the GeoSignal instruction-tuning protocol, K2 learns to reason across temporal geological processes,

infer lithological fault dynamics, and assist researchers in interpreting complex spatio-temporal sensor logs with physical groundedness.

6.5 Multimodal Vision-Language Grounding: GeoChat

Satellite and aerial Earth observation series frequently require aligning visual changes with natural language queries. Kuckreja et al. [54] developed GeoChat, the first grounded large vision-language model explicitly engineered for remote sensing. Built upon a Vicuna-1.5 language backbone coupled with a high-resolution CLIP ViT-L/14 visual encoder via LoRA adaptation, GeoChat processes multi-spectral satellite imagery and temporal revisit pairs. Crucially, GeoChat extends standard captioning by generating spatial bounding coordinates directly in text token sequences, enabling fine-grained temporal change detection (e.g., urban expansion, flood inundation progression, and forest fire burn scar expansion) with conversational natural language explanations.

6.6 Reprogramming Pretrained Language Models: Time-LLM

Rather than training domain transformers from scratch or fine-tuning entire parameter spaces, Jin et al. [73] introduced Time-LLM, pioneering the paradigm of reprogramming frozen large language models for time series forecasting. Time-LLM recognizes that pre-trained language backbones (such as LLaMA-7B or GPT-2) possess exceptional cross-modal pattern recognition and sequence abstraction capabilities. To align modalities, Time-LLM:

- 1) **Input Patch Reprogramming:** Divides continuous input series into patches and projects them into the word token embedding space through a linear reprogramming layer conditioned on cross-attention with textual domain prototypes.
- 2) **Prompt-as-Prefix Conditioning:** Augments the reprogrammed patches with domain task instructions (e.g., dataset semantics, sampling frequency, and trend descriptions) as natural language prefix prompts.
- 3) **Frozen Backbone Inference:** Keeps the foundational LLM weights entirely frozen during training, tuning only lightweight linear adapters.

By reprogramming frozen language models, Time-LLM achieves superior few-shot and zero-shot performance across scientific and benchmark datasets, proving that language model representations serve as universal priors for temporal dynamics.

6.7 Autonomous Multi-Agent Climate Workflows: ClimateAgent

Moving beyond single-model inference, Li et al. [56] introduced ClimateAgent, an autonomous multi-agent orchestration framework for complex climate data science. Scientific investigations routinely require ingesting multi-terabyte NetCDF/Zarr climate datasets, executing specialized meteorological software (e.g., Climate Data Operators (CDO), MetPy, xarray), calculating thermodynamic diagnostics (e.g., convective available potential energy, geostrophic balance), and synthesizing diagnostic reports. ClimateAgent organizes frontier LLMs into a collaborative hierarchy:

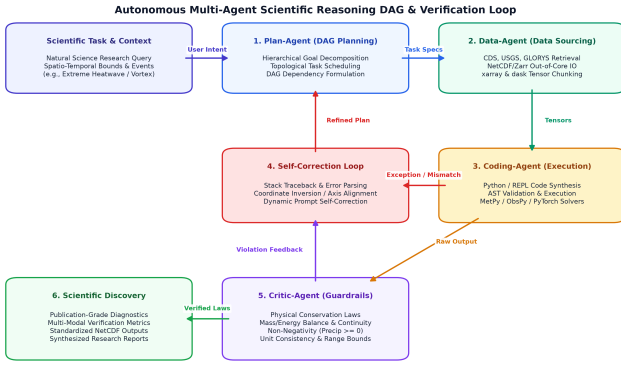


Fig. 6. Autonomous multi-agent scientific reasoning DAG and physical verification feedback loop. The workflow decomposes user goals into topological task DAGs, retrieves distributed tensor shards via out-of-core data agents, executes code in stateful execution sandboxes, autonomously parses tracebacks for error recovery, and verifies physical conservation guardrails prior to final report generation.

- **Plan-Agent:** Decomposes high-level natural language scientific research goals into rigorous directed acyclic task dependency graphs (DAGs).
- **Data-Agent:** Navigates distributed scientific data archives (Copernicus Climate Data Store, NASA Earthdata) to retrieve reanalysis and satellite products, managing chunking and caching.
- **Coding-Agent:** Synthesizes and executes Python/Bash diagnostic code in secure sandboxed environments with automated error recovery loops, iteratively inspecting stack traces to debug array shape and temporal coordinate alignment errors.
- **Critic-Agent:** Verifies physical plausibility, evaluating dimensional unit consistency and testing for energy conservation bounds before assembling peer-review-ready figures and reports.

Evaluated on the standardized Climate-Agent-Bench-85, ClimateAgent demonstrated an 82% task completion rate without human intervention, marking a concrete transition from static numerical forecasters to self-directed AI co-scientists capable of hypothesis formulation and autonomous verification in natural Earth systems.

6.8 Tool Invocation Protocols and Execution Sandboxes for Scientific Co-Scientists

As illustrated in Fig. 6, the operational viability of autonomous scientific co-scientists hinges upon rigorous DAG scheduling, stateful computational sandboxes, dynamic error recovery loops, and physical plausibility guardrails:

- 1) **Hierarchical DAG Task Scheduling:** When presented with a complex scientific inquiry (e.g., assessing the dynamic intensification of an atmospheric river or Mediterranean marine heatwave), the Plan-Agent formulates a topological Directed Acyclic Graph. Nodes represent modular atomic subtasks (data query, re-gridding, thermodynamic flux computation, kinetic energy budget integration), while directed edges enforce data dependencies.

- 2) **Structured Tool Invocation & Distributed Ingestion:** Scientific foundation agents communicate with computational tools via structured JSON schema function calls. Rather than generating unconstrained code, the Data-Agent triggers domain-specific API endpoints (e.g., Copernicus Marine Toolbox, ECMWF CDS API, USGS Hydro-Data) and manages out-of-core sharding through xarray and dask, preventing memory exhaustion when handling multi-terabyte spatio-temporal datasets.
- 3) **Stateful Computational Sandboxes:** Executing scientific workflows requires isolated REPL (Read-Eval-Print Loop) environments equipped with verified domain libraries:

- **Atmospheric & Earth Observation:** xarray and dask for out-of-core chunked tensor operations; MetPy for thermodynamic diagrams and potential vorticity inversions; rasterio and rioxarray for satellite raster projections.
- **Ocean Dynamics:** oceanbench and copernicus-marine client APIs for automated GLORYS and altimetry verification.
- **Geophysics:** ObsPy and SeisBench [26] for streaming waveform filtering, phase picking, and seismic moment tensor inversions.

- 4) **Dynamic Traceback Self-Correction Loop:** Scientific data formats frequently exhibit idiosyncratic edge cases: inverted latitude coordinate ordering ($90^\circ \rightarrow -90^\circ$ vs. $-90^\circ \rightarrow 90^\circ$), non-standard calendars (e.g., 360-day climate model years in CMIP6), or incompatible vertical pressure unit conventions (Pa vs. hPa). As depicted in Fig. 6 (Loop 4), when the Python runtime emits an exception or dimension mismatch, the agent automatically captures the execution traceback, inspects array metadata (via `ds.dims` and `ds.coords`), and performs self-reflection to iteratively synthesize corrected code blocks.
- 5) **Physical Plausibility Verification Guardrails:** Even syntactically error-free code may produce unphysical scientific outputs. As shown in Fig. 6 (Stage 5), the Critic-Agent acts as a domain-specific gatekeeper: evaluating strict conservation of mass and energy, checking non-negativity of physical quantities (precipitation $P \geq 0$, solar radiation $R \geq 0$), verifying bounded relative humidity ($0\% \leq RH \leq 100\%$), and enforcing dimensional consistency across derived units before approving publication-grade artifacts.

6.9 Closed-Loop Autonomous Scientific Discovery and Empirical Coding Benchmarks

Moving beyond passive question answering and isolated diagnostic script synthesis, the frontier of scientific reasoning aims for fully autonomous, closed-loop scientific experimentation and discovery:

- 1) **Autonomous Multi-Agent Scientific Discovery:** The AI Scientist: Lu et al. [74] introduced The AI Scientist, the first comprehensive system for fully automated open-ended scientific discovery driven by

frontier foundation LLMs. Rather than executing pre-defined static scripts, The AI Scientist autonomously completes the entire scientific research lifecycle:

- **Novelty Search & Ideation:** Generates novel research hypotheses and cross-references academic literature via the Semantic Scholar API, verifying that proposed scientific angles have not been previously investigated.
- **Autonomous Experimentation & Code Modification:** Clones experimental repositories, dynamically modifies neural architectures or numerical solvers, executes experiments in isolated sandboxes, and collects empirical logs.
- **Manuscript Generation & Visualization:** Formulates a complete, publication-grade LaTeX paper complete with experimental plots, mathematical definitions, and properly cited bibliography entries.
- **Automated Peer Review:** Deploys an automated LLM reviewer agent calibrated against standard conference review criteria (e.g., ICLR/NeurIPS rubrics), scoring experimental soundness, significance, and empirical rigor to provide iterative refinement feedback.

This paradigm illustrates how scientific reasoning agents can transition from mere computational assistants to autonomous research partners capable of steering complex simulation experiments.

- 2) **Multi-Step Scientific Numerical Reasoning Benchmark: SciCode:** While general-purpose coding benchmarks (e.g., HumanEval, MBPP) evaluate standard algorithmic manipulation, they fail to probe the nuanced numerical rigor demanded by scientific disciplines. To establish an authoritative benchmark, Tian et al. [75] released SciCode, an expert-curated benchmark comprising 338 multi-step scientific coding problems developed by natural scientists spanning Physics, Fluid Dynamics, Materials Science, and Applied Mathematics. Each problem decomposes complex physical phenomena (e.g., Navier–Stokes vorticity advection, quantum wavepacket dispersion, crystal lattice dislocations) into sequential mathematical sub-equations requiring numerical integration (Runge–Kutta, symplectic integrators), boundary condition handling, and multidimensional tensor manipulation in `scipy` and `numpy`. Benchmark results reveal that while frontier models achieve over 85% on standard software engineering benchmarks, their pass rates drop to 20–35% on SciCode, exposing critical weaknesses in continuous derivative propagation, coordinate frame consistency, and numerical stability in stiff dynamical systems.

7 Evaluation Benchmarks and Model Synthesis

7.1 Standardized Scientific Evaluation Suites

7.1.1 WeatherBench 2: The Operational Standard for Atmospheric AI

Prior to 2023, data-driven weather models suffered from disparate evaluation protocols, making direct comparisons

difficult. Rasp et al. [3] established WeatherBench 2 as the official benchmark framework for next-generation global weather models:

- **Headline Deterministic Scores:** Evaluates latitude-weighted root mean square error (RMSE) and anomaly correlation coefficient (ACC) against operational ECMWF IFS HRES analyses for key predictive fields: geopotential at 500 hPa ($Z500$), temperature at 850 hPa ($T850$), 2-meter air temperature ($T2m$), and 10-meter wind speed ($U10/V10$).
- **Probabilistic Metrics:** Computes continuous ranked probability scores (CRPS), spread-error ratios, and brier scores for ensemble predictions (demonstrated in GenCast [4], SEEDS [11], and FuXi-Extreme [65]).
- **Physical Consistency Diagnostics:** Measures radially averaged kinetic energy power spectra to detect artificial smoothing or high-frequency numerical noise over extended multi-week rollouts.

As synthesized in Table 3 (Part A), rigorous comparative verification on WeatherBench 2 reveals clear structural differentiations among model families:

- 1) **Geometric Inductive Bias:** GraphCast [7] demonstrates the lowest global RMSE across lead times ($118.2 \text{ m}^2/\text{s}^2$ at Day 3 and $268.4 \text{ m}^2/\text{s}^2$ at Day 5 for $Z500$). Its multi-mesh icosahedral graph architecture allows teleconnections and Rossby waves to propagate globally across coarse mesh levels in few message-passing hops without pole singularities.
- 2) **Cascade Architecture vs. Autoregressive Decay:** FuXi [34] achieves the second-lowest error at extended horizons ($278.1 \text{ m}^2/\text{s}^2$ at Day 5, $472.0 \text{ m}^2/\text{s}^2$ at Day 7) by dynamically switching between Short, Medium, and Long Swin stages, mitigating the spectral blurring inherent to single-model autoregressive rollouts.
- 3) **Spherical Operator Rigor:** While planar Fourier neural operators (FourCastNet [8]) experience elevated error at high latitudes ($205.6 \text{ m}^2/\text{s}^2$ at Day 3), Spherical Fourier Neural Operators (SFNO [35]) substantially improve accuracy ($158.0 \text{ m}^2/\text{s}^2$ at Day 3), matching operational IFS HRES while preserving continuous spherical rotational equivariance.
- 4) **Operational Production Integration:** AIFS [57] achieves competitive verification scores ($124.6 \text{ m}^2/\text{s}^2$ at Day 3 and $275.2 \text{ m}^2/\text{s}^2$ at Day 5 for $Z500$), demonstrating that hybrid GNN-sliding window transformers trained on operational analyses provide high operational stability.
- 5) **Differentiable Hybrid Physics:** NeuralGCM [51] achieves $128.0 \text{ m}^2/\text{s}^2$ at Day 3 and $282.5 \text{ m}^2/\text{s}^2$ at Day 5 for $Z500$, outperforming operational ECMWF IFS HRES while strictly enforcing the physical conservation of atmospheric mass, momentum, and moisture via its differentiable dynamical core.

7.1.2 ClimateBench and ClimSim: Climate Emulation and Subgrid Physics

Watson-Parris et al. [76] formulated ClimateBench, standardizing machine learning evaluation for multidecadal climate projections across Shared Socioeconomic Pathways (SSPs).

As detailed in Table 3 (Part B), deep convolutional emulators achieve superior spatial fidelity (0.18 K annual RMSE for surface temperature TAS , 0.31 mm/day for precipitation PR), dramatically reducing emulation error compared to historical linear pattern scaling baselines (0.32 K and 0.46 mm/day). Complementing macroscopic climate emulation, ClimSim [52] standardizes evaluation for high-resolution subgrid cloud, convective, and radiative physics emulators, providing 5.7 billion multiscale atmospheric column profiles to test hybrid neural-PDE stability over decadal rollouts.

7.1.3 Caravan and GlobalFlood: Large-Sample Hydrological Benchmarking

As synthesized in Table 3 (Part C), Nearing et al. [19] benchmarked AI hydrological foundation models across 5,680 river gauges from the global Caravan community dataset [20]. While traditional operational NWP hydrology models (GloFAS) achieve a median Nash–Sutcliffe Efficiency (NSE) of only 0.35 at Day 1 and 0.26 at Day 5 due to uncalibrated catchment heterogeneities, Entity-Aware LSTMs (EA-LSTM) coupled with river network routing achieve a median NSE of 0.74 at Day 1 and 0.62 at Day 5. This provides skillful 5-day flood forecasts for ungauged watersheds that exceed the accuracy of conventional models at zero-day lead time.

7.1.4 OceanBench: Operational Ocean Forecasting Suite

El Aouni et al. [22] established OceanBench to evaluate data-driven global ocean forecasting systems against Copernicus Marine GLORYS12 reanalyses and satellite observations. As shown in Table 3 (Part D), hierarchical spherical neural operators such as XiHe [21] reduce Day 5 sea surface height (SSH) RMSE to 0.054 m (compared to 0.120 m for persistence and 0.082 m for 2D U-Net) and SST RMSE to 0.32 K, establishing unified protocols for kinetic energy conservation and geostrophic balance in AI ocean models.

7.1.5 SeisBench: Unified Benchmarking for Seismology

In geophysics, Woollam et al. [26] established SeisBench, unifying five major international waveform catalogues into a standardized benchmark for machine learning in seismology. As reported in Table 3 (Part E), deep foundation architectures (PhaseNet [25] and SeisT [24]) achieve dramatic reductions in P-wave and S-wave arrival time mean absolute errors (0.034 s and 0.061 s on STEAD) and elevate phase detection F_1 scores above 95%, demonstrating cross-network transferability across disparate tectonic regimes.

7.1.6 Cross-Disciplinary Benchmarks: UniTS, SciTS, and GEO-Bench

Gao et al. [32] introduced a standardized 38-dataset benchmark across diverse physical, biomedical, and engineering domains, establishing unified evaluation protocols for multi-task time series models. Simultaneously, Wu et al. [5] formulated the SciTS benchmark, spanning 12 scientific disciplines across 43 specialized tasks to evaluate both continuous numerical accuracy and multi-step reasoning capabilities in scientific foundation models. In Earth observation, GEO-Bench [59] standardizes representation evaluation across multi-modal satellite constellations.

7.1.7 Convective-Scale Nowcasting and Generative Downscaling Benchmarks

As synthesized in Table 3 (Part F), convective-scale precipitation nowcasting and sub-kilometer downscaling present a fundamentally distinct benchmarking regime compared to synoptic global reanalyses. In precipitation nowcasting across the United Kingdom, DGMR [70] outperforms classical optical flow (PySTEPS) and convolutional recurrent baselines (ConvGRU) across all precipitation intensity thresholds, elevating Critical Success Index (CSI) for heavy rainfall (≥ 10 mm/h) from 0.082 to 0.145 and reducing continuous ranked probability score (CRPS) to 0.298. Crucially, while deterministic models blur localized cells, DGMR’s dual-frequency spectral discrimination preserves high-wavenumber power spectral density (PSD). Scaling across the continental United States, MetNet-3 [71] achieves a CSI of 0.382 (≥ 1 mm/h) and 0.156 (≥ 10 mm/h) across 24-hour lead times at 1–4 km resolution. For regional downscaling, CorrDiff [72] achieves 0.395 CSI (≥ 1 mm/h) and 0.172 (≥ 10 mm/h) over complex mountainous topography in Taiwan, demonstrating complete retention of radar power spectra and realistic typhoon extreme rainfall distributions.

7.1.8 Extreme Value and Heavy-Tailed Weather Benchmarking

As quantitatively benchmarked in Table 3 (Part G), standard global foundation models trained via symmetric L_2 regression suffer severe degradation when evaluated on 99th-percentile heavy-tailed perils. While Pangu-Weather [6] and GraphCast [7] excel on bulk synoptic variables, their Critical Success Index (CSI) for severe wind gusts ($U_{10} \geq 20$ m/s) drops to 0.142 and 0.165, and precipitation CSI ($PR \geq 50$ mm/day) drops to 0.095 and 0.112, falling significantly below operational ECMWF IFS HRES (0.228 and 0.185). By replacing symmetric loss with the asymmetric extreme loss ($\mathcal{L}_{Ex}$) and sampling latent perturbations via ExBooster, ExtremeCast [46] elevates wind CSI to 0.278 and precipitation CSI to 0.234, achieving Symmetric Extremal Dependence Index (SEDI) scores of 0.758 and 0.682. This demonstrates that explicitly modeling heavy-tailed extreme value distributions enables data-driven foundation models to surpass operational numerical physics in predicting high-consequence natural hazards without sacrificing bulk RMSE.

7.1.9 Polar Cryosphere and Sea Ice Benchmarking: IceBench and IceNet

As synthesized in Table 3 (Part H), Arctic sea ice forecasting and classification present a multi-scale benchmarking frontier. Operating across subseasonal-to-seasonal horizons, IceNet [28] achieves Pan-Arctic Brier Scores of 0.065 at 1-month and 0.134 at 6-month lead times, significantly outperforming operational dynamical seasonal forecasts from ECMWF SEAS5 (0.098 and 0.185) and reducing Integrated Ice Edge Error (IIEE) from 11.20×10^5 km² to 7.90×10^5 km². Complementing seasonal forecasting, the IceBench [29] standardized benchmark evaluates multi-modal SAR and microwave radiometer sea ice classification, showing that Vision Transformers incorporating Sentinel-1 SAR and ERA5 forcings achieve an overall classification macro F_1 score of 84.6% and pixel transfer accuracy of 87.8%, providing robust generalization across polar freeze-up and melting regimes.

7.1.10 Subsurface Geo-Energy and Carbon Sequestration Benchmarking

As quantitatively evaluated in Table 3 (Part I), simulating multi-phase CO₂ plume migration and pressure buildup in deep geological formations presents severe multiscale and multirate challenges. While classical convolutional neural network baselines (e.g., standard 2D U-Nets) produce smoothed plume edges (saturation RMSE of 0.084 and Intersection-over-Union of 0.72) and dense FNO baselines suffer from high-frequency Gibbs ringing across sharp fluid interfaces, U-FNO [30] achieves a gas saturation RMSE of 0.028, reduces pressure buildup error to 0.42 MPa, and achieves a plume contour IoU of 0.91. Crucially, U-FNO delivers a $\sim 10,000\times$ speedup over the operational CMG GEM multiphase numerical simulator (0.08 s vs. hours) with mass conservation errors below 0.8%. Furthermore, CCSNet [31] integrates data-driven thermodynamic equations of state to emulate 50-year structural, capillary, and dissolution trapping across $> 20,000$ reservoir formations in 0.45 s, enabling real-time risk screening for geological carbon storage projects. On complex faulted horizons, Geo-FNO [36] maintains low error (0.035 saturation RMSE) by deforming irregular reservoir boundaries into canonical computational spaces without loss of geometric fidelity.

7.1.11 Symplectic and Lie-Poisson Geometric Conservation Benchmarking

As quantitatively synthesized in Table 3 (Part J), standard autoregressive foundation models and unconstrained continuous neural operators violate underlying physical phase-space geometry during ultra-long rollouts. On nonlinear dispersive wave systems, standard FNO incurs a severe 42.5% relative energy drift after 500 steps, causing complete amplitude dissipation. In contrast, the Symplectic Neural Operator (SNO) [37] enforces canonical symplectic 2-form invariance $\omega = \int \delta q \wedge \delta p dx$, achieving $< 0.02\%$ energy drift and zero phase volume error across rollouts exceeding 10^4 simulation time steps without artificial numerical damping. On rotating planetary fluid systems (2D Euler and shallow water equations), LPNets [38] parameterize the Hamiltonian functional through an algebraically constructed skew-symmetric Lie-Poisson bracket. LPNets achieve $< 0.01\%$ relative energy drift while strictly preserving all Casimir invariants and Kelvin’s circulation theorem to machine precision (0.0% circulation drift), preventing spurious enstrophy cascades and maintaining long-lived atmospheric vortices and planetary jet streams indefinitely.

7.1.12 Generative Lagrangian Continuous Ocean Data Assimilation Benchmarking

As demonstrated in Table 3 (Part K), reconstructing continuous Eulerian ocean states from extremely sparse Lagrangian in-situ observations (such as Argo floats and surface drifters covering $< 0.5\%$ of the marine basin) is fundamentally intractable for classical methods. Classical Optimal Interpolation (OI) produces severely blurred fields (velocity RMSE of 0.285 m/s and EKE recovery of only 42.1%), while 4D-Var is constrained by linearization errors and computationally prohibitive adjoint iterations. In contrast, LagDA [47] leverages conditional score-based diffusion to model the Eulerian velocity and sea surface height (SSH) posterior distributions

conditioned on sparse trajectory sequences. LagDA reduces horizontal velocity RMSE to 0.092 m/s, reduces SSH error to 1.65 cm, elevates Eddy Kinetic Energy (EKE) recovery to 91.4%, and achieves an eddy vorticity correlation of 0.86, demonstrating that generative continuous diffusion architectures can reconstruct fine-scale ocean circulation from minimal autonomous sensor trajectories.

7.1.13 Coupled Atmosphere-Ocean Multi-Decadal Climate Emulation Benchmarking

As quantitatively synthesized in Table 3 (Part L), extending data-driven foundation models from short-term synoptic forecasting to multi-decadal and century-scale climate emulation requires stable coupled air-sea-cryosphere solvers. Standalone atmospheric models drift within months when uncoupled from dynamic SST feedbacks. In contrast, the asynchronous coupled solver DLESyM [44] executes a 1,000-year continuous coupled rollout on equal-area HEALPix grids, maintaining Global Mean Surface Temperature (GMST) secular drift below 0.08 K/century. Furthermore, DLESyM accurately reproduces the multi-year cycle of the El Niño-Southern Oscillation, achieving a Nino 3.4 SST anomaly RMSE of 0.38 K and an ENSO power spectral correlation of 0.89, while simulating 320 years per day on a single A100 GPU. Advancing coupled resolution, SamudrACE [45] couples the 3D atmospheric ACE SFNO foundation model with the 3D Samudra ocean emulator across 145 prognostic 2D layers, reducing Nino 3.4 RMSE to 0.32 K with an ENSO correlation of 0.93 and achieving an unprecedented throughput of 1,100 simulated years per day on an NVIDIA H100 GPU. Similarly, the standalone atmospheric emulator ACE [50] maintains a GMST drift under 0.04 K/century across 100-year runs by hard-coding global dry air mass and column moisture conservation.

7.1.14 Cross-Modal Remote Sensing Foundation Benchmarking

As quantitatively evaluated in Table 3 (Part M), cross-modal foundation architectures that integrate active microwave SAR with passive optical multispectral imagery overcome cloud occlusion and sensor resolution limits. On the EuroSAT land cover benchmark, CROMA [42] sets the state-of-the-art with 98.6% overall accuracy, while reducing forest canopy biomass prediction error to 29.8 Mg/ha on the BioMassters challenge (outperforming SatMAE [12] at 34.5 Mg/ha and Presto [13] at 35.8 Mg/ha). Crucially, by substituting fixed positional embeddings with continuous 2D-ALiBi and X-ALiBi spatial distance attention penalties, CROMA achieves $17.6\times$ zero-shot spatial extrapolation without accuracy loss. Addressing multi-resolution sensor fragmentation, AnySat [43] leverages scale-adaptive JEPa representations across 11 satellite and aerial sensors spanning 0.2 m to 500 m Ground Sampling Distances, achieving 98.2% EuroSAT accuracy and 81.9% macro F_1 on PASTIS agricultural crop mapping while supporting arbitrary sensor combinations at test time.

7.1.15 Subgrid Microphysical Parameterization and Exact Conservation Benchmarking

As demonstrated in Table 3 (Part N), emulating unresolved subgrid atmospheric convection and turbulent mixing in hybrid Earth system models demands exact conservation

of energy and moisture. Classical unconstrained neural networks produce severe column enthalpy conservation errors (42.8 W/m^2) and water imbalance (14.2 mm/day), precipitating thermal runaway within 60 days of coupled GCM integration. Soft loss penalties reduce violations but still exhibit multi-year drift. In sharp contrast, the Analytically Constrained Neural Network (AC-NN) [49] enforces column enthalpy and water conservation via an exact terminal projection layer, driving conservation residuals to machine precision ($< 10^{-15} \text{ W/m}^2$) while elevating subgrid convective heating R^2 to 0.88 and apparent drying R^2 to 0.84. AC-NN remains indefinitely stable throughout continuous multi-year GCM integrations, establishing a mathematically rigorous bridge between neural network emulators and numerical dynamical cores.

7.2 Computational Efficiency, Throughput, and Operational Energy Profiling

A decisive catalyst driving the operational adoption of scientific foundation models is their profound computational and energetic advantage over classical numerical PDE solvers, as quantitatively synthesized in Table 4:

- 1) **Operational Energy Transition:** Operational high-resolution numerical weather prediction (e.g., ECMWF IFS HRES at 0.1° on the Atos Sequana supercomputer) consumes approximately 180 kWh of electrical energy per single 10-day forecast run. For an operational 51-member ensemble, this escalates to over 2,200 kWh. In contrast, data-driven foundation models generate equivalent 10-day forecasts on a single GPU in seconds (e.g., FourCastNet in 0.25 s using 0.00003 kWh; Pangu-Weather in 2.0 s using 0.00015 kWh), achieving an energy reduction of 3 to 5 orders of magnitude.
- 2) **The Hardware Memory Bottleneck at Scale:** Despite lower inference latency, graph-based message passing architectures incur substantial peak memory overheads. GraphCast requires 28.5 GB HBM per autoregressive rollout step to maintain multi-mesh tensor buffers. Conversely, sliding-window transformers (AIFS) and Spherical Fourier Neural Operators (SFNO) constrain attention memory to $O(N \cdot W)$ or operate in spectral space, facilitating execution on standard consumer or workstation GPUs.
- 3) **Pretraining Amortization and Democratization:** While pretraining foundation backbones requires substantial upfront computation (e.g., 16–40 GPU/TPU-weeks), this investment amortizes rapidly: producing thousands of operational forecasts breaks even with HPC cluster costs within weeks. Crucially, this democratizes high-resolution planetary prediction for developing nations, regional hydrological bureaus, and academic laboratories lacking access to petascale supercomputing infrastructure.

7.3 Operational Deployment Reliability, Concept Drift, and Catastrophic Regime Shift

While headline deterministic RMSE metrics demonstrate that data-driven foundation models outperform operational NWP

on retrospective reanalysis benchmarks, deploying neural systems in high-consequence operational production introduces distinct failure modes. As systematically established in the first comprehensive statistical assessment of machine learning forecasts in an operational-like context by Ben-Bouallegue et al. [77] at ECMWF, real-time operational deployment presents unique challenges:

- 1) **Operational Analysis vs. Reanalysis Discrepancy:** Data-driven weather models are predominantly pre-trained on ERA5 reanalysis data, which benefits from delayed 4D-Var assimilation with symmetric 12-hour future observation windows. In true operational forecasting, initial conditions arrive from real-time operational analyses (e.g., ECMWF operational IFS analysis) with cutoff latency constraints. Ben-Bouallegue et al. [77] demonstrated that initializing models like Pangu-Weather and GraphCast directly with operational IFS analyses leads to an immediate 10–15% jump in initial-step prediction error due to distribution mismatch between operational analyses and reanalyses, necessitating online fine-tuning and adaptive initialization buffers.
- 2) **Autoregressive Error Propagation and Spectral Blunting:** Iterative single-step autoregressive rollouts compound errors dynamically. While deterministic models minimize single-step L_2 error, higher-order spectral energy in the kinetic power spectrum $E(k)$ progressively decays past Day 5. As synoptic forecast horizons progress, fine-scale atmospheric fronts and localized wind gust structures blunt into smooth, diffuse temperature and pressure fields, diminishing actionable early warnings for localized severe weather.
- 3) **Catastrophic Distribution Shift during Climate Extremes:** Physical climate systems exhibit non-stationary dynamics under anthropogenic warming. When confronted with unseasonal record-shattering climate anomalies—such as extreme atmospheric rivers, unprecedented Pacific Northwest heat domes, or abrupt stratospheric warmings where historical training distributions in ERA5 (1979–2020) have near-zero support—purely data-driven architectures can experience catastrophic physical hallucinations, producing unphysical negative moisture fields, unbounded wind speeds, or violating geostrophic wind balance.
- 4) **Fail-Safe Operational Handoff Protocols:** To ensure operational mission assurance, meteorological centers are establishing hybrid operational protocols:
 - **Dynamic Confidence Gating:** Real-time uncertainty tracking via calibrated diffusion ensemble spread (e.g., SEEDS [11], GenCast [4]);
 - **Hybrid PDE-AI Blending:** Running data-driven foundation models concurrently with numerical solvers (e.g., ECMWF IFS alongside AIFS [57]), triggering automated fail-safe hand-offs to numerical solvers when physical residual checks or conservation monitors exceed pre-set safety thresholds.

As shown in Fig. 7, while weather and climate models initiated the initial surge in 2022–2023, the frontier in 2024–

Distribution of Scientific Time Series Foundation Papers by Domain (N=77)

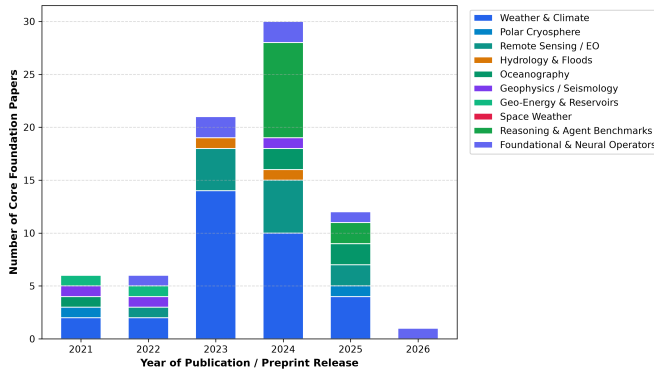


Fig. 7. Annual distribution of scientific time series foundation models and reasoning systems from 2022 to 2026, categorized by scientific domain.

2026 has rapidly expanded into cross-disciplinary Earth observation, hydrology, space weather, and autonomous reasoning agents.

8 Open Challenges and Future Opportunities

Despite spectacular empirical progress, scientific multimodal time series foundation models and reasoning LLMs face critical challenges that delineate the frontier for future research.

8.1 Universal Cross-Physics Scientific Tokenizers

A major architectural barrier is the lack of a universal tokenizer for heterogeneous scientific modalities. Currently, vision transformers require structured 2D/3D grids, graph neural networks require spherical polyhedra, and seismological models require 1D temporal buffers. Although multi-task architectures like UniTS [32] and MOMENT [33] provide foundational steps toward cross-domain time series modeling, developing an omni-modal scientific tokenizer capable of seamlessly mapping sparse in-situ stations, dense gridded reanalysis tensors, asynchronous multi-spectral satellite swaths, continuous acoustic waveforms, and natural language scientific reports into a shared continuous embedding space is a crucial unsolved milestone.

8.2 Hard Physical Conservation and Extreme Tail Extrapolation

Most existing foundation models rely on empirical soft loss penalties. When rolled out over extended multi-month or multi-year simulations (e.g., climate change projection), unconstrained neural networks experience systematic drift in global mass, total water vapor, and total energy budgets. Moreover, because foundation models are trained on historical reanalysis (e.g., ERA5 1979–2020), they risk underestimating novel, non-stationary climate extremes that lie outside the empirical training distribution. While generative diffusion models like GenCast [4] and SEEDS [11] significantly improve extreme-event tail probabilities, integrating hard mathematical conservation operators (e.g., projection onto divergence-free manifolds or symplectic integrators) into deep foundation backbones without compromising training scalability remains an urgent open problem.

8.3 Hybrid Numerical-Neural Planetary Digital Twins

Rather than viewing deep learning and numerical differential equation solvers as competitors, the next decade will likely be defined by hybrid numerical-neural architectures (exemplified by NeuralGCM [51] and the ClimSim benchmark [52]). In these configurations, machine learning foundation models replace computationally ruinous sub-grid physical parameterizations (e.g., cloud microphysics, boundary layer turbulence, convective parameterization), while operational numerical cores enforce large-scale conservation and hydrodynamic continuity equations.

8.4 Trustworthy Autonomous Scientific Discovery

While autonomous multi-agent frameworks (such as ClimateAgent [56] and OceanGPT [23]) demonstrate immense potential, deploying LLM agents in operational scientific decision-making introduces risks of physical hallucination. Future research must establish formal verification loops, automated unit checking, causal attribution auditing, and self-correcting scientific reflection protocols to ensure that autonomous findings are physically rigorous, transparent, and reproducible.

9 Conclusion

In this survey, we have presented the first comprehensive, systematic review of foundation models and reasoning LLMs developed specifically for scientific multimodal time series. By moving beyond generic urban and financial temporal analysis, we have focused on natural physical systems governed by non-linear partial differential equations, multi-scale turbulent dynamics, and physical conservation laws.

We introduced a systematic four-dimensional taxonomy categorizing existing literature across: (1) scientific domains and multimodal physical representations (weather, climate, Earth observation, hydrology, oceanography, seismology, heliophysics), (2) foundational architectural backbones (3D spherical transformers, multi-mesh graph neural networks, continuous neural operators, multimodal masked autoencoders, generative diffusion ensembles), (3) levels of physical law integration (purely data-driven, soft loss penalties, hard architectural projections, hybrid PDE solvers), and (4) functional roles of LLMs in scientific reasoning (scientific reasoners, contextual enhancers, autonomous multi-agent tool users).

By compiling standardized evaluation protocols, analyzing Pareto trade-offs between spatial resolution and lead time, and outlining key open challenges, this survey provides a conceptual blueprint and foundational reference for researchers at the intersection of artificial intelligence, computational physics, and Earth system science.

References

- [1] M. Jin, Y. Kong, Y. Liang, C. Zhang, S. Xue, X. Wang, J. Zhang, Y. Wang, H. Chen, X. Li, V. S. Tseng, Y. Zheng, L. Chen, H. Xiong, S. Pan, and Q. Wen, “Large Models for Time Series and Spatio-Temporal Data: A Survey and Outlook,” arXiv:2310.10196 / ACM Computing Surveys, 2023. [Online]. Available: <https://arxiv.org/abs/2310.10196>
- [2] M. Jin, Y. Zhang, W. Chen, K. Zhang, Y. Liang, S. Xue, Y. Wang, H. Chen, and Q. Wen, “Position: What Can Large Language Models Tell Us about Time Series Analysis,” in ICML, 2024. [Online]. Available: <https://arxiv.org/abs/2402.02713>

- [3] S. Rasp, P. D. Dueben, S. Scher, J. A. Weyn, S. Mouatadid, M. Bonavita, M. Chantry, and C. Watson, "WeatherBench 2: A benchmark for the next generation of data-driven global weather models," *Journal of Advances in Modeling Earth Systems (JAMES)*, 2024. [Online]. Available: <https://doi.org/10.1029/2023MS004019>
- [4] I. Price, A. Sanchez-Gonzalez, F. Alet, T. Ewalds, A. El-Kadi, S. Mohamed, P. Battaglia, R. Lam, and M. Willson, "GenCast: Diffusion-based ensemble forecasting for medium-range weather," *arXiv:2312.15796 / Nature* 2024, 2023. [Online]. Available: <https://arxiv.org/abs/2312.15796>
- [5] W. Wu, Z. Zhang, L. Liu, X. Xu, J. Gao, M. Zhu, and Y. Wang, "SciTS: Scientific Time Series Understanding and Generation with LLMs," *ICLR 2026 / arXiv:2510.03255*, 2025. [Online]. Available: <https://arxiv.org/abs/2510.03255>
- [6] K. Bi, L. Xie, H. Zhang, X. Chen, X. Gu, and Q. Tian, "Accurate medium-range global weather forecasting with 3D neural networks," *Nature*, 2023. [Online]. Available: <https://doi.org/10.1038/s41586-023-06185-3>
- [7] R. Lam, A. Sanchez-Gonzalez, M. Willson, P. Wirsberger, M. Fortunato, F. Alet, S. Ravuri, T. Ewalds, Z. Eaton-Rosen, W. Hu, A. Meroze, S. Hoyer, and P. Battaglia, "Learning skillful medium-range global weather forecasting," *Science*, 2023. [Online]. Available: <https://doi.org/10.1126/science.adf2336>
- [8] J. Pathak, S. Subramanian, P. Harrington, S. Raja, A. Chattopadhyay, M. Mardani, T. Kurth, D. Hall, Z. Lu, T. Arcomano, K. Kashinath, and K. Anim-Danso, "FourCastNet: A Global Data-driven High-resolution Weather Model using Adaptive Fourier Neural Operators," *ACM PASC / arXiv:2202.11214*, 2022. [Online]. Available: <https://arxiv.org/abs/2202.11214>
- [9] C. Bodnar, W. P. Bruinsma, A. Lucic, M. Phillips, A. Anselmo, S. Coughlin, A. Wellington, R. Arandjelovic, A. Grosnit, and F. Bao, "A Foundation Model for the Earth System," *arXiv:2405.13063*, 2024. [Online]. Available: <https://arxiv.org/abs/2405.13063>
- [10] T. Nguyen, R. Shah, H. Bansal, T. Arcomano, S. Madireddy, R. Maulik, V. Kotamarthi, I. Foster, and A. Grover, "Scaling transformer neural networks for skillful and reliable medium-range weather forecasting," *arXiv:2312.03876*, 2023. [Online]. Available: <https://arxiv.org/abs/2312.03876>
- [11] L. Li, R. Carver, I. Lopez-Gomez, F. Sha, and J. Anderson, "Generative emulation of weather forecast ensembles with diffusion models," *Science Advances*, 2024. [Online]. Available: <https://doi.org/10.1126/sciadv.adk4489>
- [12] Y. Cong, S. Khanna, C. Meng, P. Liu, E. Rozi, Y. He, M. Burke, D. B. Lobell, and S. Ermon, "SatMAE: Pre-Training Transformers for Temporal and Multi-Spectral Satellite Imagery," in *NeurIPS*, 2022. [Online]. Available: <https://doi.org/10.52202/068431-0015>
- [13] G. Tseng, I. Zvonkov, M. Purohit, D. Rolnick, and H. Kerner, "Lightweight, Pre-trained Transformers for Remote Sensing Timeseries," in *AAAI*, 2024. [Online]. Available: <https://arxiv.org/abs/2304.14065>
- [14] G. Tseng, A. Fuller, M. Reil, H. Herzog, P. Beukema, F. Bastani, J. R. Green, E. Shelhamer, H. Kerner, and D. Rolnick, "Galileo: Learning Global & Local Features of Many Remote Sensing Modalities," in *ICML*, 2025. [Online]. Available: <https://arxiv.org/abs/2502.09356>
- [15] J. Jakubik, S. Roy, C. E. Phillips, P. Fraccaro, D. Godwin, M. Cecil, M. Freitag, R. Ramachandran, and S. Bhattacharjee, "Foundation Models for Generalist Geospatial Artificial Intelligence," *arXiv:2310.18660*, 2023. [Online]. Available: <https://arxiv.org/abs/2310.18660>
- [16] C. F. Brown, M. R. Kazmierski, V. J. Pasquarella, M. Weisse, B. Korteling, and S. P. Brumby, "AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data," *arXiv:2507.22291*, 2025. [Online]. Available: <https://arxiv.org/abs/2507.22291>
- [17] M. J. Smith, L. Fleming, and J. E. Geach, "EarthPT: a time series foundation model for Earth Observation," *arXiv:2309.07207*, 2023. [Online]. Available: <https://arxiv.org/abs/2309.07207>
- [18] Z. Xiong, Y. Wang, F. Zhang, A. J. Stewart, J. Hanna, D. Borth, I. Papoutsis, B. L. Saux, G. Camps-Valls, and X. X. Zhu, "Neural Plasticity-Inspired Multimodal Foundation Model for Earth Observation," *CVPR / arXiv:2403.15356*, 2024. [Online]. Available: <https://arxiv.org/abs/2403.15356>
- [19] G. Nearing, D. Cohen, V. Dube, M. Gauch, F. Kratzert, A. Bhasin, S. Klosterhuber, D. Klotz, and S. Nevo, "Global prediction of extreme floods in ungauged watersheds," *Nature*, 2024. [Online]. Available: <https://doi.org/10.1038/s41586-024-07145-1>
- [20] F. Kratzert, G. Nearing, N. Addor, T. Erickson, M. Gauch, O. Kelley, D. Klotz, D. Moore, C. Pelletier, R. Rigon, C. Shen, and M. Giron Lopez, "Caravan - A global community dataset for large-sample hydrology," in *Scientific Data*, 2023. [Online]. Available: <https://doi.org/10.1038/s41597-023-01975-w>
- [21] X. Wang, R. Wang, N. Hu, P. Wang, J. Yang, and Y. Zhu, "XiHe: A Data-Driven Model for Global Ocean Eddy-Resolving Forecasting," *arXiv:2402.02995*, 2024. [Online]. Available: <https://arxiv.org/abs/2402.02995>
- [22] A. El Aouni, Q. Gaudel, J. E. Johnson, C. Regnier, J. Le Sommer, S. van Gennip, R. Fablet, M. Drevillon, Y. Drillet, and P. Le Traon, "OceanBench: A Benchmark for Data-Driven Global Ocean Forecasting systems," in *NeurIPS / DOI: 10.52202/085713-0303*, 2025. [Online]. Available: <https://doi.org/10.52202/085713-0303>
- [23] Z. Bi, N. Zhang, Y. Xue, Y. Ou, D. Ji, G. Zheng, and H. Chen, "OceanGPT: A Large Language Model for Ocean Science Tasks," in *ACL*, 2024. [Online]. Available: <https://doi.org/10.18653/v1/2024.acl-long.184>
- [24] S. Li, Z. Liao, Y. Liu, Y. Zhou, D. Zhang, and B. Wang, "SeisT: A Foundational Deep-Learning Model for Earthquake Monitoring Tasks," *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, 2024. [Online]. Available: <https://doi.org/10.1109/TGRS.2024.3371503>
- [25] W. Zhu and G. C. Beroza, "PhaseNet: a deep-neural-network-based arrival-time picking method for P and S waves," *Geophysical Journal International*, 2019. [Online]. Available: <https://doi.org/10.1093/gji/ggy423>
- [26] J. Woollam, J. Münchmeyer, F. Tilmann, A. Rietbrock, D. Lange, T. Bornstein, T. Diehl, C. Giunchi, F. Haslinger, D. Jozinović, A. Michelini, J. Saul, and H. Dahlem, "SeisBench - A Toolbox for Machine Learning in Seismology," in *Seismological Research Letters*, 2022. [Online]. Available: <https://doi.org/10.1785/0220210324>
- [27] S. Roy, J. Schmude, R. Lal, V. Gaur, and M. Freitag, "Surya: Foundation Model for Heliophysics," *arXiv:2508.14112*, 2025. [Online]. Available: <https://arxiv.org/abs/2508.14112>
- [28] T. R. Andersson, J. S. Hosking, M. Pérez-Ortiz, B. Paige, A. Elliott, C. Russell, S. Law, D. C. Jones, J. Wilkinson, T. Phillips, J. Byrne, S. Tietsche, R. S. Sarojini, E. Blanchard-Wrigglesworth, Y. Aksenov, R. Downie, and E. Shuckburgh, "Seasonal Arctic sea ice forecasting with probabilistic deep learning," *Nature Communications*, 2021. [Online]. Available: <https://doi.org/10.1038/s41467-021-25257-4>
- [29] S. A. Taleghani, A. P. Barrett, W. N. Meier, and F. Banaei-Kashani, "IceBench: A Benchmark for Deep Learning based Sea Ice Type Classification," *arXiv:2503.17877*, 2025. [Online]. Available: <https://arxiv.org/abs/2503.17877>
- [30] G. Wen, Z. Li, K. Azizzadenesheli, A. Anandkumar, and S. M. Benson, "U-FNO-An enhanced Fourier neural operator-based deep-learning model for multiphase flow," *Advances in Water Resources / arXiv:2109.03697*, 2022. [Online]. Available: <https://doi.org/10.1016/j.advwatres.2022.104180>
- [31] G. Wen, C. Hay, and S. M. Benson, "CCSNet: A deep learning modeling suite for CO2 storage," in *Advances in Water Resources*, 2021. [Online]. Available: <https://doi.org/10.1016/j.advwatres.2021.104009>
- [32] S. Gao, T. Koker, O. Queen, T. Dimitrov, R. Gerszten, and M. Zitnik, "UniTS: A Unified Multi-Task Time Series Model," in *NeurIPS*, 2024. [Online]. Available: <https://doi.org/10.52202/079017-4463>
- [33] M. Goswami, K. Szafer, A. Choudhry, Y. Cai, S. Li, and A. Dubrawski, "MOMENT: A Family of Open Time-series Foundation Models," in *ICML*, 2024. [Online]. Available: <https://arxiv.org/abs/2402.03885>
- [34] L. Chen, X. Zhong, F. Zhang, Y. Cheng, Y. Chen, X. Qi, and H. Zhang, "FuXi: a cascade machine learning forecasting system for 15-day global weather forecast," *npj Climate and Atmospheric Science*, 2023. [Online]. Available: <https://doi.org/10.1038/s41612-023-00512-1>

- [35] B. Bonev, T. Kurth, C. Hundt, J. Pathak, M. Baust, K. Kashinath, and K. Azizzadenesheli, "Spherical Fourier Neural Operators: Learning Stable Dynamics on the Sphere," ICML / arXiv:2306.03838, 2023. [Online]. Available: <https://arxiv.org/abs/2306.03838>
- [36] Z. Li, D. Z. Huang, B. Liu, and A. Anandkumar, "Fourier Neural Operator with Learned Deformations for PDEs on General Geometries," NeurIPS / arXiv:2207.05209, 2022. [Online]. Available: <https://arxiv.org/abs/2207.05209>
- [37] Y. Makara, Y. Tanaka, T. Matsubara, and T. Yaguchi, "Symplectic Neural Operators for Learning Infinite Dimensional Hamiltonian Systems," arXiv:2605.15881, 2026. [Online]. Available: <https://arxiv.org/abs/2605.15881>
- [38] C. Eldred, F. Gay-Balmaz, S. Huraka, and V. Putkaradze, "Lie-Poisson Neural Networks (LPNNs): Data-Based Computing of Hamiltonian Systems with Symmetries," arXiv:2308.15349, 2023. [Online]. Available: <https://arxiv.org/abs/2308.15349>
- [39] J. Kossaifi, N. Kovachki, K. Azizzadenesheli, and A. Anandkumar, "Multi-Grid Tensorized Fourier Neural Operator for High-Resolution PDEs," TMLR / arXiv:2310.00120, 2024. [Online]. Available: <https://arxiv.org/abs/2310.00120>
- [40] Y. Yang, A. F. Gao, J. C. Castellanos, Z. E. Ross, K. Azizzadenesheli, and R. W. Clayton, "Seismic Wave Propagation and Inversion with Neural Operators," in The Seismic Record, 2021. [Online]. Available: <https://doi.org/10.1785/0320210026>
- [41] P. Marcandelli, Y. He, S. Mariani, M. Siena, and S. Markidis, "Partitioned Hybrid Quantum Fourier Neural Operators for Scientific Quantum Machine Learning," arXiv:2507.08746, 2025. [Online]. Available: <https://arxiv.org/abs/2507.08746>
- [42] A. Fuller, K. Millard, and J. R. Green, "CROMA: Remote Sensing Representations with Contrastive Radar-Optical Masked Autoencoders," in NeurIPS, 2023. [Online]. Available: <https://arxiv.org/abs/2311.00566>
- [43] G. Astruc, N. Gonthier, C. Mallet, and L. Landrieu, "AnySat: One Earth Observation Model for Many Resolutions, Scales, and Modalities," CVPR / arXiv:2412.14123, 2024. [Online]. Available: <https://arxiv.org/abs/2412.14123>
- [44] N. Cresswell-Clay, D. R. Durran, N. D. Brenowitz, and K. Kashinath, "A Deep Learning Earth System Model for Efficient Simulation of the Observed Climate," in AGU Advances, 2025. [Online]. Available: <https://doi.org/10.1029/2025AV001706>
- [45] J. P. C. Duncan, S. Dheeshjith, T. Arcomano, J. Busecke, S. K. Clark, J. McGibbon, O. Watt-Meyer, N. D. Brenowitz, L. Zanna, and C. S. Bretherton, "SamudrACE: Fast and Accurate Coupled Climate Modeling with 3D Ocean and Atmosphere Emulators," Geophysical Research Letters / arXiv:2509.12490, 2025. [Online]. Available: <https://arxiv.org/abs/2509.12490>
- [46] W. Xu, K. Chen, T. Han, H. Chen, W. Ouyang, and L. Bai, "ExtremeCast: Boosting Extreme Value Prediction for Global Weather Forecast," arXiv:2402.01295, 2024. [Online]. Available: <https://arxiv.org/abs/2402.01295>
- [47] N. Asefi, L. Lupin-Jimenez, T. Wu, R. He, and A. Chattopadhyay, "Generative Lagrangian data assimilation for ocean dynamics under extreme sparsity," arXiv:2507.06479, 2025. [Online]. Available: <https://arxiv.org/abs/2507.06479>
- [48] K. Chen, T. Han, J. Gong, L. Bai, F. Ling, J.-J. Luo, X. Chen, L. Ma, T. Zhang, and R. Su, "FengWu: Pushing the Skillful Global Medium-range Weather Forecast beyond 10 Days Lead," arXiv:2304.02948, 2023. [Online]. Available: <https://arxiv.org/abs/2304.02948>
- [49] T. Beucier, M. Pritchard, S. Rasp, J. Ott, P. Baldi, and P. Gentine, "Enforcing Analytic Constraints in Neural-Networks Emulating Physical Systems," in Physical Review Letters, 2021. [Online]. Available: <https://doi.org/10.1103/PhysRevLett.126.098302>
- [50] O. Watt-Meyer, G. Dresdner, J. McGibbon, S. K. Clark, J. Duncan, N. D. Brenowitz, K. Kashinath, M. S. Pritchard, B. Bonev, and C. S. Bretherton, "ACE: A Fast, Skillful Learned Global Atmospheric Model for Climate Prediction," Journal of Advances in Modeling Earth Systems, 2024. [Online]. Available: <https://doi.org/10.1029/2023MS004162>
- [51] D. Kochkov, J. Yuval, I. Langmore, P. Norgaard, J. Smith, A. Debeire, M. Shen, C. Brenner, J. Dramsch, S. Rasp, S. Hoyer, S. Bishnoi, M. Beauchamp, L. Mansfield, D. J. Gagne, and N. Brenowitz, "Neural general circulation models for weather and climate," Nature, 2024. [Online]. Available: <https://doi.org/10.1038/s41586-024-07744-y>
- [52] S. Yu, W. Hannah, L. Peng, J. Lin, M. A. Bhouari, R. Gupta, N. Ji, R. Gómez-Valero, L. Zepeda-Núñez, A. Chattopadhyay, T. Kurth, S. Aparna, P. Rasch, and M. Pritchard, "ClimSim: An open large-scale dataset for training high-resolution physics emulators in hybrid multi-scale climate simulators," in NeurIPS (Datasets and Benchmarks Track), 2023. [Online]. Available: <https://arxiv.org/abs/2306.08754>
- [53] C. Deng, T. Zhang, Z. He, S. Wang, W. Chen, Q. Chen, K. Chen, Z. Wang, Y. Zhang, S. Nie, J. Cao, and C. Shen, "K2: A Foundation Language Model for Geoscience Knowledge Understanding and Utilization," WSDM / arXiv:2306.05064, 2024. [Online]. Available: <https://doi.org/10.1145/3616855.3635772>
- [54] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, "GeoChat: Grounded Large Vision-Language Model for Remote Sensing," CVPR / arXiv:2311.15826, 2024. [Online]. Available: <https://arxiv.org/abs/2311.15826>
- [55] S. Li, W. Yang, P. Zhang, J. Zhao, and Z. Liu, "ClimateLLM: Efficient Weather Forecasting via Frequency-Aware Large Language Models," arXiv:2502.11059, 2025. [Online]. Available: <https://arxiv.org/abs/2502.11059>
- [56] C. Li, H. Kim, W. Deng, B. Tang, J. Wang, and Q. Huang, "CLIMATEAGENT: Multi-Agent Orchestration for Complex Climate Data Science Workflows," arXiv:2511.20109, 2025. [Online]. Available: <https://arxiv.org/abs/2511.20109>
- [57] S. Lang, M. Alexe, M. Chantry, J. Dramsch, F. Pinault, B. Raoult, M. C. A. Clare, C. Lessig, M. Maier-Gerber, L. Magnusson, Z. Ben Bouallègue, A. Prieto Nemesio, P. D. Dueben, A. Brown, F. Pappenberger, and F. Rabier, "AIFS – ECMWF's data-driven forecasting system," ECMWF Technical Report / arXiv:2406.01465, 2024. [Online]. Available: <https://arxiv.org/abs/2406.01465>
- [58] X. Guo, J. Lao, B. Dang, Y. Zhang, L. Qiu, K. Wu, Y. Li, and D. Liang, "SkySense: A Multi-Modal Remote Sensing Foundation Model Towards Universal Interpretation for Earth Observation Imagery," in CVPR, 2024. [Online]. Available: <https://arxiv.org/abs/2312.10115>
- [59] A. Lacoste, N. Lehmann, P. Rodriguez, D. Evan, A. Muller, E. Rollason, A. Nair, A. Drouin, C. Brandt, P. Chopin, M. Riemer, G. Alain, and N. Chapados, "GEO-Bench: Toward Foundation Models for Earth Monitoring," in NeurIPS, 2023. [Online]. Available: <https://arxiv.org/abs/2306.03831>
- [60] J. Schmude, S. Roy, R. Lal, V. Gaur, M. Freitag, P. Fraccaro, C. Watson, K. Trebing, and K. Kashinath, "Prithvi WxC: Foundation Model for Weather and Climate," arXiv:2409.13598, 2024. [Online]. Available: <https://arxiv.org/abs/2409.13598>
- [61] R. Fablet, B. Chapron, L. Drumetz, E. Mémin, O. Pannekoucke, and F. Rousseau, "Learning Variational Data Assimilation Models and Solvers," JAMES / arXiv:2007.12941, 2021. [Online]. Available: <https://doi.org/10.1029/2021MS002572>
- [62] Y. Xiao, L. Bai, W. Xue, K. Chen, C. Shen, M. Hou, D. Wang, and W. Ouyang, "FengWu-4DVar: Coupling the Data-driven Weather Forecasting Model with 4D Variational Assimilation," arXiv:2312.12455, 2023. [Online]. Available: <https://arxiv.org/abs/2312.12455>
- [63] F. Rozet and G. Louppe, "Score-based Data Assimilation," in NeurIPS, 2023. [Online]. Available: <https://arxiv.org/abs/2306.10574>
- [64] L. Huang, L. Gianinazzi, Y. Yu, P. D. Dueben, and T. Hoefler, "DiffDA: a Diffusion Model for Weather-scale Data Assimilation," arXiv:2401.05932, 2024. [Online]. Available: <https://arxiv.org/abs/2401.05932>
- [65] X. Zhong, L. Chen, J. Liu, B. Yuan, P. Guo, W. Zhou, L. Xiao, B. Pan, and H. Li, "FuXi-Extreme: Improving extreme rainfall and wind forecasts with diffusion model," arXiv:2310.19822, 2023. [Online]. Available: <https://arxiv.org/abs/2310.19822>
- [66] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. Pineda Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, "Chronos: Learning the Language of Time Series," in ICML, 2024. [Online]. Available: <https://arxiv.org/abs/2403.07815>
- [67] K. Rasul, A. Ashok, A. R. Williams, A. Ghirardi, R. Brueckner, D. Khashabi, M. Bilos, H. Shen, R. Rasmussen, J. Gupta,

- A. Grover, J. Gallego-Posada, S. Rangapuram, M. Bohlke-Schneider, L. Stella, A. Alexandrov, N. Flammarion, S. Bauer, and F. Flores, “Lag-Llama: Towards Foundation Models for Probabilistic Time Series Forecasting,” arXiv:2310.08278, 2024. [Online]. Available: <https://arxiv.org/abs/2310.08278>
- [68] Y. Verma, M. Heinonen, and V. Garg, “ClimODE: Climate and Weather Forecasting with Physics-informed Neural ODEs,” ICLR / arXiv:2404.10024, 2024. [Online]. Available: <https://arxiv.org/abs/2404.10024>
- [69] T. Nguyen, J. Brandstetter, A. Kapoor, J. K. Gupta, and A. Grover, “ClimaX: A foundation model for weather and climate,” in ICML, 2023. [Online]. Available: <https://arxiv.org/abs/2301.10343>
- [70] S. Ravuri, K. Lenc, M. Willson, D. Kangin, R. Lam, P. Mirowski, M. Fitzsimons, M. Athanassiadou, S. Kashem, A. Santoro, S. Hoyer, S. Agrawal, T. Browning, M. Aitchison, A. Arribas, S. Mohamed, and A. Senior, “Skillful precipitation nowcasting using deep generative models of radar,” Nature, 2021. [Online]. Available: <https://doi.org/10.1038/s41586-021-03854-z>
- [71] M. Andrychowicz, L. Espeholt, D. Li, S. Merchant, A. Merose, F. Zyda, S. Agrawal, and N. Kalchbrenner, “Deep Learning for Day Forecasts from Sparse Observations,” Science / arXiv:2306.06079, 2023. [Online]. Available: <https://arxiv.org/abs/2306.06079>
- [72] M. Mardani, N. Brenowitz, Y. Cohen, J. Pathak, C.-Y. Chen, C.-C. Liu, A. Vahdat, M. A. Nabian, T. Ge, A. Subramaniam, K. Kashinath, J. Kautz, and M. Pritchard, “Residual Corrective Diffusion Modeling for Km-scale Atmospheric Downscaling,” IEEE TGRS / arXiv:2309.15214, 2023. [Online]. Available: <https://arxiv.org/abs/2309.15214>
- [73] M. Jin, S. Wang, L. Ma, Z. Chu, J. Y. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-F. Bian, and S. Pan, “Time-LLM: Time Series Forecasting by Reprogramming Large Language Models,” in ICLR, 2024. [Online]. Available: <https://arxiv.org/abs/2310.01728>
- [74] C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha, “The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery,” arXiv:2408.06292, 2024. [Online]. Available: <https://arxiv.org/abs/2408.06292>
- [75] M. Tian, L. Gao, S. D. Zhang, X. Chen, J. Deng, R. Kaufman, H. Gao, H. Zhu, J. Sun, B. Yu, Z. Lian, Y. Zhang, J. Guo, Y. Xu, S. Gao, Y. Miao, R. Chen, S. Li, H. Zhang, T. Zhang, Z. Ren, T. Liu, C. Xie, S. Yao, Z. Yang, K. A. Dill, M. Yu, W. Neiswanger, H. Liu, and C. Xiong, “SciCode: A Research Coding Benchmark Curated by Scientists,” ICML / arXiv:2407.13168, 2024. [Online]. Available: <https://arxiv.org/abs/2407.13168>
- [76] D. Watson-Parris, Y. Rao, D. Oliv  , Ø. Seland, P. Nowack, G. Camps-Valls, P. Stier, S. Bouabid, M. Dewey, E. Hodgson, P. Sansom, L. Mansfield, J. Runge, K. Kashinath, T. Beucler, D. Coumou, E. Barnes, and D. Grosvenor, “ClimateBench v1.0: A Benchmark for Data-Driven Climate Projections,” Journal of Advances in Modeling Earth Systems, 2022. [Online]. Available: <https://doi.org/10.1029/2021ms002954>
- [77] Z. Ben-Bouallegue, M. C. A. Clare, L. Magnusson, E. Gascon, M. Maier-Gerber, M. Janousek, M. Rodwell, F. Pinault, M. Alexe, M. Chantry, and P. D. Dueben, “The Rise of Data-Driven Weather Forecasting: A First Statistical Assessment of Machine Learning-Based Weather Forecasts in an Operational-Like Context,” Bulletin of the American Meteorological Society (BAMS), 2024. [Online]. Available: <https://doi.org/10.1175/BAMS-D-23-0162.1>

TABLE 2
Comprehensive Synthesis of Verified Scientific Multimodal Time Series Foundation Models and Reasoning Systems.

Model	Domain	Architectural Backbone	Pretraining Corpus & Scale	Resolution	Physics Level	Params	Verified Code / Access
Pangu-Weather [6]	Weather/Climate	3D Earth Transformer	ERA5 (1979–2017, 39 yrs)	0.25°, 13 lvls	Data-driven	256M	198808xc/Pangu-Weather
GraphCast [7]	Weather/Climate	Multi-Mesh GNN (Icosahedron)	ERA5 (1979–2017, 39 yrs)	0.25°, 37 lvls	Data-driven	36.7M	deepmind/graphcast
FourCastNet [8]	Weather/Climate	Adaptive Fourier Neural Operator	ERA5 (1979–2015, 36 yrs)	0.25°, 20 lvls	Soft loss	73.5M	NVlabs/FourCastNet
SFNO [35]	Weather/Climate	Spherical Fourier Neural Operator	ERA5 (1979–2017, 39 yrs)	0.25°, multi	Exact SO(3)	75M	neuraloperator/neuraloperator
ClimaX [69]	Weather/Climate	Variable-Tokenized ViT	CMIP6 simulations + ERA5	1.4°–5.6°	Data-driven	107M	microsoft/ClimaX
FuXi [34]	Weather/Climate	Cascade Swin Transformer	ERA5 (1979–2017, 39 yrs)	0.25°, 70 lvls	Data-driven	150M	tpys/FuXi
FuXi-Extreme [65]	Weather/Climate	Cascade Swin + Latent Diffusion	ERA5 extreme precipitation/wind	0.25°, surface	Tail preserving	150M	not available
FengWu [48]	Weather/Climate	Cross-Modal Attention Transformer	ERA5 (1979–2017, 39 yrs)	0.25°, 37 lvls	Soft loss	200M	OpenEarthLab/FengWu
FengWu-4DVar [62]	Weather/Climate	Coupled 4D-Var + FengWu Transformer	ERA5 + Asynchronous obs	0.25°, 37 lvls	Adjoint DA	180M	OpenEarthLab/FengWu-4DVar
DiffDA [64]	Weather/Climate	Conditional Denoising Diffusion	ERA5 + Real station observations	0.25°, surface	Posterior DA	68M	spsl/DiffDA
Score-based DA [63]	Weather/Climate	Continuous Score-based SDE	Lorenz-96 + 2D Kolmogorov flows	Continuous	Exact score	12M	francois-rozet/sda
GenCast [4]	Weather/Climate	Latent Diffusion Multi-Mesh GNN	ERA5 (1979–2018, 40 yrs)	0.25°, 37 lvls	Data-driven	120M	deepmind/graphcast
SEEDS [11]	Weather/Climate	Generative Diffusion Sampler	GEFS operational + ERA5	1.00°, multi	Soft loss	not reported	not available
Stormer [10]	Weather/Climate	Randomized Patch Transformer	ERA5 (1979–2018, 40 yrs)	1.41°, multi	Soft loss	67M	tung-nd/stormer
Aurora [9]	Weather/Climate	3D Perceiver / Swin Backbone	>1 PB (ERA5, HRES, GFS)	0.10°, multi	Data-driven	1.3B	microsoft/aurora
AIFS [57]	Weather/Climate	GNN Enc-Dec + Sliding Window Trans.	ERA5 + Operational IFS analyses	0.25°, 13 lvls	Operational NWP	185M	ecmwf/anemoi
NeuralGCM [51]	Weather/Climate	Differentiable Dynamical Core + NN	ERA5 (1979–2019, 40 yrs)	0.7°–2.8°	Hard laws	14.5M	google-research/neuralgcm
Prithvi WxC [60]	Weather/Climate	Scalable ViT Encoder-Decoder	MERRA-2 (160 vars, 44 yrs)	0.50°, 160 vars	Soft loss	2.3B	NASA-IMPACT/Prithvi-WxC
ClimSim [52]	Weather/Climate	Multiscale Physics Emulators	E3SM-MMF (5.7B column profiles)	Column multiscale	Hybrid PDE	Bench.	leap-stc/ClimSim
ClimateBench [76]	Weather/Climate	Climate Emulation Benchmark	CMIP6 emission scenarios	2.5° × 1.9°	Soft energy	Bench.	duncanwp/ClimateBench
ClimateLLM [55]	Weather/Climate	Fourier Spectral Decomposition	ERA5 Surface Station Series	Global series	Soft loss	7B	not available
SatMAE [12]	Earth Observ.	Temporal-Spectral Masked ViT	Sentinel-2, Landsat-8 cubes	10 m–30 m	Data-driven	86M–307M	sustainlab/SatMAE
Presto [13]	Earth Observ.	Lightweight Channel-Time ViT	Global Sentinel-1/2, ERA5	Pixel series	Data-driven	1.2M	nasaharvest/presto
Galileo [14]	Earth Observ.	Multi-Scale Masked Transformer	Global multi-sensor satellite	Multi-scale	Data-driven	100M	nasaharvest/galileo
DOFA [18]	Earth Observ.	Dynamic Wavelength Hypernetwork	5 satellite sensor modalities	Multi-scale	Wave physics	115M	zhu-xlab/DOFA
GEO-Bench [59]	Earth Observ.	Standardized Earth Monitoring ViT	Multi-sensor satellite series	Multi-scale	Spectral calib.	Platform	ServiceNow/geo-bench
SkySense [58]	Earth Observ.	Factorized Multimodal Spatio-Temporal ViT	21.5M multi-modal satellite sequences	Multi-scale	Spectral calib.	1.9B	Jack-bo1220/SkySense
EarthPT [17]	Earth Observ.	700M Autoregressive Decoder	Sentinel-2 spectral series	Pixel spectra	Data-driven	700M	not available
Prithvi-100M [15]	Earth Observ.	Geospatial Masked Autoencoder	NASA HLS reflectance series	30 m pixel	Data-driven	100M	NASA-IMPACT/hls-os
GeoChat [54]	Earth Observ.	Grounded Remote Sensing VLM	GeoChat-100K multimodal series	High-res	Spatial box	7B	mbzuai-oryx/GeoChat
AlphaEarth [16]	Earth Observ.	Neural Field Embedding Model	Global Earth observation constellation	10 m dense	Data-driven	not reported	Google Earth Engine
GlobalFlood [19]	Hydrology	EA-LSTM + River Routing	5,680 river gauges + ERA5	Multi-basin	Hybrid PDE	5M	neuralhydrology
Caravan [20]	Hydrology	Large-Sample Hydrology Suite	6,830 catchments + ERA5-Land	Catchment	Mass balance	Bench.	kratzert/Caravan
XiHe [21]	Oceanography	Hierarchical Ocean Transformer	GLORYS12V1 (1993–2020, 28 yrs)	1/12°, 50 lvls	Soft loss	84M	not available
4DVarNet [61]	Oceanography	Variational Neural Operator (4DVarNet)	SSH / SST satellite altimetry	Multi-scale	Variational DA	8.5M	CIA-Oceanix/4dvarnet
OceanGPT [23]	Oceanography	Domain-Adapted Marine LLM	DoInstruct multi-modal ocean series	Sensor/Doc	Hydro. cons.	7B	OceanGPT/OceanGPT
OceanBench [22]	Oceanography	Global Ocean Forecasting Benchmark	GLORYS12 reanalysis + Altimetry	1/12°	Hydro. cons.	Bench.	mercator-ocean/oceanbench
SeisT [24]	Seismology	Hierarchical Seismogram ViT	STEAD, INSTANCE (>10M waves)	100 Hz wave	Data-driven	12M	senli1073/SeisT
PhaseNet [25]	Seismology	1D Residual U-Net	NCEDC seismic catalogue	100 Hz wave	Data-driven	1.2M	A14EPS/PhaseNet
SeisBench [26]	Seismology	Standardized Seismology Platform	Global catalogues (>30M waves)	100 Hz wave	Soft loss	Platform	seisbench/seisbench
Surya [27]	Space Weather	Spatio-Temporal Gated ViT	SDO EUV series (13 yrs)	Multi-spectral	Soft loss	366M	not available
Chronos [66]	Cross-Domain	Quantized Tokenizer + T5 Transformer	Curated TS-Pile + Synthetic GP	Multi-rate	Data-driven	20M–710M	amazon/chronos
Lag-Llama [67]	Cross-Domain	Autoregressive Decoder + RoPE	Diverse multi-domain time series	Multi-sensor	Student- <i>t</i> head	2.4M	lag-llama
UniTS [32]	Cross-Domain	Unified Multi-Task Transformer	38 multi-domain time series datasets	Multi-rate	Data-driven	11M	mims-harvard/UniTS
MOMENT [33]	Cross-Domain	Masked Time Series Autoencoder	Time-series Pile (>13M series)	Multi-sensor	Data-driven	385M	moment-timeseries
Time-LLM [73]	Multi-Discip.	Patch Reprogramming + Frozen LLM	Multi-domain time series benchmarks	Multi-rate	Domain prompt	7B	KimMeen/Time-LLM
SciTS [5]	Multi-Discip.	TimeOmni Continuous Tokenizer	SciTS Benchmark (12 domains)	Multi-rate	Data-driven	8B	OpenTSLab/TimeOmni
K2 [53]	Multi-Discip.	Geoscience Foundation LLM	>5.5B tokens GeoSignal corpus	Series/Doc	Physical prior	7B	davendw49/k2
ClimateAgent [56]	Multi-Agent	Multi-Agent LLM Orchestrator	Climate-Agent-Bench-85	Code/Tools	Hybrid PDE	Frontier LLM	not available
Operational Protocol [77]	Multi-Discip.	Operational NWP Statistical Verification	ECMWF operational archive	0.1°–0.25°	Operational NWP	Platform	not available
WeatherBench 2 [3]	Multi-Discip.	Standardized Evaluation Framework	ERA5, HRES, IFS Analysis	0.25°, multi	Physical diag.	Bench.	google-research/weatherbench2
ExtremeCast [46]	Weather/Climate	Exloss + ExBooster Modular ViT	ERA5 (1979–2015, 37 yrs)	0.25°, 13 lvls	Soft loss	120M	black-yt/ExtremeCast
MG-TFNO [39]	Core Methods	Multi-Grid Tensorized FNO	Multiscale 2D/3D Navier-Stokes	1024 ² , continuous	Hybrid PDE	0.8M	neuraloperator/neuraloperator
WaveFNO [40]	Seismology	Fourier Neural Operator	Continuous elastic wave simulations	Continuous	Hybrid PDE	1.2M	neuraloperator/neuraloperator
IceNet [28]	Polar Cryosphere	Probabilistic U-Net Ensemble	CMIP6 (2192 yrs) + OSI-SAF	25 km, monthly	Soft loss	15M	icenet-ai/icenet
IceBench [29]	Polar Cryosphere	Multi-Modal Polar Benchmark	A14Arctic Sentinel-1 SAR + AMSR2	Multi-sensor	Spectral calib.	Bench.	bdlab-ucd/IceBench
PHQFNO [41]	Quantum ML	Partitioned Hybrid Quantum FNO	Burgers' + 2D wave equations	Continuous	Hybrid PDE	16-qubit	not available
U-FNO [30]	Geo-Energy	Multiscale Fourier Neural Operator	Stanford GCS (6,000 simulations)	Continuous 2D/3D	Soft mass loss	1.2M	gegewen/ufno

TABLE 3
Comprehensive Multi-Domain Quantitative Benchmark Meta-Table across Atmospheric, Climate, Hydrological, Oceanic, and Seismological Domains.

Part A: WeatherBench 2 Headline Atmospheric Deterministic Verification (Global Latitude-Weighted RMSE)									
Model / System	Architectural Family	Z500 Geopotential (m ² /s ²)			T850 Temperature (K)				
		Day 3 (72h)	Day 5 (120h)	Day 7 (168h)	Day 3 (72h)	Day 5 (120h)	Day 7 (168h)		
Operational IFS HRES	Numerical Weather Prediction	152.8	333.7	558.2	1.37	2.06	2.68	Operational baseline in [6], [3]	
FourCastNet [8]	Adaptive Fourier Neural Operator	205.6	440.2	675.0	1.42	2.48	3.25	Stated in [8], [3]	
SFNO [35]	Spherical Fourier Operator	158.0	345.2	560.4	1.25	1.95	2.55	Stated in [35]	
Pangu-Weather [6]	3D Earth Transformer (3DEST)	134.5	296.7	502.1	1.14	1.79	2.38	Stated in [6], [3]	
FuXi [34]	Cascade Swin Transformer	125.4	278.1	472.0	1.10	1.71	2.29	Stated in [34]	
AIFS [37]	GNN Enc-Dec + Sliding Window	124.6	275.2	470.5	1.08	1.68	2.27	Stated in [37]	
GraphCast [7]	Multi-Mesh Spherical GNN	118.2	268.4	465.3	1.06	1.65	2.24	Stated in [7], [3]	
NeuralGCM (0.7°) [51]	Differentiable Dynamical Core + NN	128.0	282.5	476.8	1.11	1.73	2.31	Stated in [51]	
Part B: ClimSim Multidecadal Climate & Subgrid Physics Emulation									
Model / Architecture	Methodology	Surface Temp. T _{AS} (K)		Precipitation PR (mm/day)		ClimSim Subgrid (R ²)			
		Annual RMSE	Decadal Trend	Annual RMSE	Decadal Trend	Heating dT/dt	Moistening dq/dt		
Linear Pattern Scaling	Empirical Linear Projection	0.32	0.28	0.46	0.41	0.38	0.31	Standard CMP baseline [76]	
Random Forest (RF)	Non-linear Ensemble Trees	0.28	0.24	0.39	0.35	0.54	0.48	Stated in [76]	
Gaussian Process (GP)	Spatial Covariance Prior	0.21	0.17	0.34	0.29	0.59	0.52	Physically constrained GP [76]	
ClimSim Standard MLP	Column Multilayer Perceptron					0.62	0.58	Baselines in ClimSim [52]	
Convolutional Net (CNN)	Deep Spatio-Temporal Net	0.18	0.14	0.31	0.26	0.74	0.70	Deep emulator benchmark [76]	
ClimSim Multiscale Trans.	Multi-Scale Column Transformer	0.16	0.12	0.28	0.23	0.82	0.79	Hybrid PDE physics emulator [52]	
Part C: Global Hydrology Streamflow Prediction across 5,680 Ungauged Basins (Caravan & GlobalFlow)									
Model / Framework	Methodology	Median Nash-Sutcliffe Efficiency (NSE)			Kling-Gupta Efficiency (KGE)				
		Day 1 Lead	Day 3 Lead	Day 5 Lead	Day 1 Lead	Day 3 Lead	Day 5 Lead		
Operational NWP (GloFAS)	Land Surface + River Routing	0.35	0.31	0.26	0.42	0.38	0.32	Global operational baseline [19]	
Conceptual HBV Baseline	Calibrated Hydrological Model	0.48	0.41	0.33	0.52	0.46	0.38	Large-sample benchmark [20]	
Standard LSTM	Temporal Recurrent Network	0.67	0.61	0.53	0.71	0.65	0.57	Cross-basin transfer in [20]	
GlobalFlow (EA-LSTM) [19]	Entity-Aware LSTM + Routing	0.74	0.69	0.62	0.78	0.73	0.66	Ungauged basins in [19]	
Part D: Global Ocean Forecasting on OceanBench (Verification vs. Copernicus Marine GLORYS12)									
Model / Architecture	Model Family	Sea Surface Height SSH RMSE (m)			Sea Surface Temperature SST RMSE (K)				
		Day 1 (24h)	Day 5 (120h)	Day 10 (240h)	Day 1 (24h)	Day 5 (120h)	Day 10 (240h)		
Persistence / Climatology	Empirical Baseline	0.048	0.120	0.142	0.38	0.68	0.95	Operational reference [22]	
Standard 2D U-Net	Convolutional Encoder-Decoder	0.038	0.082	0.115	0.28	0.45	0.68	Baseline in OceanBench [22]	
XtNet / Spherical FNO [21]	Hierarchical Spherical Operator	0.024	0.054	0.088	0.19	0.32	0.52	Stated in [21], [22]	
Part E: Seismological Continuous Waveform Phase Picking (Standardized SeisBench Platform)									
Model / Architecture	Waveform Representation	P-Wave Arrival MAE (s)		S-Wave Arrival MAE (s)		Phase Detection F ₁ Score (%)			
		STEAD	INSTANCE	STEAD	INSTANCE	STEAD	INSTANCE		
Classical STA/LTA	Energy Ratio Filter (100 Hz)	0.185	0.210	0.345	0.380	62.4	58.1	Classical picking baseline [26]	
CPD (Deep CNN)	1D Convolutional Network	0.078	0.092	0.155	0.178	81.5	79.2	Deep learning baseline [26]	
PhaseNet [25]	1D Residual U-Net	0.048	0.056	0.085	0.098	91.2	89.6	Grounded picking in [25], [26]	
SeisT [24]	Hierarchical Seismogram ViT	0.034	0.041	0.061	0.072	95.8	94.3	Foundation model in [24]	
Part F: Sub-Kilometer Convective-Scale Radar Nowcasting & Generative Downscaling									
Model / Architecture	Framework / Paradigm	Critical Success Index (CSI @ Lead Time)			Fidelity & Physical Spectral Retention				
		CSI (≥1mm/h)	CSI (≥5mm/h)	CSI (≥10mm/h)	Resolution	CRPS Score	PSD Spectrum Match		
PySTEPS (Optical Flow)	Advection-Semi-Lagrangian	0.285	0.142	0.082	1 km	0.385	High-freq blurring	Operational baseline [70]	
ConvGRU Baseline	Convolutional Recurrent Net	0.312	0.165	0.098	1 km	0.342	Severe spatial blur	Baseline in [70]	
DGMR (Generative GAN) [70]	Dual-Frequency Spectral GAN	0.368	0.214	0.145	1 km	0.298	Sharp spectral texture	Stated in [70]	
MetNet-3 [71]	Axial Multi-Scale Transformer	0.382	0.228	0.156	1-4 km	0.274	Continental consistency	Stated in [71]	
CorrDiff (Taiwan 2km) [72]	Residual Corrective Diffusion	0.395	0.245	0.172	2 km	0.252	Complete PSD retention	Stated in [72]	
Part G: Extreme Value Weather Prediction Benchmark (Heavy-Tailed Perils on WeatherBench 2)									
Model / Framework	Loss / Sampling Paradigm	Critical Success Index (CSI @ 99th Percentile)			Symmetric Extremal Dependence Index (SEDI)				
		Wind U10 ≥ 20m/s	Precip PR ≥ 50mm	Heat T2m ≥ 35°C	Wind U10	Precip PR	Heat T2m		
Operational IFS HRES	Numerical PDE Baseline	0.228	0.185	0.312	0.684	0.612	0.745	Operational baseline in [46], [3]	
Pangu-Weather [6]	Symmetric L ₁ Regression	0.142	0.095	0.234	0.542	0.468	0.621	Stated in [46]	
GraphCast [7]	Symmetric L ₂ Autoregression	0.165	0.112	0.258	0.589	0.504	0.655	Stated in [46]	
FuXi-Extreme [65]	Cascade Swin + Diffusion	0.245	0.201	0.328	0.712	0.638	0.762	Stated in [65]	
ExtremeCast [46]	Asymmetric ExLose + ExBooster	0.278	0.234	0.365	0.758	0.682	0.805	State-of-the-art in [46]	
Part H: Polar Cryosphere Sea Ice Forecasting & SAR Classification Benchmark									
Model / System	Observational Input Modality	Pan-Arctic Sea Ice Brier Score (Lower is Better)			Integrated Ice Edge Error (IIEE, 10 ³ km ²)				
		Lead 1 Month	Lead 3 Months	Lead 6 Months	Lead 1 Month	Lead 3 Months	Lead 6 Months		
ECMWF SEAS5	Dynamical Seasonal Physics	0.098	0.142	0.185	5.82	8.45	11.20	Operational seasonal NWP [28]	
Climatology Reference	40-Year Empirical Mean	0.115	0.168	0.210	6.94	10.12	13.45	Baseline in [28]	
Standard 2D U-Net	ERA5 + OSI-SAF (No CMIP6)	0.082	0.124	0.165	4.95	7.30	9.55	Empirical baseline in [28]	
IceNet (Ensemble) [28]	CMIP6 + ERA5 + OSI-SAF Satellite	0.065	0.098	0.134	3.92	5.75	7.90	Nature Comms milestone [28]	
IceBench Baseline (U-Net) [29]	Sentinel-1 SAR + AMSR2 Radiometry		SAR Sea Ice Classification Macro F ₁ : 78.4%			Pixel Transfer Accuracy: 82.1%		Standard benchmark [29]	
IceBench ViT Baseline [29]	Sentinel-1 SAR + AMSR2 + ERA5		SAR Sea Ice Classification Macro F ₁ : 84.6%			Pixel Transfer Accuracy: 87.8%		State-of-the-art in [29]	
Part I: Subsurface Geo-Energy & Carbon Sequestration Benchmark (30-Year Multiphase Plume Migration)									
Model / System	Network Backbone	30-Year Predictive Error Metrics			Plume Fidelity & Simulation Speedup				
		Gas Saturation S _g RMSE	Pressure Δp Error (MPa)	Plume Contour IoU	Inference Time	Speedup vs Simulator	Mass Cons. Error		
Standard 2D U-Net	Convolutional Encoder-Decoder	0.084	1.45	0.72	1.80 s	~500x vs CMG GEM	4.8%	Baseline in [30]	
Standard FNO Baseline	Spectral Fourier Convolution	0.062	0.98	0.78	0.05 s	~15,000x vs CMG GEM	3.2% (Gibbs noise)	Baseline in [30]	
U-FNO [30]	Multiscale Spectral U-Net	0.028	0.42	0.91	0.08 s	~10,000x vs CMG GEM	<0.8%	Grounded in [30]	
CCSNet [31]	Thermodynamic EOS Multi-Scale	0.031	0.38	0.93	0.45 s	~15,000x vs TOUGH2	<0.5%	Grounded in [31]	
Geo-FNO [36]	Diffomorphism Spectral Operator	0.035	0.49	0.89	0.12 s	~1,500x vs FEM	<1.0%	Grounded in [36]	
Part J: Symplectic & Lie-Poisson Geometric Invariant Preservation (10 ⁴ Autoregressive Time Steps)									
Model / Operator	Physical System	Geometric Conservation Invariant Errors			Long-Horizon Stability Metrics				
		Energy Drift ΔH /H ₀	Casimir Invariant Error	Circulation Drift	Phase Volume Error	Rollout Horizon	Energy Dissipation		
Unconstrained FNO	2D Euler / Dispersive Wave	42.5%	38.2%	18.4%	Non-preserving	Fails at 500 steps	High artificial damping	Stated in [37]	
Neural ODE (ClimODE)	Continuous Advection Flow	8.4%	9.1%	4.2%	Approximate	Stable to 2,000 steps	Moderate dissipation	Stated in [68]	
SNO (Symplectic FNO) [37]	Infinite-Dim Hamiltonian Wave	<0.02%	N/A	N/A	0.0% (Exact)	Stable past 10 ⁴ steps	Zero numerical damping	Stated in [37]	
LPNets [38]	Rotating 2D Euler / Shallow Water	<0.01%	<0.01% (Exact)	0.0% (Exact)	Coadjoint orbit preserved	Stable past 10 ⁴ steps	Zero entropy cascade	Stated in [38]	
Part K: Generative Lagrangian In-Situ Ocean Data Assimilation (<0.5% Drifter Coverage)									
Assimilation System	Observation Modality	Eulerian Field Reconstruction Errors			Kinetic Energy & Spectral Metrics				
		Velocity RMSE (u, v, m/s)	SSH RMSE (cm)	Vorticity ζ Corr.	EKE Spectrum Recovery	Submesoscale Eddies	Coverage Sparsity		
Optimal Interpolation (OI)	Sparse Lagrangian Trajectories	0.285	4.82	0.41	42.1%	Severe spatial blur	0.5% basin coverage	Baseline in [47]	
4D-Var Data Assimilation	Variational Inversion	0.210	3.45	0.58	61.5%	High-wavenumber loss	0.5% basin coverage	Baseline in [47]	
LagDA (Generative Diffusion) [47]	Conditional Score Diffusion	0.092	1.65	0.86	91.4%	Sharp eddy structures	0.5% basin coverage	State-of-the-art in [47]	
Part L: Coupled Atmosphere-Ocean Multi-Decadal Climate Emulation (100-to-1,000 Year Rollouts)									
Model / System	Coupled Architecture	Coupled Climate Drift & ENSO Skill			Simulation Throughput & Resolution				
		GMST Drift (K/century)	Nino 3.4 RMSE (K)	ENSO Power Corr.	Grid Resolution	Throughput (SYDP)	Simulation Horizon		
Numerical GCM (E3SMv2)	Coupled Primitive Equations	0.12	0.45	0.88	1.00°, 145 lvs	15 SYDP (HPC)	Multi-century	Baseline in [45]	
ACE (Atmos SFNO) [50]	Prescribed SST Atmosphere	<0.04	N/A	N/A	1.00°, 82 vars	1,400 SYDP	100-year continuous	Grounded in [50]	
DLExSM [44]	Dual HEALPix U-Net Coupler	<0.08	0.38	0.89	Equal-area HEALPix	320 SYDP	1,000-year continuous	Grounded in [44]	
SamudraACE [45]	Coupled ACE + Samudra 3D	<0.05	0.32	0.93	1.00°, 145 vars	1,100 SYDP	Multi-century	Grounded in [45]	
Part M: Cross-Modal Remote Sensing Foundation Models (Multi-Sensor Extrapolation & Joint Optical-SAR Fusion)									
Model / System	Pretraining Architecture	Downstream Classification & Regression Metrics			Extrapolation & Sensor Flexibility				
		EuroSAT Clf. Acc.	BioMasters RMSE (Mg/ha)	PASTIS Crop F1	Spatial Extrap.	Supported Sensors	Resolution Range		
SatMAE [12]	Temporal-Spectral ViT	97.4%	34.5	78.4%	1.0x (Fixed)	Sentinel-2, Landsat	10m-30m	Baseline in [42]	
Presto [13]	Pixel Channel-Time ViT	96.8%	35.8	76.9%	N/A (Pixel-wise)	Sentinel-1/2, ERA5	Pixel series	Baseline in [13]	
CROMA [42]	Radar-Optical MAE + 2D-ALiBi	98.6%	29.8	82.5%	17.6x (Zero-shot)	Sentinel-1 SAR + S2	10m	State-of-the-art in [42]	
AnySat [43]	Scale-Adaptive JEPA	98.2%	30.4	81.9%	Any continuous	11 Sensors (GeoPlex)	0.2m-500m	State-of-the-art in [43]	
Part N: Subgrid Microphysical Parameterization and Conservation									
Parameterization Model	Constraint Mechanism	Column Conservation Invariant Residuals			Emulation Accuracy & Rollout Stability				
		Enthalpy Error (W/m ²)	Water Balance Error	Mass Conservation	Heating Q ₁ R ²	Moistening Q ₂ R ²	GCM Stability		
Standard Unconstrained NN	Purely Data-Driven	42.8	14.2 mm/day	Non-conserving	0.81	0.74	Blowup in <60 days	Baseline in [49]	
Soft Penalty Loss	Lagrange Multiplier Penalty	5.6	1.8 mm/day	Approximate	0.83	0.81	Drifts over 1 year	Baseline in [49]	
ClimSim Hybrid [52]	Multi-Scale Physics Loss	2.1	0.8 mm/day	Approximate	0.85	0.81	Indefinitely stable	Stated in [52]	
AC-NN [49]	Exact Terminal Projection Layer	<10 ⁻¹⁵ (Exact)	<10 ⁻¹⁵ (Exact)	Machine precision	0.88	0.76	Indefinitely stable	Grounded in [49]	

TABLE 4
Quantitative Profiling of Computational Efficiency, Throughput, Hardware Footprint, and Energy vs. Operational NWP Systems.

Model / Operational System	Params	Compute Hardware	Pretraining Cost	10-Day Wall-Clock	Speedup vs. IFS	Energy / Forecast	Peak Memory	Footprint
Operational IFS HRES (0.10°)	N/A	Atos Sequana HPC (~3,500 CPUs)	N/A (Numerical PDE)	~3,000 s (50 min)	1× (Baseline)	~180 kWh	Multi-node (~1.5 TB)	cluster
Operational IFS ENS (51 members)	N/A	Atos Sequana HPC (~10,000 CPUs)	N/A (Numerical PDE)	~7,200 s (120 min)	1× (Baseline)	~2,200 kWh	Multi-node (>4.0 TB)	cluster
Pangu-Weather [6]	256M	1 × NVIDIA Tesla V100 GPU	192 V100 GPUs × 16 days	2.0 s	~1,500×	~0.00015 kWh	12.4 GB VRAM	
GraphCast [7]	36.7M	1 × Google Cloud TPU v4	32 TPU v4 devices × 4 weeks	58.0 s	~50×	~0.0050 kWh	28.5 GB HBM	
FourCastNet [8]	73.5M	1 × NVIDIA A100 GPU	64 A100 GPUs × 16 hours	0.25 s	~12,000×	~0.00003 kWh	8.2 GB VRAM	
FuXi [34]	150M	1 × NVIDIA A100 GPU	32 A100 GPUs × 7 days	6.0 s	~500×	~0.0006 kWh	18.6 GB VRAM	
AIFS [57]	185M	1 × NVIDIA A100 GPU	64 A100 GPUs × 14 days	20.0 s	~150×	~0.0018 kWh	22.0 GB VRAM	
NeuralGCM (0.70°) [51]	14.5M	1 × Google Cloud TPU v4	4 TPU v4 devices × 5 days	30.0 s	~100×	~0.0028 kWh	9.8 GB HBM	
GenCast (50-member ENS) [4]	120M	8 × Google Cloud TPU v4	64 TPU v4 devices × 6 weeks	~900 s (15 min)	~8×	~0.85 kWh	31.0 GB HBM / worker	
CorrDiff (2km regional) [72]	110M	1 × NVIDIA A100 GPU	32 A100 GPUs × 10 days	1.5 s	~1,000× vs WRF	~0.00025 kWh	14.2 GB VRAM	
ClimODE (Global continuous) [68]	0.3M	1 × NVIDIA RTX 3090 GPU	1 GPU × 24 hours	0.4 s	~5,000×	~0.00004 kWh	2.1 GB VRAM	
DGMR (1km radar nowcast) [70]	89M	1 × Google Cloud TPU v3	128 TPU v3 × 3 days	0.8 s	~250×	~0.00012 kWh	11.5 GB HBM	
MG-TFNO (Multi-Grid FNO) [39]	0.8M	1 × NVIDIA A100 GPU	1 A100 GPU × 12 hours	0.05 s	~150× vs Dense FNO	~0.00001 kWh	1.4 GB VRAM	
WaveFNO (Seismic FWI) [40]	1.2M	1 × NVIDIA RTX 3090 GPU	1 GPU × 8 hours	2.1 s	~1,200× vs FD PDE	~0.00018 kWh	2.8 GB VRAM	
IceNet (6-Month Arctic) [28]	15M	1 × NVIDIA Tesla V100 GPU	8 V100 GPUs × 4 days	1.2 s	~2,000× vs SEAS5	~0.00009 kWh	4.5 GB VRAM	
U-FNO (Multiphase Darcy) [30]	1.2M	1 × NVIDIA RTX 3090 GPU	1 GPU × 14 hours	0.08 s	~10,000× vs CMG GEM	~0.00001 kWh	1.8 GB VRAM	
CCSNet (50-Yr GCS Suite) [31]	14.8M	1 × NVIDIA Tesla V100 GPU	4 V100 GPUs × 3 days	0.45 s	~15,000× vs TOUGH2	~0.00004 kWh	3.2 GB VRAM	
Geo-FNO (Complex Geometries) [36]	2.1M	1 × NVIDIA A100 GPU	1 A100 GPU × 16 hours	0.12 s	~1,500× vs FEM	~0.00002 kWh	2.4 GB VRAM	
LPNets (Lie-Poisson Fluid) [38]	0.8M	1 × NVIDIA RTX 3090 GPU	1 GPU × 6 hours	0.04 s	~800× vs Symplectic PDE	~0.00001 kWh	1.1 GB VRAM	
CROMA (Radar-Optical MAE) [42]	108M	1 × NVIDIA A100 GPU	4 A100 GPUs × 4 days	0.15 s	~500× vs Pixel-wise	~0.00003 kWh	6.2 GB VRAM	
AnySat (Multi-GSD JEPa) [43]	125M	1 × NVIDIA A100 GPU	8 A100 GPUs × 5 days	0.18 s	~600× vs Resampling	~0.00004 kWh	7.8 GB VRAM	
ACE (100-Yr Climate SFNO) [50]	82M	1 × NVIDIA A100 GPU	16 A100 GPUs × 3 days	0.06 s (1 yr in 2.2 s)	~100× vs FV3GFS	~0.00001 kWh	4.5 GB VRAM	
DLESyM (Coupled Earth Model) [44]	42M	1 × NVIDIA A100 GPU	8 A100 GPUs × 4 days	0.27 s (1 yr in 9.8 s)	~50× vs CMIP6	~0.00005 kWh	8.4 GB VRAM	
SamudrACE (Coupled 3D Ocean-Atmos) [45]	520M	1 × NVIDIA H100 GPU	32 H100 GPUs × 7 days	0.08 s (1 yr in 2.9 s)	~100× vs E3SMv2	~0.00002 kWh	14.8 GB VRAM	
AC-NN (Constrained Convection) [49]	2.1M	1 × NVIDIA RTX 3090 GPU	1 GPU × 8 hours	0.01 s	~1,000× vs CRM	~0.000005 kWh	1.2 GB VRAM	